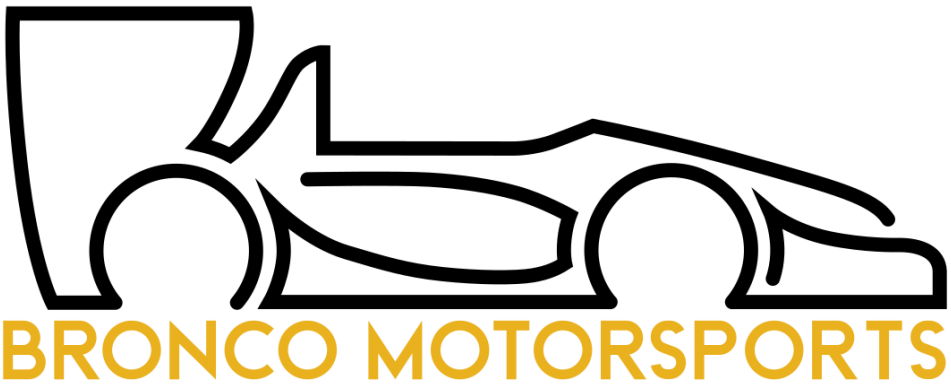


California State Polytechnic University, Pomona
Formula SAE

2025-2026 Design Binder

Hubs and Uprights

Nathan Su

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BM 26 Full Vehicle Objectives

This year, the overall vehicle goals are **refinement, reliability, and simplicity**. These three top level attributes would hopefully get us a top 10 finish (668 points) by scoring well in design, performing at a high level, and yet still be reliable so we can place high in endurance. The previous year, our main goal was simplicity and reliability because our previous cars were notoriously unreliable, causing DNFs and low points scoring. Bringing things back to basics allowed everyone to re-gather themselves by learning the fundamentals of their respective systems. This simplicity and reliability came at the cost of performance, with many people reverting back to older legacy designs, as well as making their systems have higher factors of safety. Now this year, we still obviously want to build a reliable and simple car, but now a goal is to push for more performance as well. We also wanted to take feedback from the previous design judges and implement their changes. Increasing performance and taking judge feedback both tie into the refinement goal.

Vehicle Dynamics Objectives

More specifically, the vehicle dynamics system goals are **reducing setup/assembly times, reaching ~1.4 lateral g's in steady state cornering, increasing collaboration with aerodynamics, and keeping our components reliable** throughout the testing phase and competition. This year, we accelerated our timeline to try and maximize the amount of testing time before competition. However, during vehicle dynamics testing, changing car setup parameters like camber, toe, spring rate, etc. takes time. Minimizing the time, it takes to do change setup allows us to have possibly hours of extra time to gather data and fully understand how our car and find a good setup that is fast and drivable. 1.4 lateral g's were what the previous year's top teams were hitting in their competition skid pad runs, so using this as a benchmark allows us to directly compare our car's performance to the competition's. Collaborating with the aerodynamics team was a point of feedback from one of the judges last year. Implementing his feedback should give us a boost in points.

Hubs and Uprights Subsystem Objectives

The goals for the hubs and uprights this year are to maintain a **1.5 factor of safety for stress and fatigue on all components, reduce component weights by 10% or more**, and accomplish all of the VD and full car goals. Last year, to align with vehicle goals, the hubs and uprights system used a simple, tried a true reliable, easy to manufacture legacy design. This design, while reliable, had very high factors of safety ranging from 2.5 to 6. These values are overkill since we have a very good idea of what the load cases are that the car experiences. Getting the FOS values down to 1.5 while still employing good design principles will consequently also reduce weight in the parts, increasing performance.

Purpose of Subsystem

Hubs:

• Establish a designated position for affixing the wheel, brake rotor, and wheel speed sensor gear.

- Connect the wheel securely to the vehicle structure.

• Axial constraint is achieved through the hub nuts, housing of the upright, and hub design.

- Constraints act on the inner race of wheel bearings.

• Engineering Concepts

- Transfers tire lateral, longitudinal, and vertical forces from tire contact patch to wheel bearings & uprights which support the vehicle's weight.
- Overturning moment from the tire lateral force and car weight reacted by wheel bearing radial loads as well as axial loads. Determined for bearing selection.

Upright:

• Provides physical mounting for suspension linkages, tie rods, brake caliper, wheel speed sensor, wheel bearings, half shafts, and the hub.

- Assembly with the hub is secured through concentrically constraining the bearings.

• Characterizes vehicle dynamics suspension and steering parameters.

- Kingpin inclination angle → steering stability, alignment stability, etc.
- Caster angle → steering effort, cornering stability, dynamic camber changes, etc.
- Ackermann geometry → inner and outer wheel slip angle difference

• Engineering concepts

- Supports loads transmitted through tire contact patch and distributes them throughout suspension and steering linkages.
- Axial load from wheel bearings
- Force couple from bearing radial loads.
- Mechanical linkages in tension/compression as well as overturning moments

FSAE 2025-2026 Rules

V.3.2.3 Steering systems must use a rigid mechanical linkage capable of tension and compression loads for operation

V.3.2.4 The steering system must have positive steering stops that prevent the steering linkages from locking up (the inversion of a four bar linkage at one of the pivots). The stops:

a. Must prevent the wheels and tires from contacting suspension, bodywork, or Chassis during the track events

b. May be placed on the uprights or on the rack

V.4.2 Wheel Attachment

V.4.2.1 Any wheel mounting system that uses a single retaining nut must incorporate a device to retain the nut and the wheel if the nut loosens.

A second nut (jam nut) does not meet this requirement

V.4.2.2 Teams using modified lug bolts or custom designs must provide proof that Good Engineering Practices have been followed in their design.

V.4.2.3 If used, aluminum wheel nuts must be hard anodized and in pristine condition.

Background/Theory

The hubs and uprights are an integral part of the car's suspension system. It mounts the A-arms, steering arms, push rods, and brakes. Because of this, they see a tremendous amount of load, so it is extremely important to accurately determine those loads so that the final design can withstand real life loads and have the highest achievable performance. Using IMU data from our car testing and endurance run last year, we have determined the following load cases and corresponding accelerations.

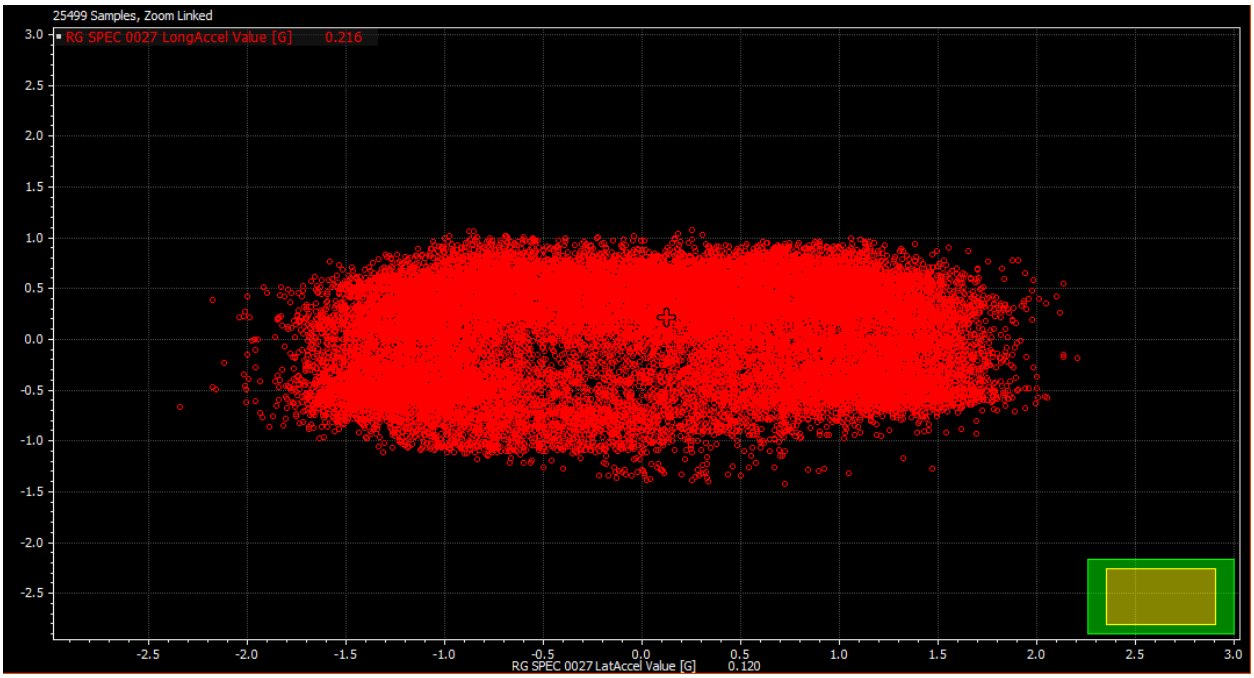


Figure 1: G-G Diagram

BRONCO MOTORSPORTS

BM26 FORCE MODULES

Set #	Name	Acceleration		
		Lateral: Y (g)	Longitudinal: X (g)	Vertical: Z (g)
0	Pure Cornering (Left Turn)	1.96	0	1
1	Pure Acceleration	0	1.16	1
2	Pure Breaking	0	-1	1
3	Cone Strike (FR)	0	0	2.58
4	Combined Corner and Brake (Left Turn)	1.25	-0.6	1
5	Combined Corner and Cone (FR)	1.96	0	2.58
6	Combined Brake and Cone (FR)	0	-1.22	2.58
7	Combined Corner and Accel	1.37	1.07	1
8	Combined Corner Brake and Cone	1.59	-0.63	2.58

Figure 2: Max Load Cases

Using these accelerations, we can then use OptimumKinematics software to calculate the loads on the suspension components. By inputting the suspension geometry used on our car, we get a force module of all of the loads on each suspension component and the tires.

Load Cases Analysis for OptK Force Module

Set #	Name	Vertical Loads [lbf] (Fz)				Lateral Loads [lbf] (Fy)			
		Front Left	Front Right	Rear Left	Rear Right	Front Left	Front Right	Rear Left	Rear Right
0	Pure Cornering (Left Turn)	24.53	274.73	11.18	319.58	48.07	538.46	21.90	626.37
1	Pure Acceleration	42.98	42.98	272.03	272.03	0.00	0.00	0.00	0.00
2	Pure Breaking	256.51	256.51	58.50	58.50	0.00	0.00	0.00	0.00
3	Cone Strike (FR)	149.63	386.03	165.38	165.38	0.00	0.00	0.00	0.00
4	Combined Corner and Brake (Left Turn)	121.92	287.72	8.09	212.29	152.39	359.64	10.11	265.36
5	Combined Corner and Cone (FR)	251.83	488.23	39.48	291.28	290.39	563.00	45.52	335.88
6	Combined Brake and Cone (FR)	256.51	492.91	58.50	58.50	0.00	0.00	0.00	0.00
7	Combined Corner and Accel (Left Turn)	(-) OFF	163.65	159.62	358.62	(-) OFF	207.14	202.04	453.93
8	Combined Corner Brake and Cone (Left Turn & FR)	121.92	524.12	8.09	212.29	140.95	605.97	9.35	245.43

Set #	Name	Longitudinal Loads [lbf] (Fx)				Self Alligning Torque [lbf*ft]			
		Front Left	Front Right	Rear Left	Rear Right	Front Left	Front Right	Rear Left	Rear Right
0	Pure Cornering (Left Turn)	0.00	0.00	0.00	0.00	N/A			
1	Pure Acceleration	49.85	49.85	315.55	315.55				
2	Pure Breaking	256.51	256.51	58.50	58.50				
3	Cone Strike (FR)	0.00	0.00	0.00	0.00				
4	Combined Corner and Brake (Left Turn)	73.15	172.63	4.85	127.37				
5	Combined Corner and Cone (FR)	0.00	0.00	0.00	0.00				
6	Combined Brake and Cone (FR)	227.55	437.27	51.89	51.89				
7	Combined Corner and Accel (Left Turn)	0.00	161.78	157.80	354.53				
8	Combined Corner Brake and Cone (Left Turn & FR)	55.85	240.10	3.70	97.25				

Figure 3: Tire Loads

BM26 Force Results [Tires:R20] UPRIGHT (lbs)

Front Left

Set #	Name	Upright to Lower A-Arm Fx	Upright to Lower A-Arm Fy	Upright to Lower A-Arm Fz	Upright to Upper A-Arm Fx	Upright to Upper A-Arm Fy	Upright to Upper A-Arm Fz	Upright to Tierod Fx	Upright to Tierod Fy	Upright to Tierod Fz
0	Pure Cornering (Left Turn)	2.731	106.861	7.048	-0.237	-41.770	20.311	-2.494	-17.020	-2.829
1	Pure Acceleration	108.806	44.041	0.754	-44.198	-2.455	54.961	-8.918	-41.586	-7.705
2	Pure Breaking	785.397	268.311	8.019	-320.997	35.956	292.817	-63.741	-304.268	-44.326
3	Cone Strike (FR)	5.468	62.994	-4.041	-3.903	-55.523	133.892	-1.565	-7.471	-0.417
4	Combined Corner and Brake (Left Turn)	159.436	451.324	22.751	-61.610	-157.617	116.928	-21.352	-126.672	-19.590
5	Combined Corner and Cone (FR)	17.009	655.804	-44.425	-4.247	-304.475	297.749	-13.017	-73.336	-3.724
6	Combined Brake and Cone (FR)	444.168	201.927	-18.692	-175.130	-8.308	278.213	-41.488	-198.234	-8.546
7	Combined Corner and Accel	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
8	Combined Corner Brake and Cone	116.803	342.491	6.010	-45.284	-126.900	128.376	-15.669	-83.208	-9.302

Front Right

Set #	Name	Upright to Lower A-Arm Fx	Upright to Lower A-Arm Fy	Upright to Lower A-Arm Fz	Upright to Upper A-Arm Fx	Upright to Upper A-Arm Fy	Upright to Upper A-Arm Fz	Upright to Tierod Fx	Upright to Tierod Fy	Upright to Tierod Fz
0	Pure Cornering (Left Turn)	-1.147	672.332	-61.193	-16.146	-253.393	322.608	17.226	-80.480	13.316
1	Pure Acceleration	108.806	-44.041	0.754	-44.198	2.455	54.961	-8.918	41.586	-7.705
2	Pure Breaking	785.397	-268.311	8.019	-320.997	-35.956	292.817	-63.741	304.268	-44.326
3	Cone Strike (FR)	14.107	-162.519	-10.425	-10.070	143.243	397.026	-4.037	19.275	-1.075
4	Combined Corner and Brake (Left Turn)	352.092	661.766	-44.845	-159.094	-260.004	341.734	-11.738	55.708	-9.169
5	Combined Corner and Cone (FR)	2.826	862.892	50.327	-17.420	-227.157	435.380	15.234	-72.734	3.585
6	Combined Brake and Cone (FR)	853.531	-388.028	-35.919	-336.537	15.962	534.615	-79.724	380.935	-16.422
7	Combined Corner and Accel	322.519	365.733	-27.769	-137.509	-129.282	209.362	-17.060	78.598	-13.118
8	Combined Corner Brake and Cone	467.968	819.486	-20.260	-208.260	306.899	551.275	-19.608	93.383	-10.394

Rear Left

Set #	Name	Upright to Lower A-Arm Fx	Upright to Lower A-Arm Fy	Upright to Lower A-Arm Fz	Upright to Upper A-Arm Fx	Upright to Upper A-Arm Fy	Upright to Upper A-Arm Fz	Upright to Tierod Fx	Upright to Tierod Fy	Upright to Tierod Fz
0	Pure Cornering (Left Turn)	3.777	46.676	4.347	-0.516	-23.499	-7.653	-5.178	11.270	1.006
1	Pure Acceleration	671.306	68.397	13.663	-253.029	-151.606	-37.943	-148.507	324.007	26.545
2	Pure Breaking	202.176	-2.383	2.968	-76.939	-32.749	-10.881	-41.690	90.053	10.115
3	Cone Strike (FR)	-6.070	192.560	-4.172	6.037	-93.174	-13.357	-20.422	45.182	-1.375
4	Combined Corner and Brake (Left Turn)	13.544	24.937	2.890	-4.589	-14.570	-4.996	-5.130	11.128	1.108
5	Combined Corner and Cone (FR)	7.451	114.682	2.420	0.905	-60.120	-8.119	-13.795	30.586	-1.060
6	Combined Brake and Cone (FR)	112.830	17.619	7.876	-39.986	-32.045	-4.994	-27.071	59.760	1.203
7	Combined Corner and Accel	404.088	390.075	45.171	-143.396	-264.490	-82.738	-120.030	261.195	23.170
8	Combined Corner Brake and Cone	9.715	20.646	1.588	-3.162	-12.034	-3.229	-4.061	8.873	0.473

Rear Right

Set #	Name	Upright to Lower A-Arm Fx	Upright to Lower A-Arm Fy	Upright to Lower A-Arm Fz	Upright to Upper A-Arm Fx	Upright to Upper A-Arm Fy	Upright to Upper A-Arm Fz	Upright to Tierod Fx	Upright to Tierod Fy	Upright to Tierod Fz
0	Pure Cornering (Left Turn)	-130.840	543.369	-55.736	27.495	-312.366	114.673	82.497	35.631	-13.931
1	Pure Acceleration	671.306	-68.397	13.663	-253.029	151.606	-37.943	148.507	324.007	26.545
2	Pure Breaking	202.176	2.383	2.968	-76.939	32.749	-10.881	41.690	90.053	10.115
3	Cone Strike (FR)	-6.070	-192.560	-4.172	6.037	93.174	-13.357	-20.422	45.182	-1.375
4	Combined Corner and Brake (Left Turn)	227.348	348.641	-37.899	-99.116	-143.208	59.060	21.877	47.345	5.204
5	Combined Corner and Cone (FR)	-75.345	174.751	-4.901	15.684	-116.767	17.236	28.330	82.553	1.643
6	Combined Brake and Cone (FR)	112.830	-17.619	7.876	-39.986	32.045	-4.994	27.071	59.760	1.203
7	Combined Corner and Accel	657.803	501.188	-46.834	-265.966	145.152	82.922	96.743	214.367	
8	Combined Corner Brake and Cone	162.683	199.992	-7.385	-70.073	-78.582	24.146	20.986	45.782	2.642

Figure 4: Upright Loads

However, it is good practice to always do hand calculations to double check to see if the software is accurate.

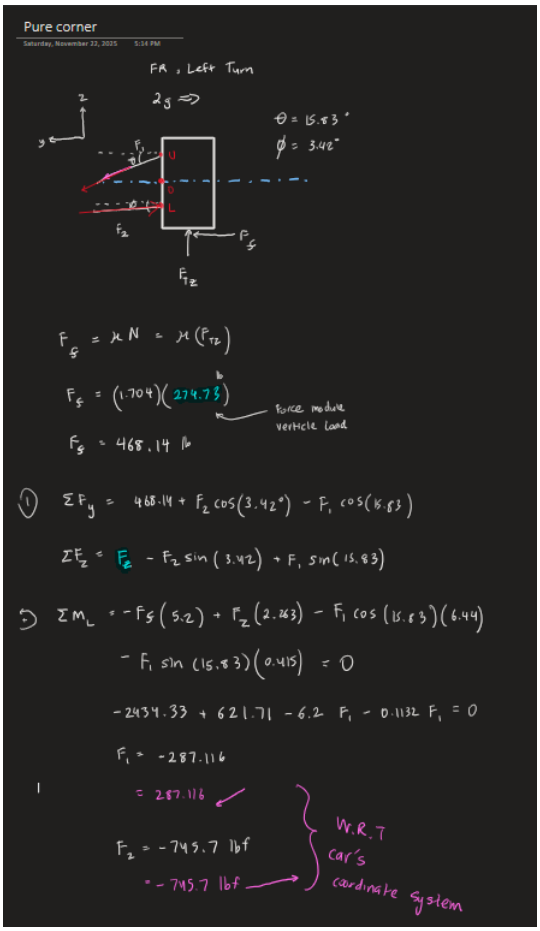
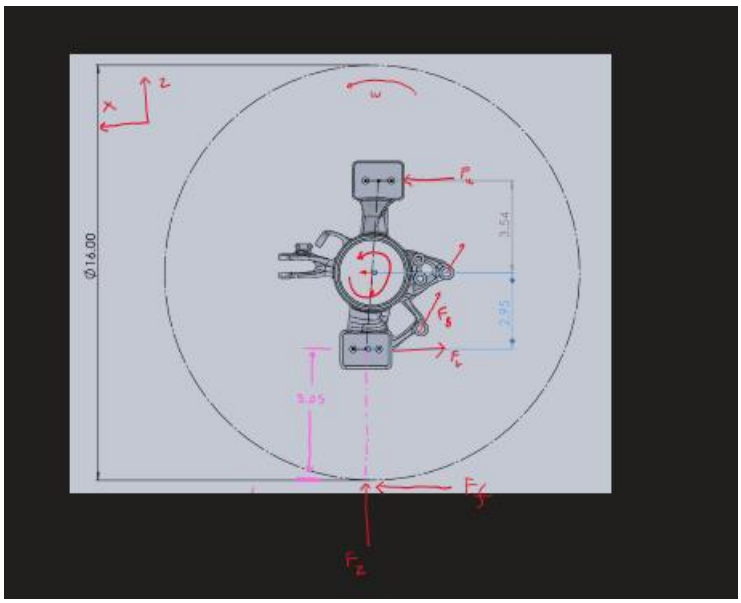


Figure 5: FBD + Hand Calculations

Purple Arrows are Actual Loading Directions



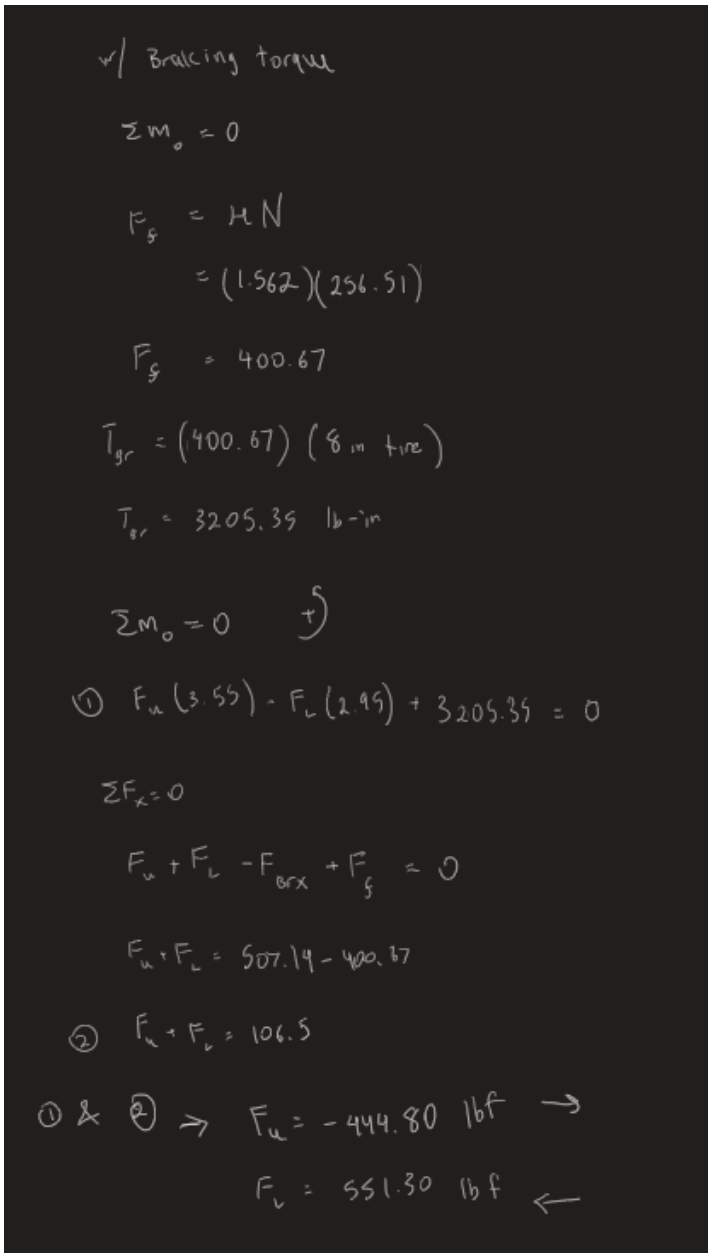


Figure 6 & 7: Longitudinal Load FBD for a Pure Braking Scenario

Although the numbers are a little off, what I felt mattered was that the directions of loading are correct. The force values depend on other factors that I did not calculate by hand such as not all forces necessarily being translated through the A-arms to the upright, neglecting external factors such as friction from bearings, and taking the motion ratio into account.

Unfortunately, OptK does not directly output force values for the hubs, which also see very high amounts of loading. Drawing a FBD for the tire forces on the hub, and the reaction forces on the two wheel bearings (SKF 61909 2RS1 front, SKF 61815 2RS1 rear):

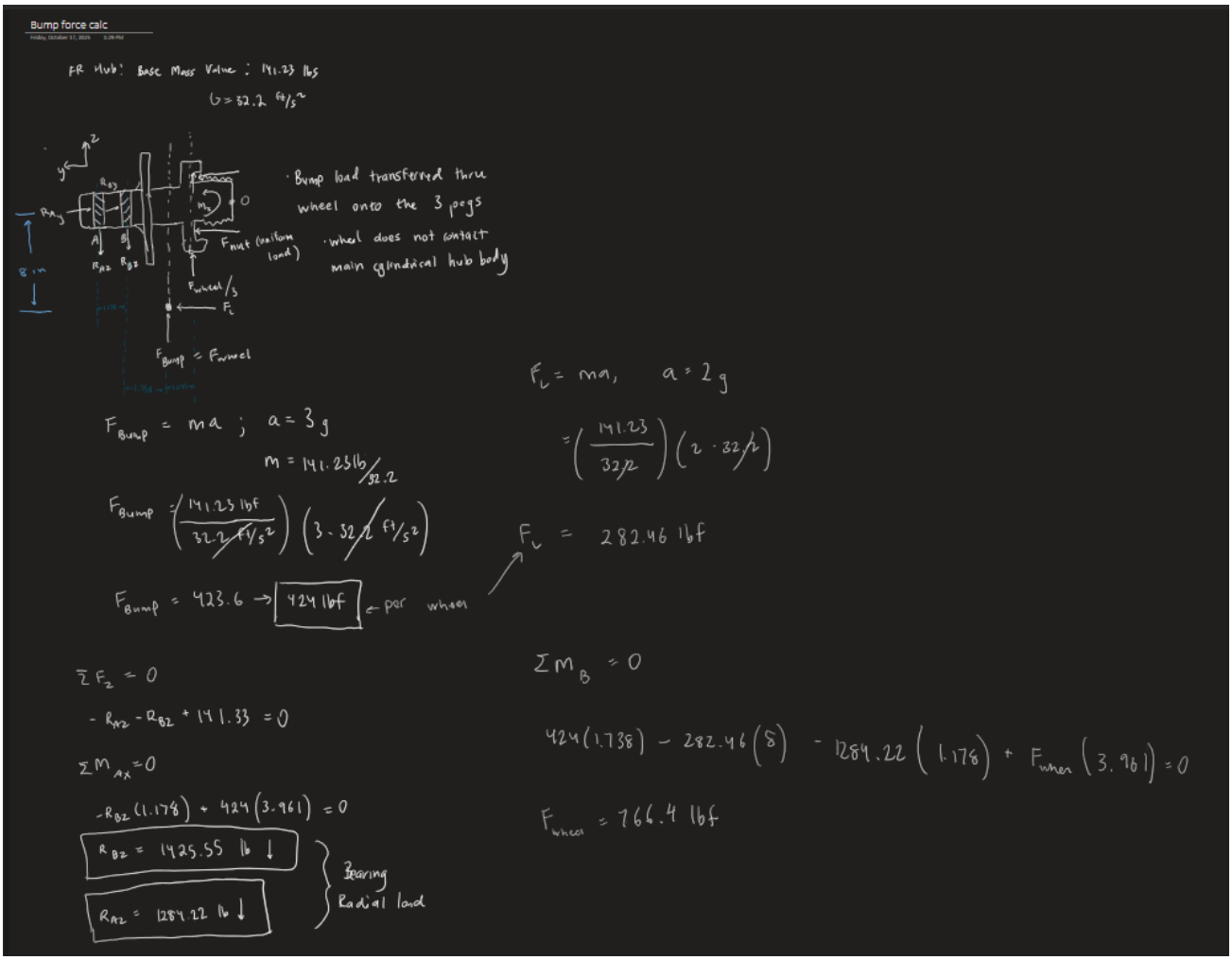
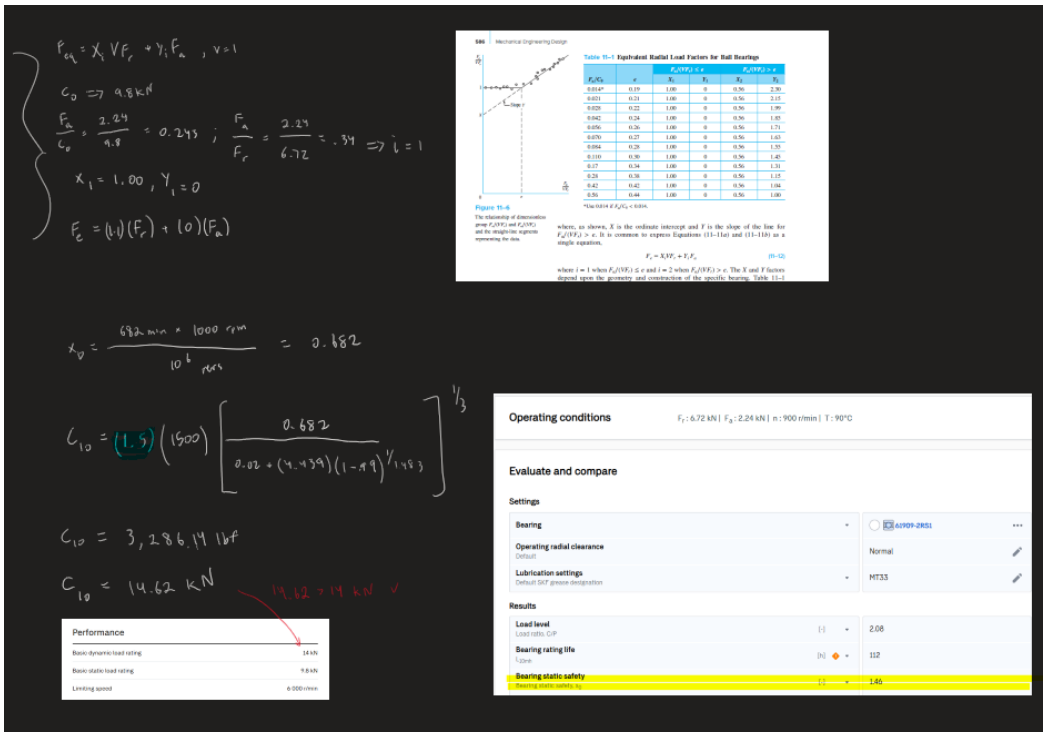
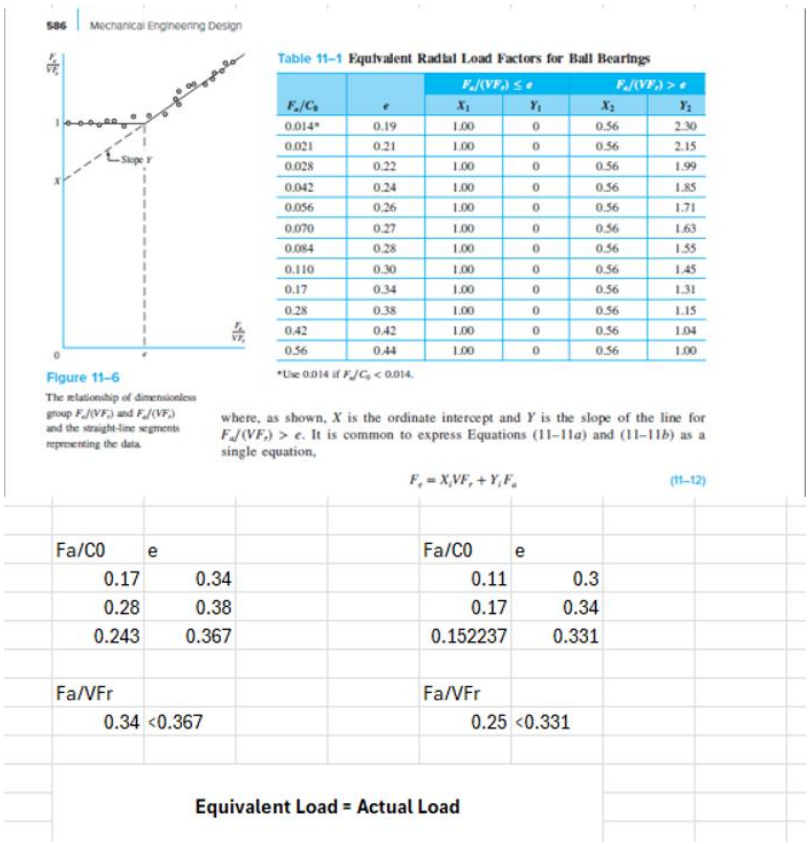


Figure 8: Hub & Wheel Bearing FBD and Load Calculations





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Figure 11–6
The relationship of dimensionless group $F_a/(VF_r)$ and $F_a/(VF_r)$ and the straight-line segments representing the data.

Table 11–1 Equivalent Radial Load Factors for Ball Bearings

F_a/C_0	e	$F_a/(VF_r) \le e$		$F_a/(VF_r) > e$	
		X_1	Y_1	X_2	Y_2
0.014*	0.19	1.00	0	0.56	2.30
0.021	0.21	1.00	0	0.56	2.15
0.028	0.22	1.00	0	0.56	1.99
0.042	0.24	1.00	0	0.56	1.85
0.056	0.26	1.00	0	0.56	1.71
0.070	0.27	1.00	0	0.56	1.63
0.084	0.28	1.00	0	0.56	1.55
0.110	0.30	1.00	0	0.56	1.45
0.17	0.34	1.00	0	0.56	1.31
0.28	0.38	1.00	0	0.56	1.15
0.42	0.42	1.00	0	0.56	1.04
0.56	0.44	1.00	0	0.56	1.00

*Use 0.014 if $F_a/C_0 < 0.014$.

where, as shown, X is the ordinate intercept and Y is the slope of the line for $F_a/(VF_r) > e$. It is common to express Equations (11–11*a*) and (11–11*b*) as a single equation,

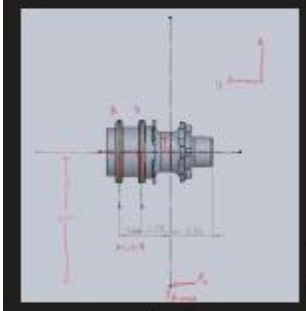
$$F_e = XVF_r + YF_a \qquad (11–12)$$

Fa/C0	e	Fa/C0	e
0.17	0.34	0.11	0.3
0.28	0.38	0.17	0.34
0.243	0.367	0.152237	0.331
Fa/VFr		Fa/VFr	
0.34	<0.367	0.25	<0.331

Equivalent Load = Actual Load

Figures 8.1 & 8.2: Calculating equivalent loads for bearing with axial and radial loads

Rear Bearings



$F = ma$

$F_r = 185.4\ lb/s^2 \cdot 3(ft/s^2)$

$F_r = 350.8\ lbf$

$\sum M_1 = 0$

$(R_{A2})(1.66) - 350.8(5) + 320(1.71) = 0$

$R_{A2} = 1575.9\ lbf$

$\sum F_z = 0$

$-R_{B2} - 1575.9 + 320 = 0$

$R_{B2} = -1,225.88\ lbf$

$\sum F_y = 0$

$R_{Ay} + R_{By} - 350.8 = 0$

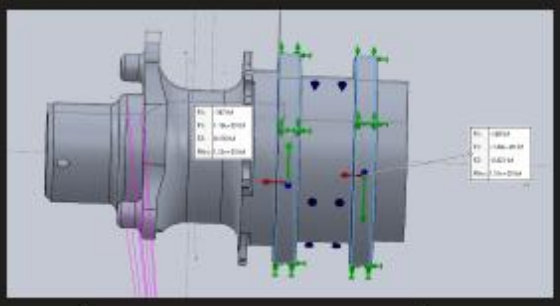
$\sum F_{1-0} = 0$

$-R_{By}(3.08) + R_{By}(3.08) + (1575.9)(3.08) - (1225.88)(1.71) = 0$

$-R_{By} + R_{By} = -350.81\ lbf$

Axial Load: 367 lbf

Radial Load: 1,480 lbf

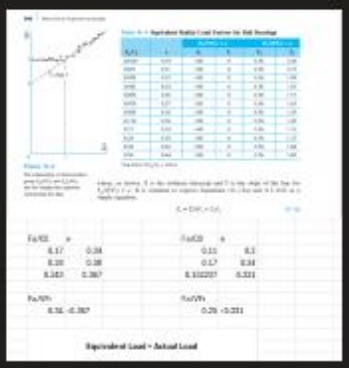


$$C_{10} = (1.5)(1490)\left[\frac{0.682}{0.02 + (4.439)(1-.99)^{9/1.483}}\right]^{1/3}$$

$C_{10} = 14.27\ kN$,

Performance

Basic dynamic load rating C 13.3 kN



Equivalent Load = Actual Load

Results

Load level

Equivalent dynamic load, P	kN	5.53
Load ratio, C/P		1.89
Minimum load, F_{min}	kN	0.0927
Minimum load met?		Yes

Bearing rating life

SKF rating life, L_{10mh}	M	542
Basic rating life, L_{10h}	M	126
SKF life modification factor, a_{SKF}		2.77
Contamination factor, η_c		0.88

Bearing static safety

Equivalent static load, P_0	kN	4.43
Static safety factor, s_0		1.63

Entering the same loading information into SKF calculator, the calculator shows a 1.63 factor of safety compared to the <1.5 value from the hand equation

One of the key feedback points from last season was the lack of fastener calculations. So, I calculated torque specifications for each bolt type used on the system. Using values equations and friction coefficients from Shigley's Mechanical Engineering design:

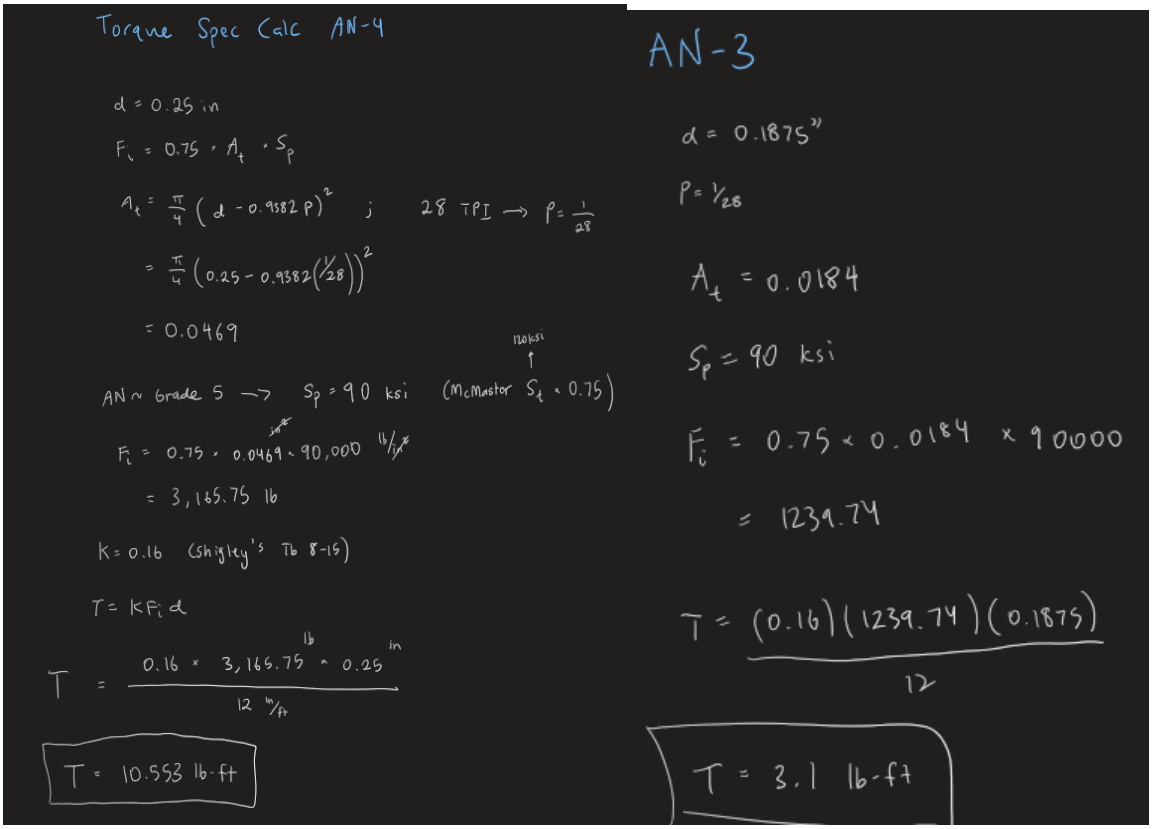


Figure 9: Torque specs for AN-4 and AN-3 Bolts Based on Proof Strength

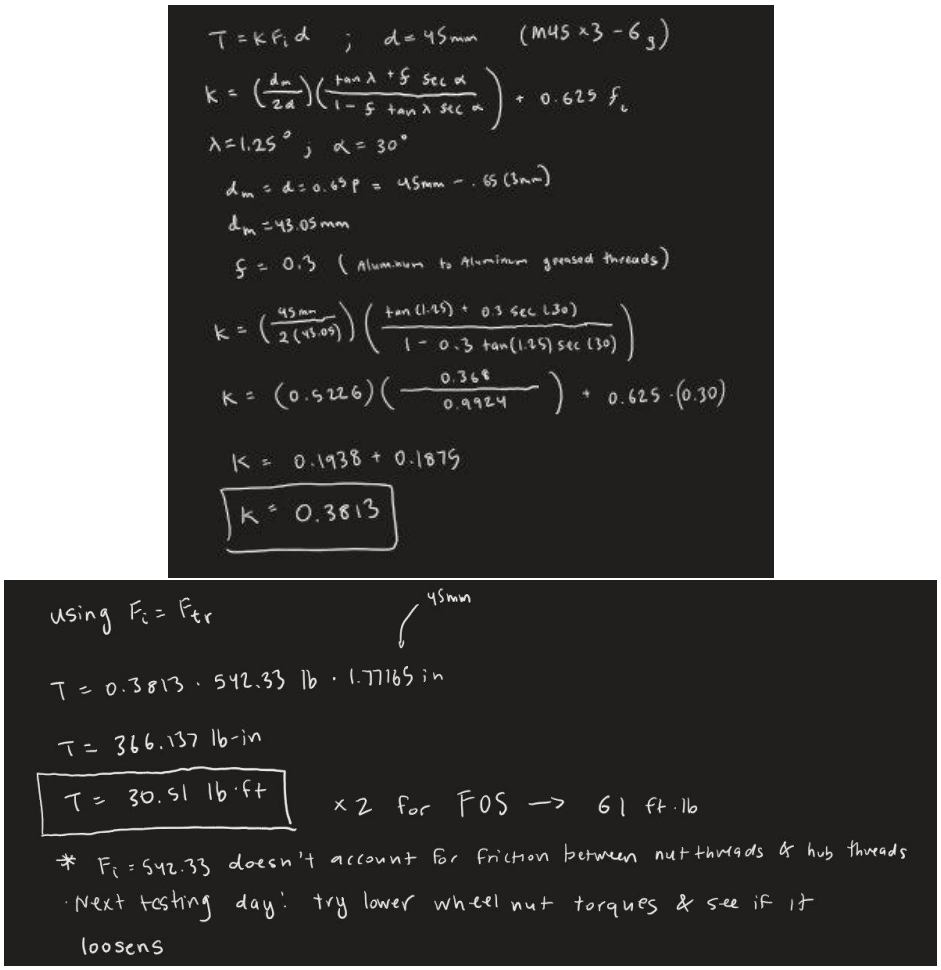


Figure 10: Torque Specs for Center lock Wheel Nut

Design Procedure

Step 1: Analyze judge feedback, vehicle goals, and previous system flaws

- What can I change this year to satisfy these metrics?

Step 2: Address each metric/KPI

- Reliability

Each component was designed to maintain a factor of safety (FOS) in stress greater than 1.5. This threshold provided sufficient margin to account for load uncertainties, irregularities, and potential anomaly conditions while remaining low enough to avoid unnecessary weight penalties. The approach

ensured an efficient structural design that balanced reliability and mass optimization without introducing excessive conservatism.

Hand calculations were performed to validate the loads and stress results obtained from simulation software. These analytical checks served as an independent verification method to confirm that applied loads, reaction forces, and resulting stress levels were reasonable and physically consistent. Finite element analysis (FEA) models were developed with carefully defined boundary conditions that accurately represented real-world constraints and load paths. Proper constraint definition and load application were used to prevent artificial stiffness, stress singularities, and unrealistic deformation behavior in the simulations.

Because the components were manufactured from aluminum, fatigue performance was evaluated explicitly, as aluminum does not exhibit a true endurance limit and is therefore more susceptible to fatigue failure. Fatigue analysis considered expected cyclic loading conditions, stress ranges, and stress concentrations to estimate component life and ensure adequate durability under service conditions. Additionally, modal analyses were conducted to identify the natural frequencies of each part. These frequencies were compared against common excitation sources from vehicle operation to prevent resonance, which could amplify stresses, accelerate fatigue damage, or compromise structural integrity.

- Refinement

All issues identified in the previous design were thoroughly reviewed to ensure they were not repeated in the updated iteration. Particular attention was given to manufacturing and assembly-related problems, including bolt interference that previously required grinding for clearance and improper tolerancing that led to fitment inconsistencies. These shortcomings were corrected through improved dimensional control, proper clearance considerations, and better integration of hardware within the assembly envelope.

The system was also evaluated for opportunities to improve overall performance and reliability. Design refinements were implemented where feasible to enhance structural efficiency, reduce unnecessary complexity, and improve serviceability. Reliability checks outlined in the earlier analysis were repeated following the implementation of design changes to verify that modifications did not introduce new failure modes or reduce structural integrity.

Performance-driven updates were aligned with specific objectives, including reduced setup time and achieving the targeted lateral acceleration capability. Structural and geometric adjustments were assessed to ensure they supported these measurable goals without compromising durability. Additionally, detailed bolt calculations were performed to confirm that all fasteners operated within allowable stress limits under the expected loading conditions, ensuring that joint integrity was maintained and fastener failure was avoided.

Step 3: Upright Design

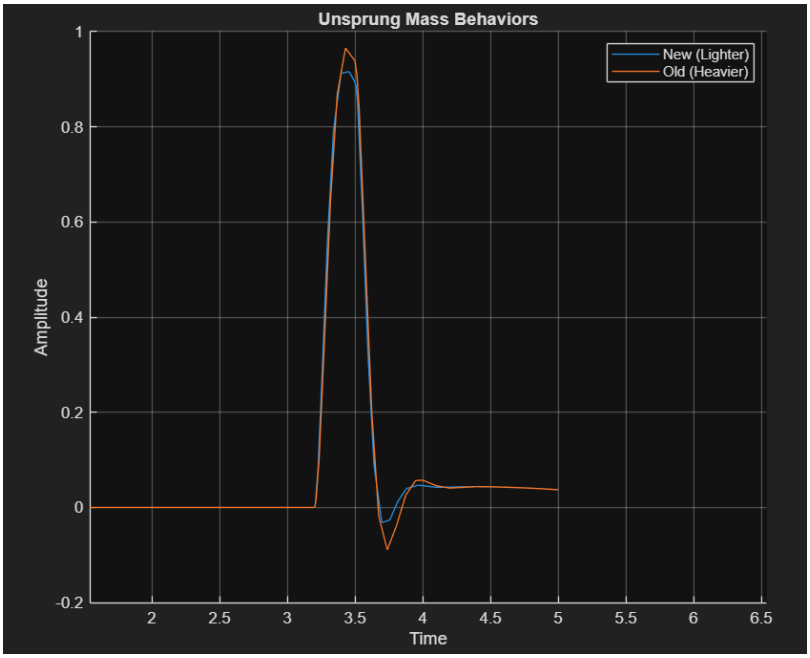
The design process began with referencing team goals and analyzing previous upright designs. The strengths and weaknesses of the earlier iterations were identified to better understand what constituted upright “performance” and how the new design could be improved. The old design binder was reviewed, and key performance indicators (KPIs) were evaluated, including factor of safety, maximum displacement (both overall and in specific areas and directions), weight, manufacturability, cost, and ease of serviceability and assembly. Each KPI was assessed to determine whether it met expectations or required improvement.

The previous uprights exhibited variable factors of safety, with minimum values of approximately 1.3 in the rear and around 2.5 in the front. Maximum displacements were low, remaining under 0.015 inches. Weight was considered decent but had clear potential for reduction. Manufacturability was excellent and could likely be achieved at no cost through a sponsor using a 3-axis mill. Cost was low, as machining was expected to be free and only raw materials were required, which could also be discounted through sponsorship. Serviceability, however, was poor. The wheel speed sensor mount blocked the brake caliper bolt, preventing installation and removal without grinding down the bolt head. Additionally, the rear lower A-arm interfered with the push rod bolt, again requiring the bolt head to be ground down. These issues were addressed by relocating sensor mounts and reducing the size of the A-arm to eliminate interference.

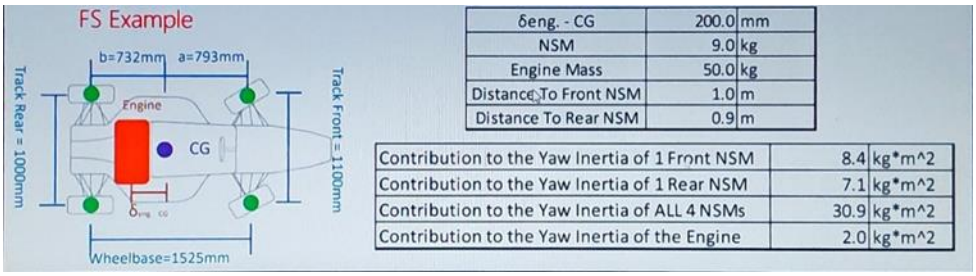
Methods for reducing high factors of safety and saving weight were explored. Logical trial-and-error iterations were performed in CAD. Topology optimization was considered using the old design as a baseline, as well as the possibility of generating a completely new design. Generative design was also evaluated as a method to create geometry from the ground up while adhering to defined parameters such as factor of safety, maximum displacement, weight targets, and manufacturing constraints. Considering the available budget, overseas manufacturing options, and prior team experience, generative design was selected as the primary development method for the new uprights.

Generative design was chosen because it produces geometry that satisfies the specified constraints and objectives. For a required factor of safety of 1.5, a defined maximum allowable displacement in any direction, manufacturing via 5-axis machining, specific applied loads and fixtures, and a goal of minimizing weight, the resulting geometry theoretically represents the lightest configuration capable of meeting all constraints simultaneously. Given the system budget, large prototyping and manufacturing services such as Xometry were available, providing automated quotes and lead times for parts.

Weight reduction was prioritized because the uprights and hubs constitute unsprung mass. Lower unsprung mass improves suspension performance by reducing the inertia that springs and dampers must control, allowing the suspension to track the ground more effectively when encountering bumps or surface undulations.



The uprights and hubs are also among the components located furthest from the vehicle's center of gravity. This gives them a very high moment of inertia.



δeng. - CG	200.0	mm
NSM	9.0	kg
Engine Mass	50.0	kg
Distance To Front NSM	1.0	m
Distance To Rear NSM	0.9	m

Contribution to the Yaw Inertia of 1 Front NSM	8.4	kg*m^2
Contribution to the Yaw Inertia of 1 Rear NSM	7.1	kg*m^2
Contribution to the Yaw Inertia of ALL 4 NSMs	30.9	kg*m^2
Contribution to the Yaw Inertia of the Engine	2.0	kg*m^2

Inputting our car's dimensional values and corresponding masses, we attain the following table, where the corner category includes all the assemblies that are on the corner (brakes, hubs + uprights, wheels + tires).

I=mr^2 (lb-in^2)	Inertia new	Inertia old	% Diff
Front Assembly	622.69	663.52	6.15
Front Corner	122.50	163.33	25.00
Rear Assembly	650.25	697.72	6.80
Rear Corner	180.68	228.15	20.81

Despite the overshadowing weight of the wheels and brakes, there is still a substantial 6% reduction for both the front and rear inertias.

A higher inertia means the car resists changes in direction more strongly, making turn-in more difficult.

The final approach satisfied the goals of refinement, performance, and reliability. The geometry could not be made lighter without violating the defined load constraints, reduced unsprung weight contributed to improved suspension performance and increased grip, and reliability was maintained by ensuring a minimum factor of safety of 1.5.

Material and manufacturing process selection were then evaluated. Options included billet 7075 aluminum, billet Grade 5 titanium, and additively manufactured titanium. Fusion generative design was used with defined input loads and constraints. Keep-out zones and preserve zones were created around mounting points and wheel bearing interfaces. Loads were applied at the A-arm mounts, steering arm interface, brake caliper bolt holes, push rod mounts, and sensor mounts. Components such as the steering arm and brake caliper were also modeled as keep-out regions to prevent interference.

Material comparisons were performed between 7075-T6 aluminum, Ti-6Al-5V billet titanium, and Ti-6Al-5V additively manufactured titanium. Based on generative design results, aluminum demonstrated better overall performance relative to cost, manufacturability, and weight, and was therefore selected.

Iterations were then conducted. Keep-out zones were adjusted based on the appearance of generated geometries and any observed interferences. Results were evaluated using Fusion FEA; however, the generative design mesh was relatively coarse and occasionally failed to capture stress concentrations. Loads were adjusted in regions exhibiting factors of safety below 1.5 to eliminate undesirable geometries such as extremely thin members or features that were difficult to machine. Iterations were repeated, modifying loads to influence geometric outcomes.

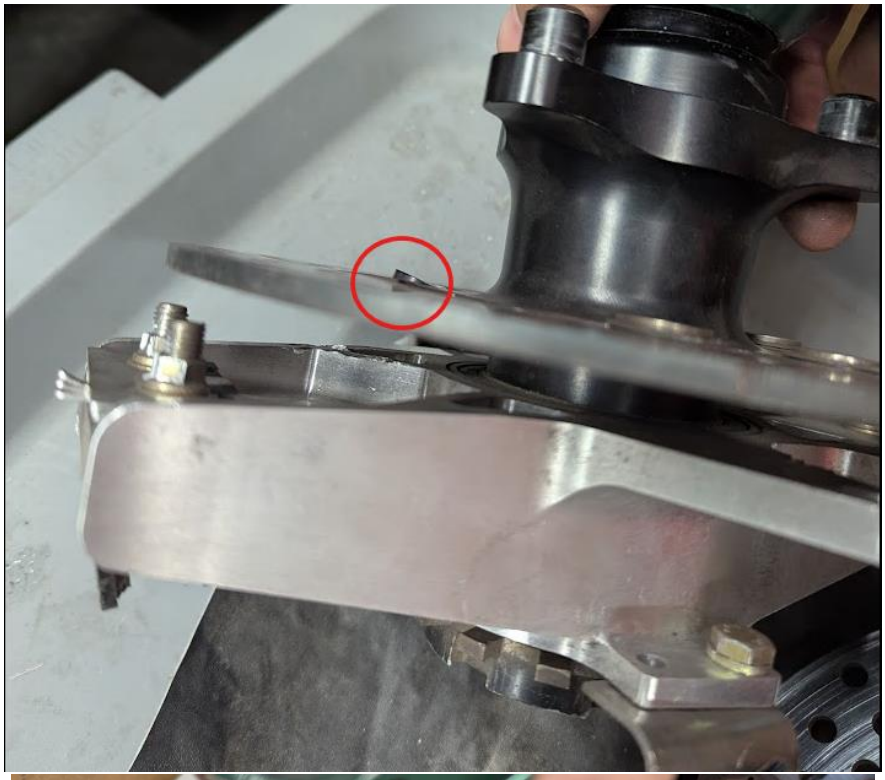
Designs that passed Fusion FEA were validated in SolidWorks FEA. A final iteration was selected and analyzed further using SolidWorks for static stress, fatigue performance, and frequency response simulations, with additional detail provided in later sections. Manufacturing files and drawings were then submitted to the selected manufacturer.

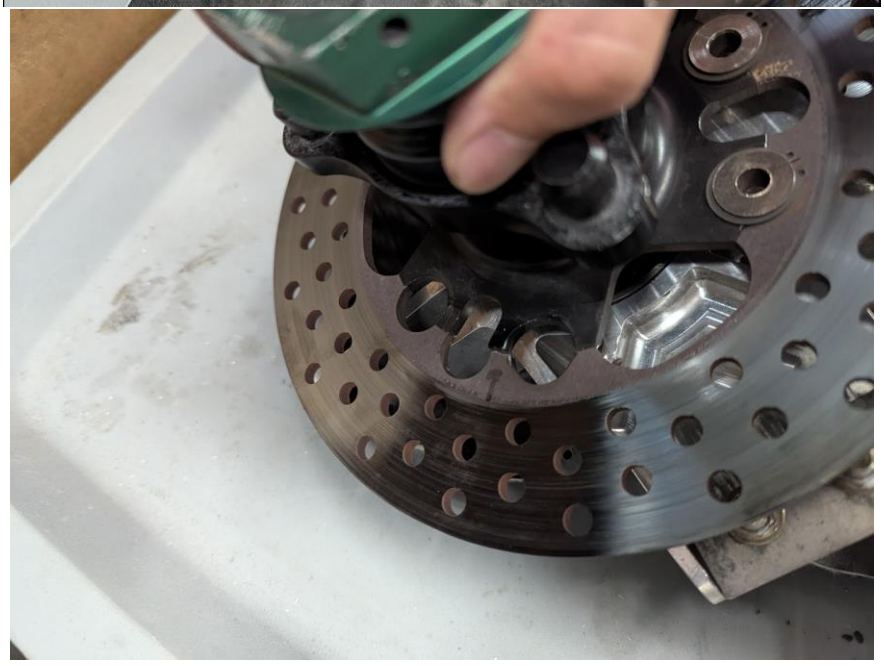
Step 4: Hub Design

The hub design process followed similar early steps as the upright development, beginning with a review of performance goals and identifying opportunities to improve hub performance while maintaining reliability. Previous hub FEA results were analyzed and showed minimum factors of safety ranging from approximately 2.5 to above 6, depending on the applied boundary conditions. This indicated excess structural margin, and material was removed from internal regions to reduce weight while maintaining a minimum factor of safety above 1.5. Simulations confirmed that the remaining wall thicknesses were still structurally adequate.

Failure analysis conducted earlier in the season identified improper tolerancing of the brake rotor button holes. The holes were manufactured undersized, requiring manual sanding to enlarge them. This process lacked precision and likely resulted in excessive material removal. As a consequence, the holes became oversized, allowing the brake rotor buttons to disengage, become lodged between the rotor and hub, and cause significant gouging damage to both the upright and rotor. The issue was corrected

by applying a unilateral positive tolerance to the hole diameter and slightly increasing the nominal dimension to properly account for the button's tolerance stack-up.





Figures: Hub brake button mounts bent out of plane

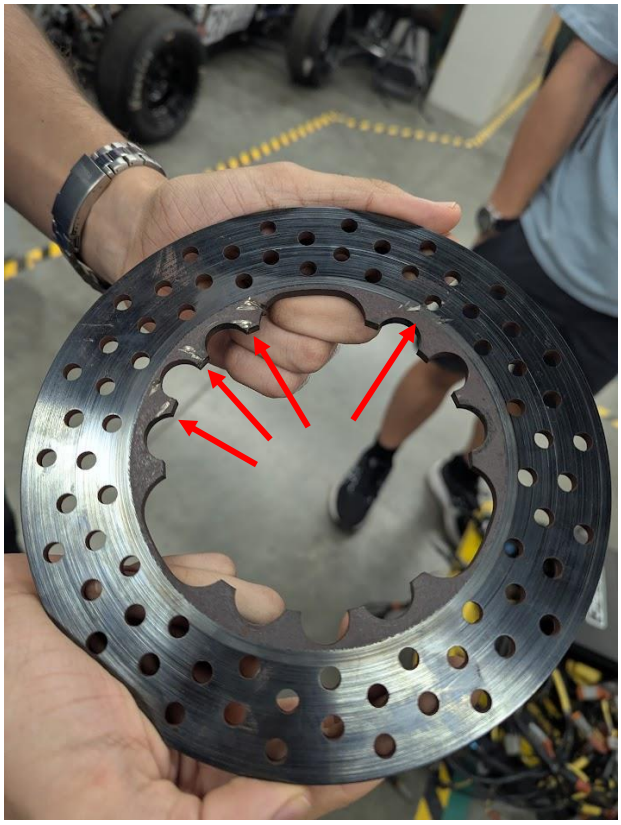


Figure: Corresponding brake rotor with gouges at brake button mounts

To improve setup time, the wheel nut was redesigned to sit behind the pin holes. The current wheel nut was made thinner to achieve this packaging requirement. Thread strength calculations were performed to determine whether the reduced thread engagement length could still support the applied loads. A second design iteration was developed that reduced weight further while incorporating additional thread engagement and still fitting behind the pin holes. This revised design required the use of a custom socket. Simulation was performed to evaluate whether the new head pattern could withstand the required break-loose torque without failure, and thread strength calculations were conducted to verify adequate load-carrying capacity.

Design Revisions

Uprights

Throughout the design process, I went through dozens of different generative design iterations. Each iteration represents the experimentation of inputting different loads and boundary conditions in order to create an upright which does not have any interference, is structurally sound, and is relatively easy to manufacture.

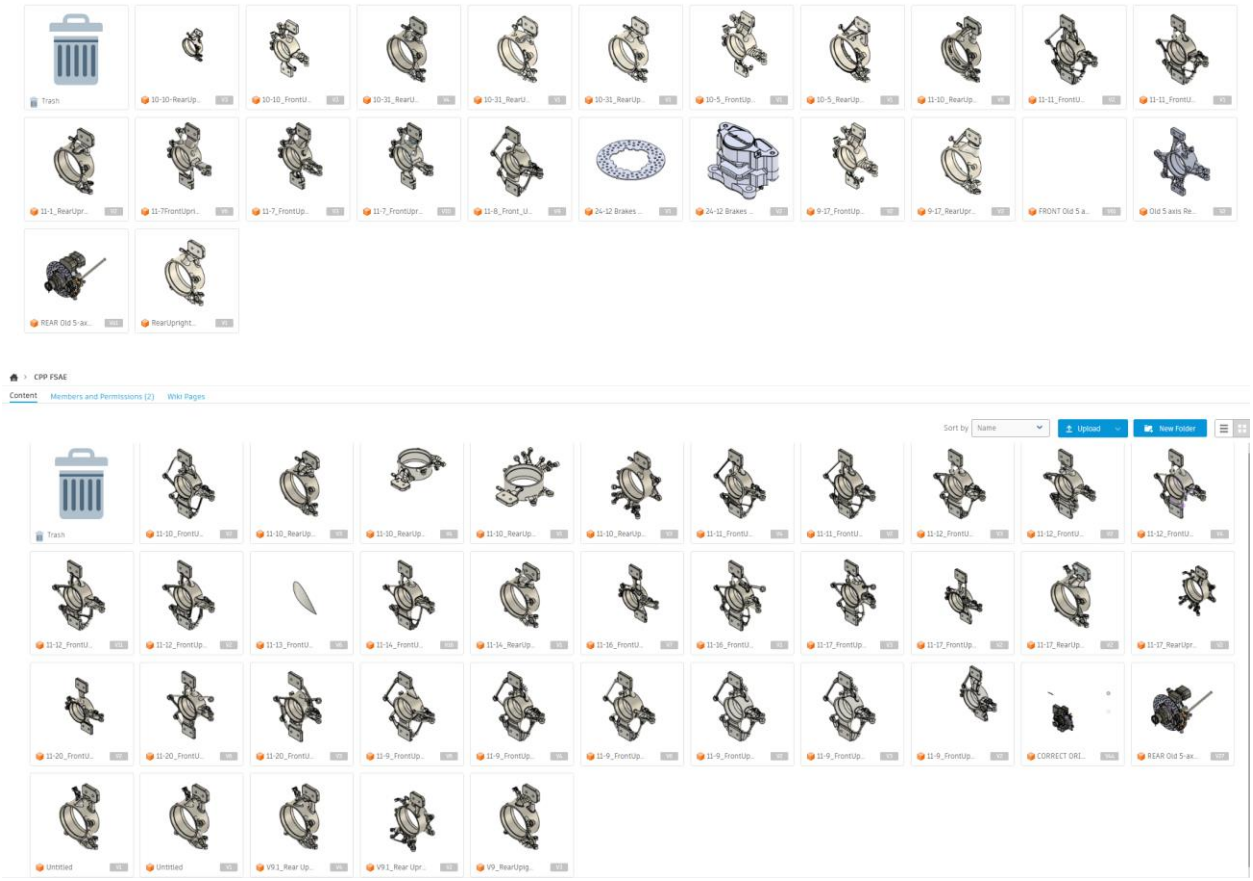


Figure 11: All upright generative design iteration outcomes

Since there are so many revisions, I will only highlight the most important ones which had significant impacts

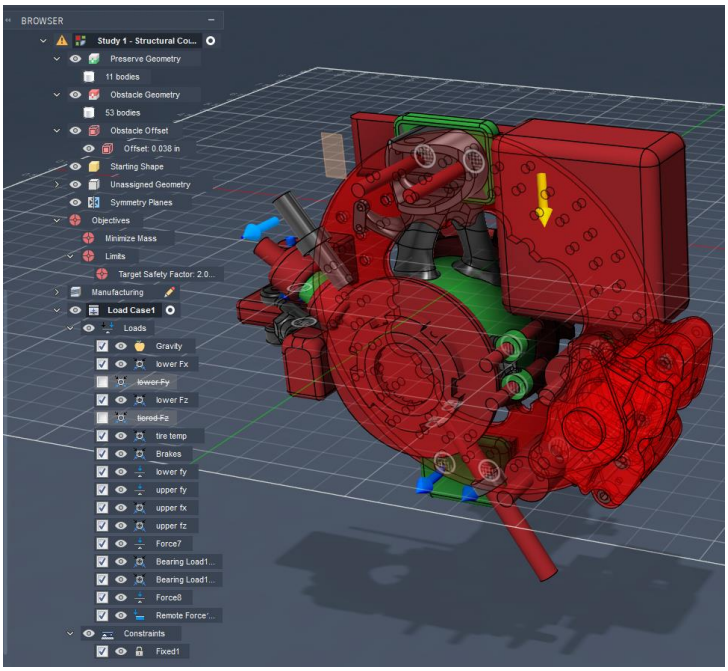
Generative Design

Generative Design in Fusion is a design exploration tool that uses predefined engineering requirements to automatically generate multiple viable design solutions. Instead of manually modeling a part, the user defines the problem by specifying constraints such as loads, supports, available design space, and performance goals (ex. minimizing weight or maximizing stiffness). The software then iterates a range of geometries that eventually converge to satisfy the initial conditions.

The workflow is centered on setting up three main elements: preserved regions such as mounting interfaces, obstacle regions (assembly interference locations), and the overall design space where material can exist. Additional inputs include material selection, manufacturing methods, maximum stresses, displacements, minimum resonances, and much more, which guide the shape and feasibility of the generated outcomes. The result is a set of design alternatives, often with organic or non-traditional forms, each accompanied by performance data for comparison.

Generative design is closely related to topology optimization but differs in scope and output. Topology optimization typically starts with a single design space and iteratively removes material to improve performance, resulting in one optimized geometry. In contrast, generative design explores a broader range of possibilities simultaneously by having the freedom to build from the ground up.

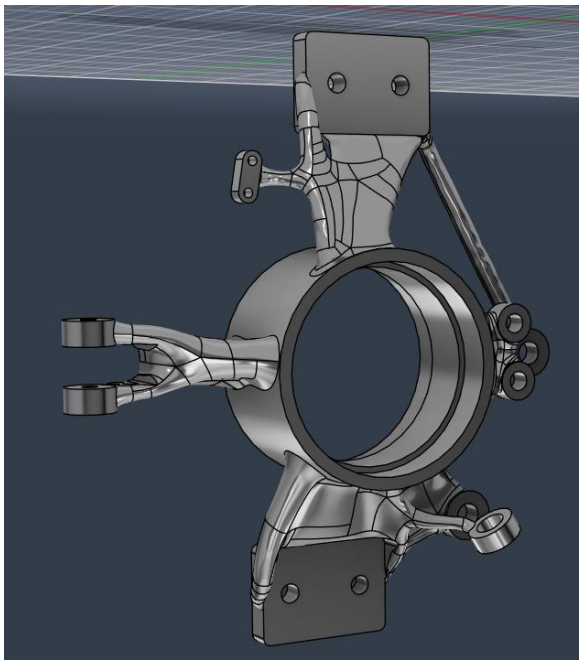
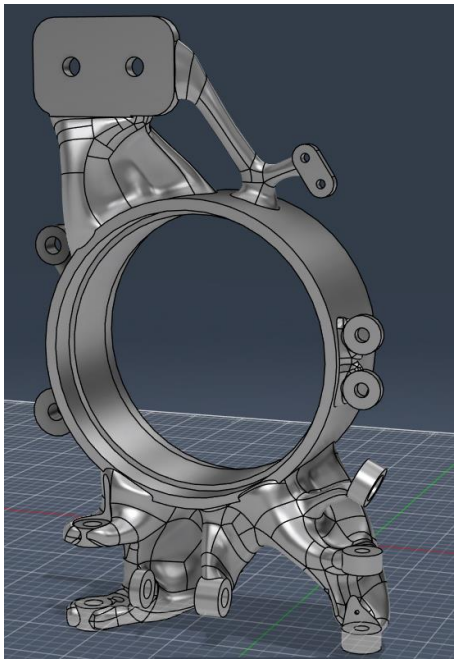
Overall, generative design shifts the designer's role from creating geometry to defining requirements and evaluating solutions, enabling faster exploration of high-performance designs.



Red: Obstacles

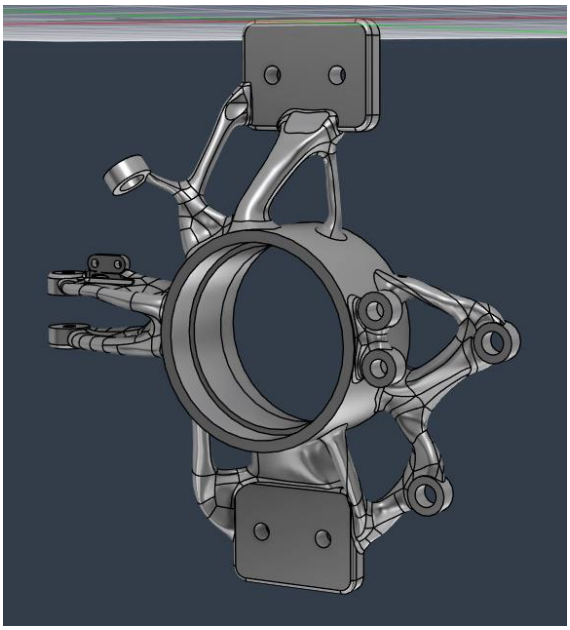
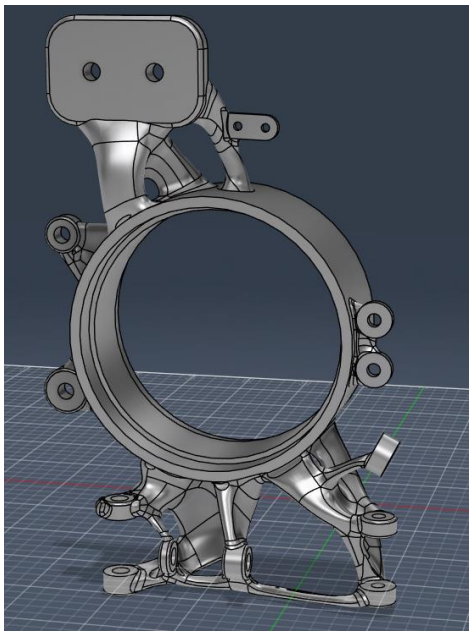
Green: Preserve Regions

Iteration 1

The first iteration used the BM-25 load case which came from the data from the BM-24 and previous cars. These loads were higher than the BM-26 ones, resulting in uprights which are heavier and more bulky than the final version. Consequently, these uprights were the heaviest out of all. They use the original sensor placement which was designed for the 3-axis uprights.

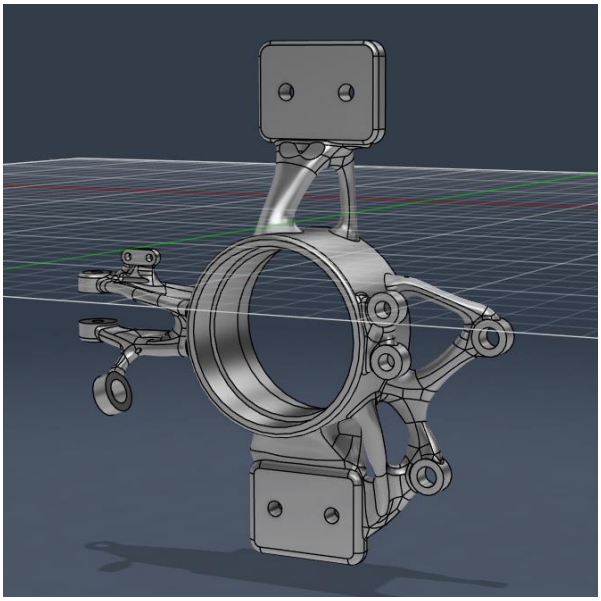
Iteration 2

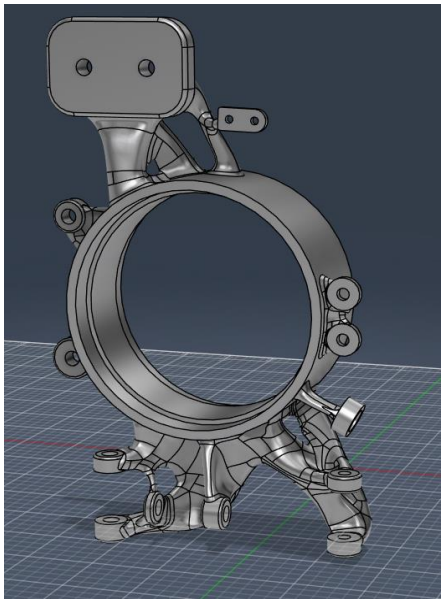



In order to eliminate stress concentrations at the junction between the preserve geometry and the generated geometry, fillets were added to the edges of the preserve regions to create smoother transitions. Sensor mounting locations were also updated. However, there are many stringy elements

which would mean they would be more difficult to machine and also have a lower resonant frequency which could lead to fatigue failure from vibrations. The loads were updated to the BM-26 loads, which are lower. However, likely due to meshing differences, the uprights would sometimes not meet the 1.5 FOS in simulations using the given loads. To remedy this, certain loads would be inflated depending on where the <1.5 FOS's were happening and the condition it was failing under.

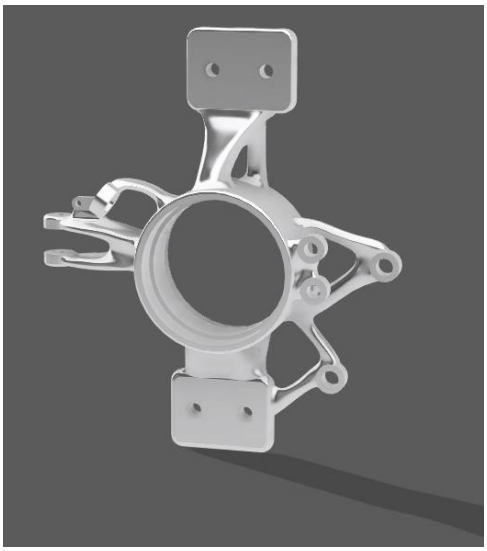
Iteration 3





The thin member connecting the upper brake caliper mount to the upper a-arm clevis interface was removed due to alumni feedback with the fear of it being prone to vibration resonance. The sensor mounts have further been adjusted to try and find an optimal spot which uses existing features where the sensor mount can be based off. This ensures an efficient use of geometry that still satisfies the specifications and requirements for the sensor locations. Extra room was made to accommodate the rear tie-rods to prevent any interference. The front steering arm mount location was also slightly changed at the request of our steering captain.

Iteration 4 (Final)





Mass:

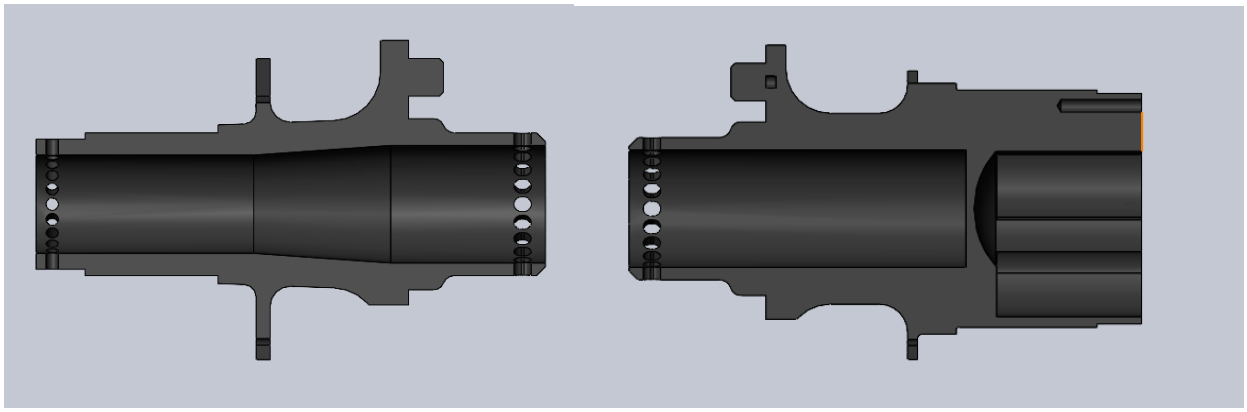
- Front: 0.734 lbs
- Rear: 0.965 lbs

Front uprights have eliminated any “excess” stringy components, making them easier to manufacture. Rear uprights got rid of some small thin elements near the lower a-arm mount which were a stress concentration in FEA. This came at the expense of added weight.

Hubs

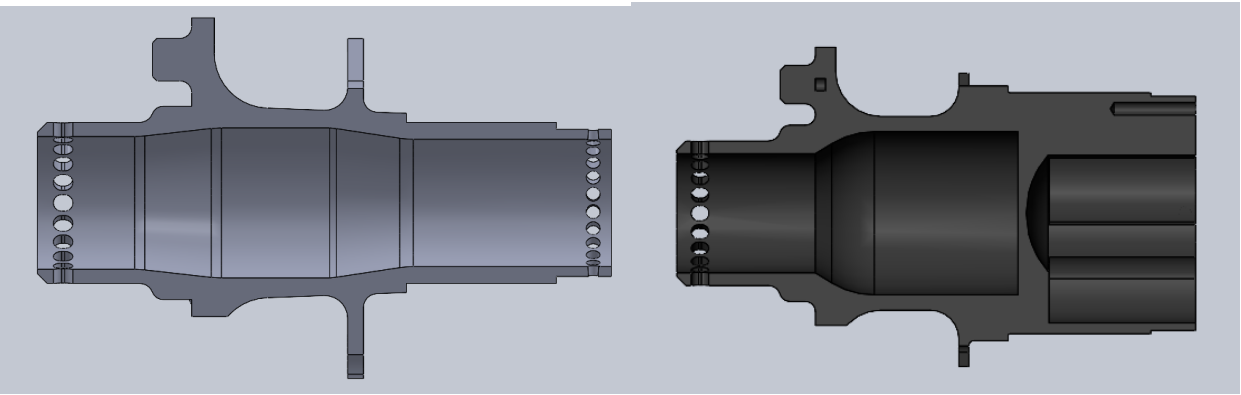
Previous Design Failure

Original



This iteration is the hub design that has been used in the past few seasons. The FEA results on this hub were a very high FOS in areas that could be thinned. Lowering the FOS to a more reasonable value of 1.5 would mean weight would be able to be decreased.

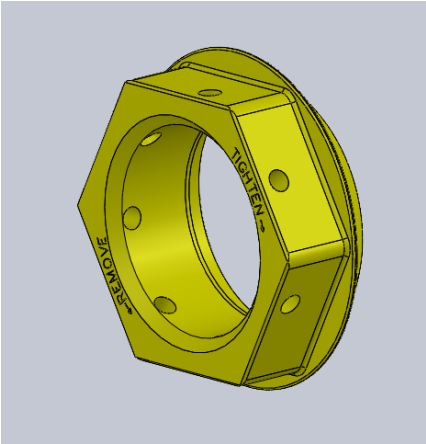
Iteration 1 (Final)



Material is removed from the inside of the hubs, thinning the walls. Also, the material that supports the pegs is thinner, further removing weight.

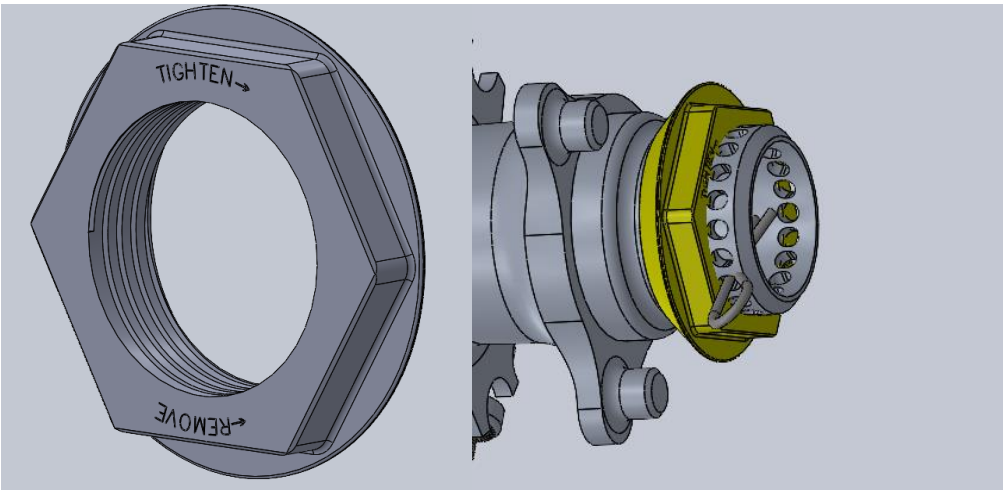
Wheel Nuts

Original



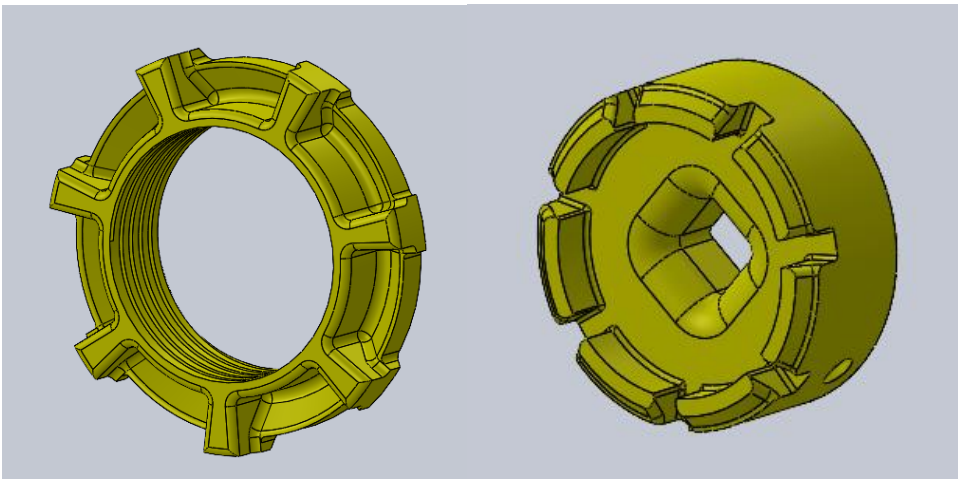
The original wheel nut design is a simple hex nut with 7 holes to incorporate the positive locking mechanism (cotter pin) required by rules. However, only one hole would be aligned at a time, so one would have to try each hole individually until the cotter pin went through. This took a bit of time if the only aligned hole was the sixth or seventh hole that had to be checked.

Iteration 1



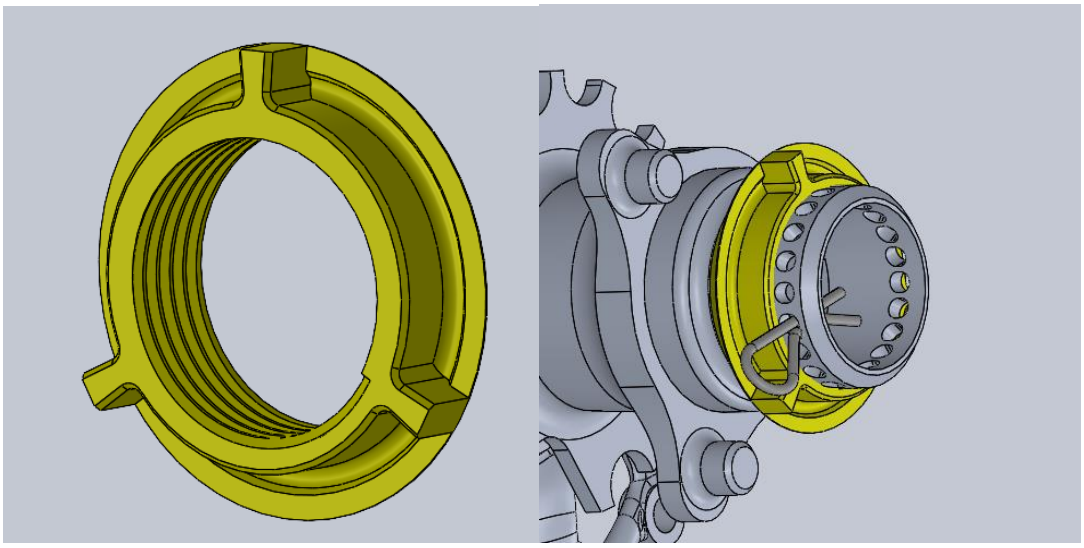
This new iteration of wheel nut retains the hex pattern so that the existing socket can be used. It is now thinner, so it sits behind the holes of the hub. However, there is the risk of the existing socket not being having a weaker grip on the nut, causing potential wear and eventual rounding of the nut.

Iteration 2



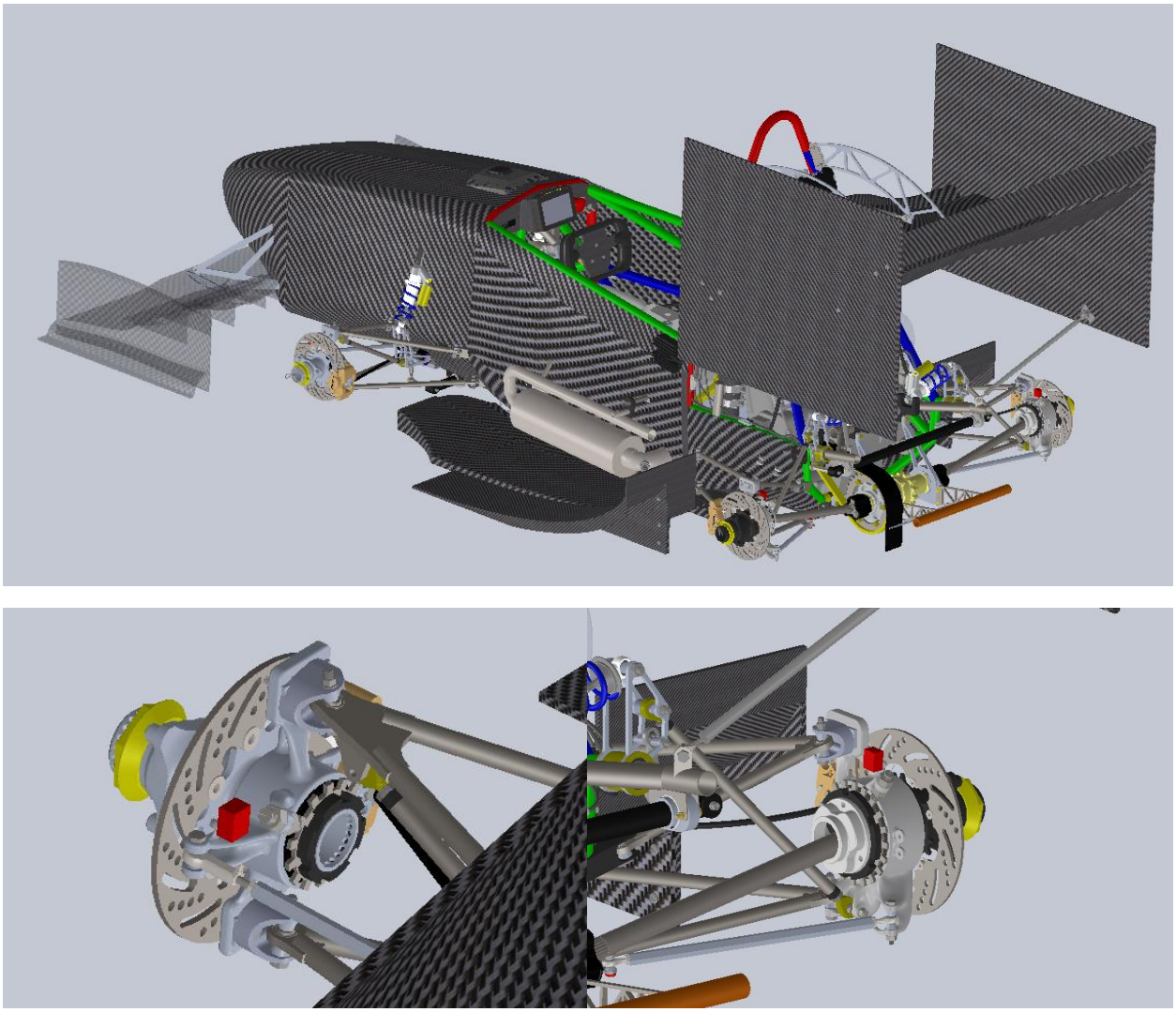
To remedy the socket having less grip on the nut, a splined nut is developed which incorporates seven splines and an indentation in order to perfectly mate with a custom-made socket.

Iteration 3



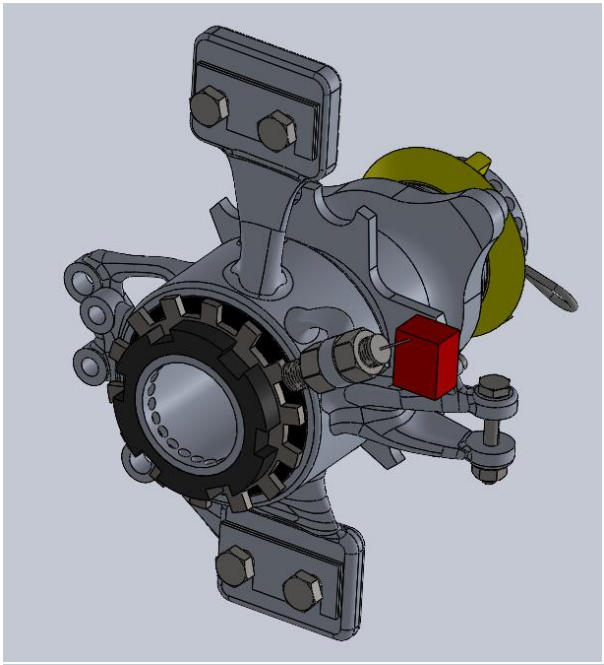
Seven splines were deemed to be excessive, so the amount is reduced to just three. This shaves weight and makes the resulting nut almost half the weight of the original nut.

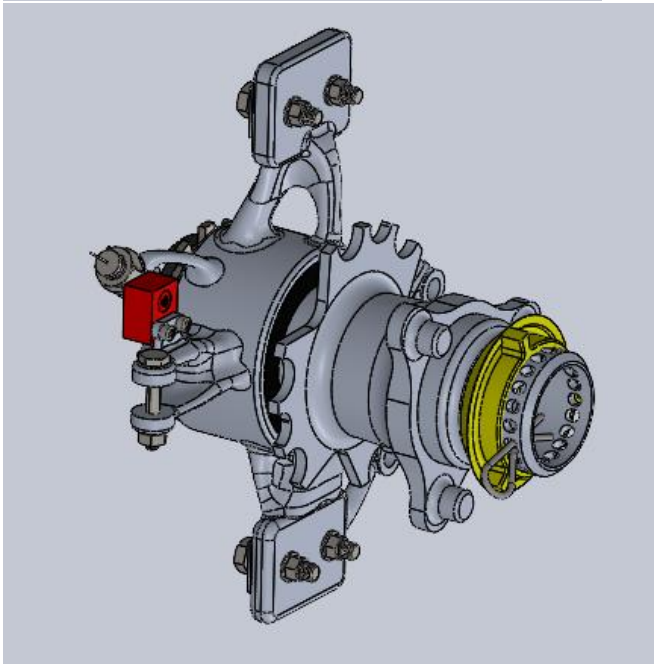
CAD Models



Assembly on Full Vehicle

Front Corner Assembly





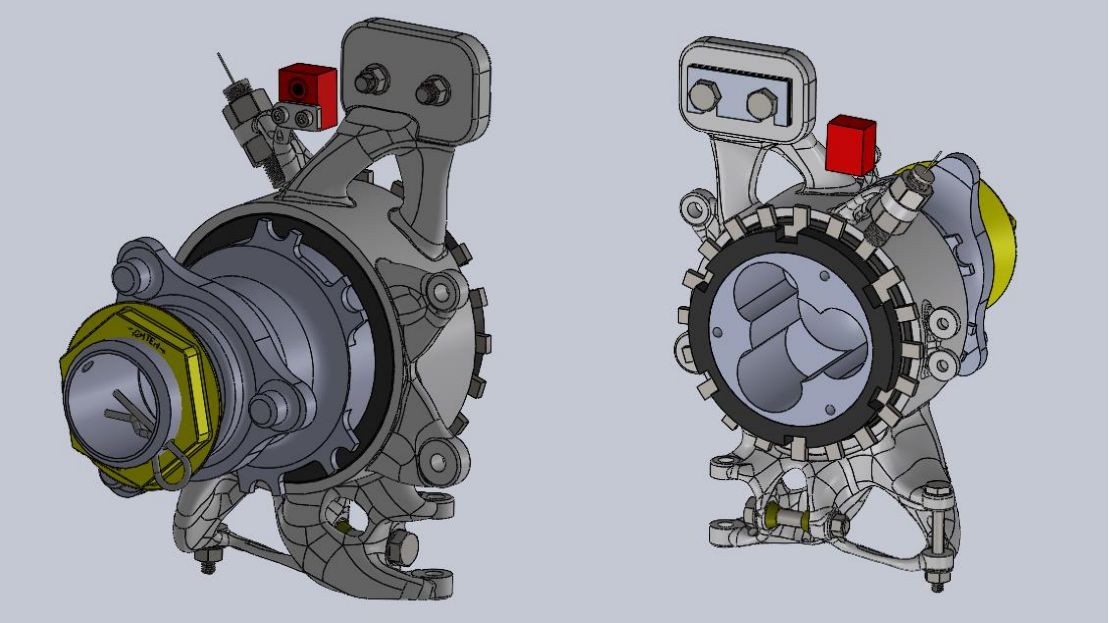
Total Mass (Including sensors and hardware): 3.51 lbs

Upright Weight: 0.735 lbs

Hub Weight: 0.921 lbs

Nut Weight: 0.09 lbs

Rear Corner Assembly



Total Weight: 3.392 lbs

Upright Weight: 0.965 lbs

Hub Weight: 1.77 lbs

Nut (Hex) Weight: 0.124 lbs

Simulations

Uprights

BM26 Force Results [Tires:R20] UPRIGHT (lbs)

Front Left

Set #	Name	Upright to Lower A-Arm Fx	Upright to Lower A-Arm Fy	Upright to Lower A-Arm Fz	Upright to Upper A-Arm Fx	Upright to Upper A-Arm Fy	Upright to Upper A-Arm Fz	Upright to Tierod Fx	Upright to Tierod Fy	Upright to Tierod Fz
0	Pure Cornering (Left Turn)	2.731	106.861	7.048	-0.237	-41.770	20.311	-2.494	-17.020	-2.829
1	Pure Acceleration	108.806	44.041	0.754	-44.198	-2.455	54.961	-8.918	-41.586	-7.705
2	Pure Breaking	785.397	268.311	8.019	-320.997	35.956	292.817	-63.741	-304.268	-44.326
3	Cone Strike (FR)	5.468	62.994	-4.041	-3.903	-55.523	133.892	-1.565	-7.471	-0.417
4	Combined Corner and Brake (Left Turn)	159.436	461.324	22.751	-61.610	-157.617	116.928	-21.352	-126.672	-19.590
5	Combined Corner and Cone (FR)	17.009	665.804	-44.425	-4.247	-304.475	297.749	-13.017	-73.336	-3.724
6	Combined Brake and Cone (FR)	444.168	201.927	-18.692	-175.130	-8.308	278.213	-41.468	-198.234	-8.546
7	Combined Corner and Accel	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
8	Combined Corner Brake and Cone	116.803	342.491	6.010	-45.284	-126.900	128.376	-15.669	-83.208	-9.302

Front Right

Set #	Name	Upright to Lower A-Arm Fx	Upright to Lower A-Arm Fy	Upright to Lower A-Arm Fz	Upright to Upper A-Arm Fx	Upright to Upper A-Arm Fy	Upright to Upper A-Arm Fz	Upright to Tierod Fx	Upright to Tierod Fy	Upright to Tierod Fz
0	Pure Cornering (Left Turn)	-1.147	872.332	-61.193	-16.146	-253.393	322.608	17.226	-80.480	13.316
1	Pure Acceleration	108.806	-44.041	0.754	-44.198	2.455	54.961	-8.918	41.586	-7.705
2	Pure Breaking	785.397	-268.311	8.019	-320.997	-35.956	292.817	-63.741	304.268	-44.326
3	Cone Strike (FR)	14.107	-162.519	-10.425	-10.070	143.243	397.026	-4.037	19.275	-1.075
4	Combined Corner and Brake (Left Turn)	352.092	661.766	-44.845	-159.094	-260.004	341.734	-11.738	55.708	-9.169
5	Combined Corner and Cone (FR)	2.826	862.892	50.327	-17.420	-227.157	435.380	15.234	-72.734	3.585
6	Combined Brake and Cone (FR)	863.631	-388.028	-35.919	-336.537	15.962	634.615	-79.724	380.935	-16.422
7	Combined Corner and Accel	322.519	265.733	-27.769	-137.509	-129.282	209.362	-17.060	78.598	-13.118
8	Combined Corner Brake and Cone	467.968	819.486	-20.260	-208.260	-306.899	551.275	-19.608	93.383	-10.394

Rear Left

Set #	Name	Upright to Lower A-Arm Fx	Upright to Lower A-Arm Fy	Upright to Lower A-Arm Fz	Upright to Upper A-Arm Fx	Upright to Upper A-Arm Fy	Upright to Upper A-Arm Fz	Upright to Tierod Fx	Upright to Tierod Fy	Upright to Tierod Fz
0	Pure Cornering (Left Turn)	3.777	46.676	4.347	-0.516	-23.499	-7.653	-5.178	11.270	1.006
1	Pure Acceleration	671.306	68.397	13.663	-253.029	-151.606	-37.943	-148.507	324.007	26.545
2	Pure Breaking	202.176	2.383	2.968	-76.939	-32.749	-10.881	-41.690	90.053	10.115
3	Cone Strike (FR)	-6.070	192.560	-4.172	6.037	-93.174	-13.357	-20.422	45.182	-1.375
4	Combined Corner and Brake (Left Turn)	13.544	24.937	2.690	-4.589	-14.570	-4.996	-5.130	11.128	1.108
5	Combined Corner and Cone (FR)	7.451	114.682	2.420	0.905	-60.120	-8.119	-13.795	30.586	-1.060
6	Combined Brake and Cone (FR)	112.830	17.619	7.876	-39.986	-32.045	-4.994	-27.071	59.760	1.203
7	Combined Corner and Accel	404.088	390.075	45.171	-143.396	-264.490	-82.738	-120.030	261.195	23.170
8	Combined Corner Brake and Cone	9.715	20.646	1.588	-3.162	-12.034	-3.229	-4.061	8.873	0.473

Rear Right

Set #	Name	Upright to Lower A-Arm Fx	Upright to Lower A-Arm Fy	Upright to Lower A-Arm Fz	Upright to Upper A-Arm Fx	Upright to Upper A-Arm Fy	Upright to Upper A-Arm Fz	Upright to Tierod Fx	Upright to Tierod Fy	Upright to Tierod Fz
0	Pure Cornering (Left Turn)	-130.840	543.369	-55.736	27.495	-312.366	114.673	82.497	135.631	-13.931
1	Pure Acceleration	671.306	-68.397	13.663	-253.029	151.606	-37.943	148.507	324.007	26.545
2	Pure Breaking	202.176	2.383	2.968	-76.939	32.749	-10.881	41.690	90.053	10.115
3	Cone Strike (FR)	-6.070	192.560	-4.172	6.037	93.174	-13.357	20.422	45.182	-1.375
4	Combined Corner and Brake (Left Turn)	227.848	348.641	-37.699	-99.116	-143.208	59.060	21.877	47.345	5.204
5	Combined Corner and Cone (FR)	-75.345	174.751	-4.901	15.684	-116.767	17.236	28.330	62.553	1.643
6	Combined Brake and Cone (FR)	112.830	17.619	7.876	-39.986	32.045	-4.994	27.071	59.760	1.203
7	Combined Corner and Accel	657.603	301.188	-46.634	-265.966	-145.152	62.922	98.743	214.367	1
8	Combined Corner Brake and Cone	162.683	199.992	-7.385	-70.073	-78.582	24.146	-20.986	-45.782	2.642

FEA Boundary Conditions:

Fixtures

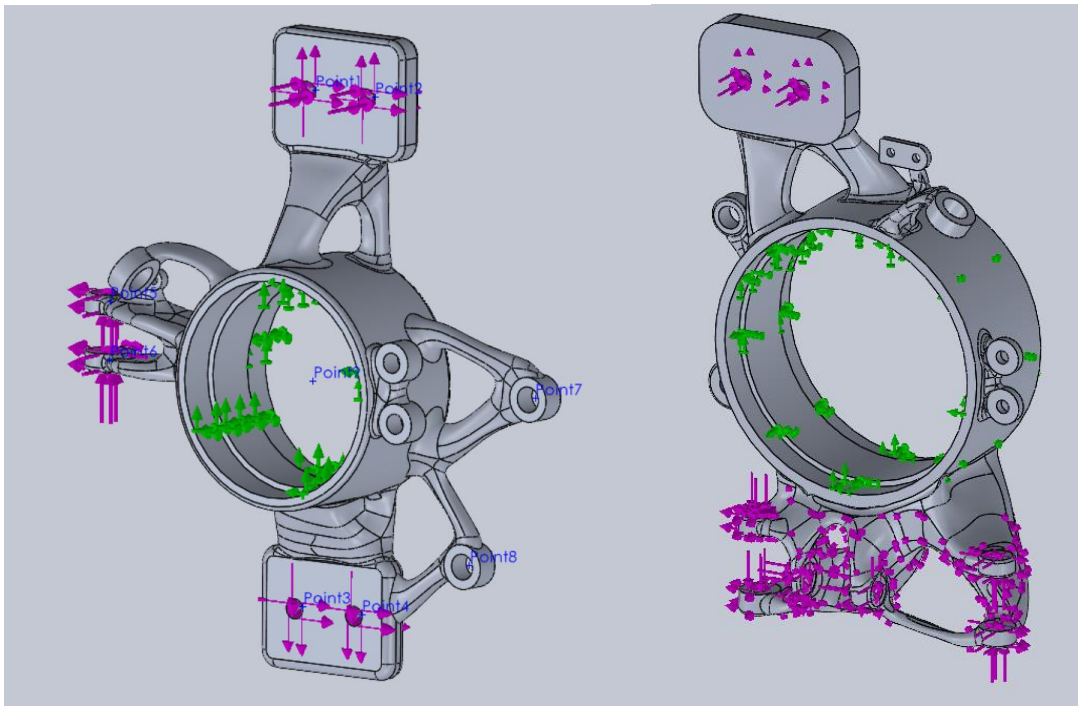


Figure: Front Upright with shown fixtures and loads

Fixed geometry is applied to the faces of the wheel bearing holders. This is because in a static loading scenario, the wheel bearings, which are press fit into the uprights, restricted all movement in the x, y, and z directions. Loads are given to each of the bolt holes where loadbearing suspension components are attached. Bearing loads are given for radial loads and force load is given for axial loads.

Mesh

In order to achieve the most accurate results, using a standard fine mesh usually results in large elements with large aspect ratios. This is undesirable, as larger aspect ratios result in less detail and less accuracy.

To mitigate this, an h-convergence mesh was used in order to find a mesh size that is fine enough to a point where further refining the mesh will yield the same results. However, setting the convergence to 98%, the solver would fixate on arbitrary stress concentrations which are not representative of real life behavior.

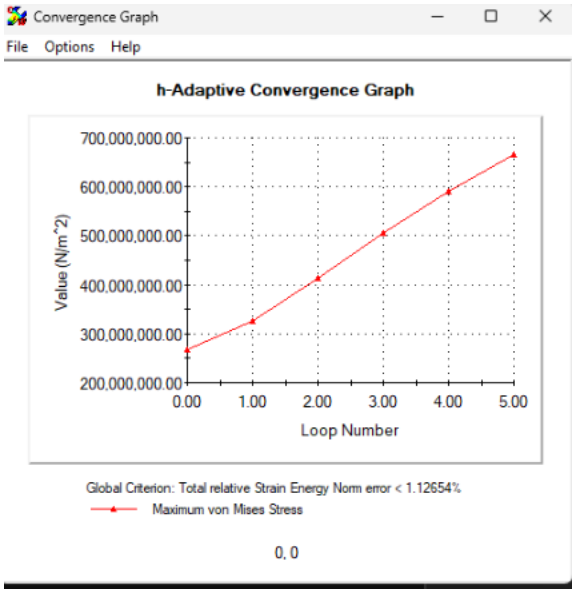
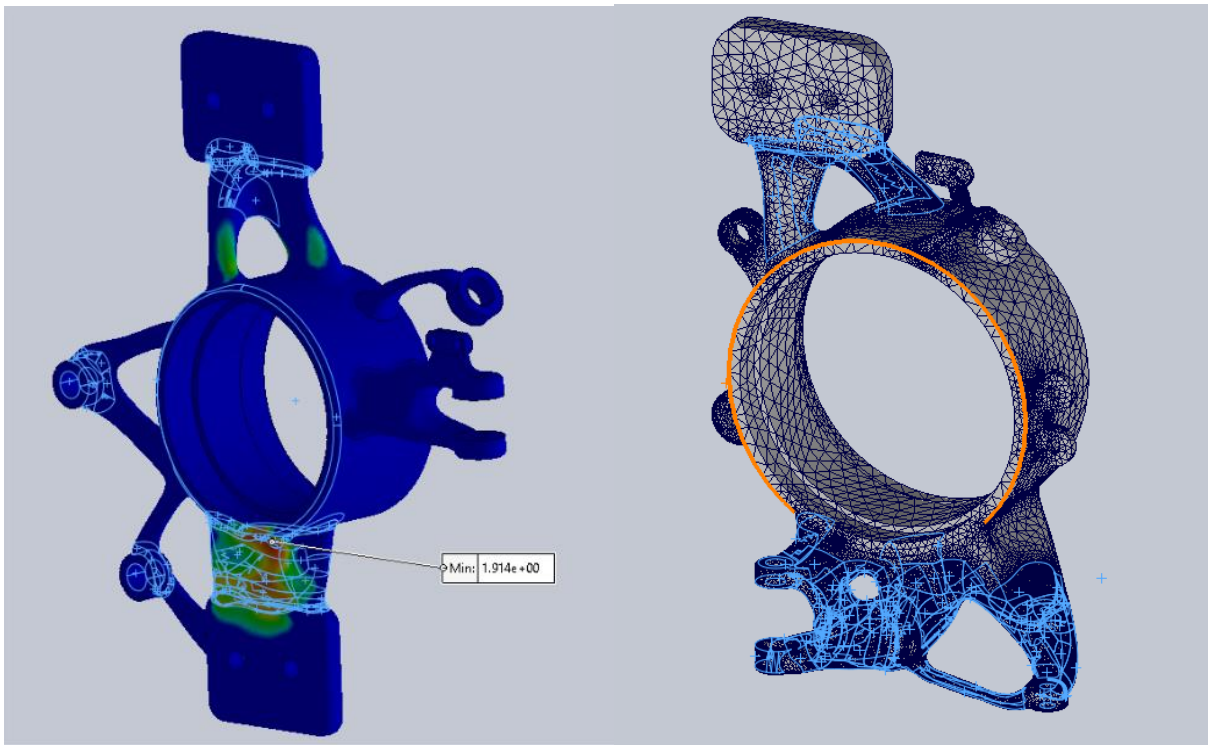
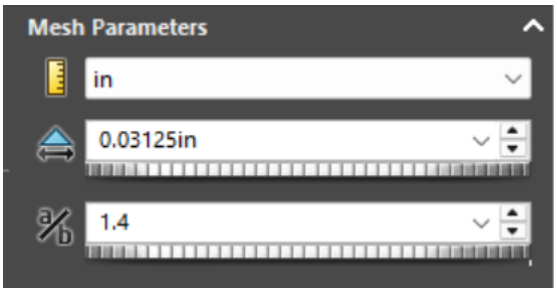


Figure: Stress concentration on outer edge of bearing holder & h-convergence plot

Instead, local refined meshes were used to get more accurate results. A mesh size of 0.03125 (1/32”) was chosen, as that was believed to be the minimum tool size that would be used to machine the uprights. As a result, there should not be any features which are smaller than the minimum mesh size.





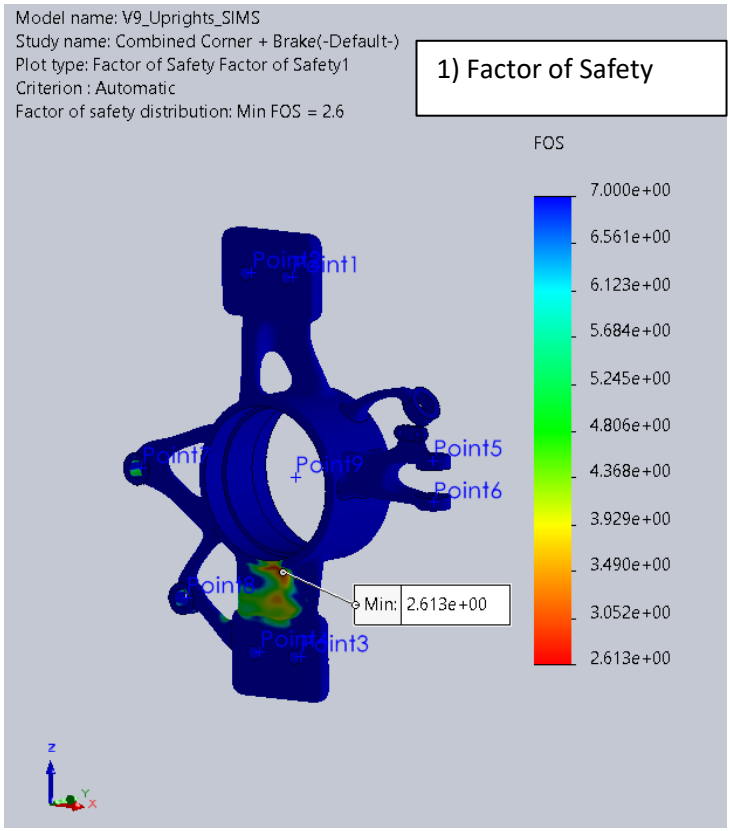
Percentage of elements with Aspect Ratio < 3	99.3
Percentage of elements with Aspect Ratio > 10	0.0234

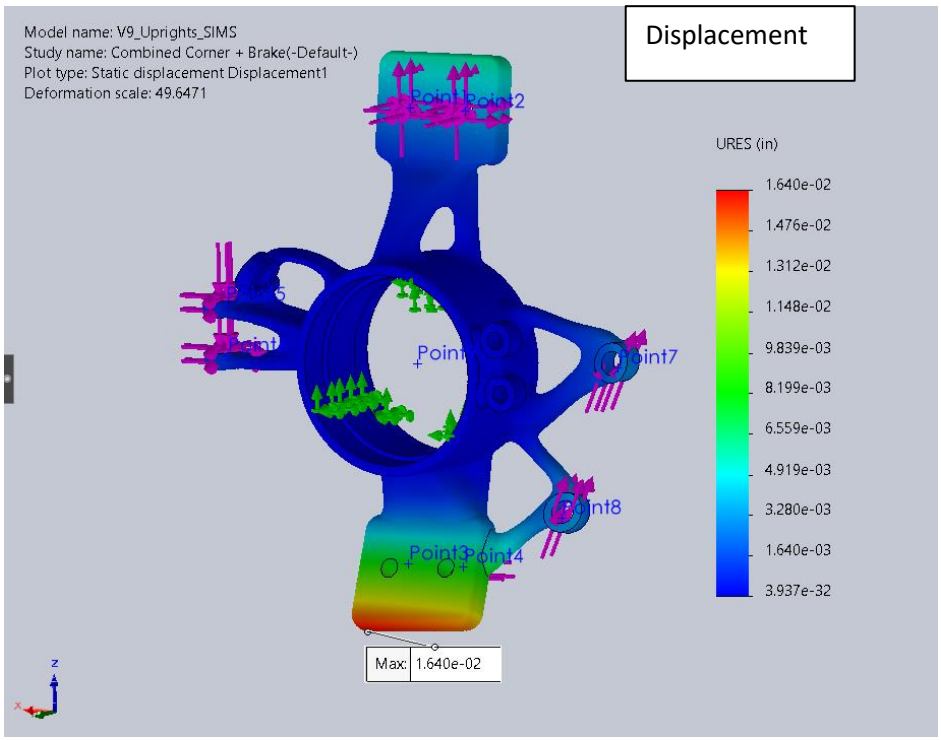
Percentage of elements with Aspect Ratio < 3	99.2
Percentage of elements with Aspect Ratio > 10	0.0323

Left: Mesh Refinement Results; Right: h-convergence

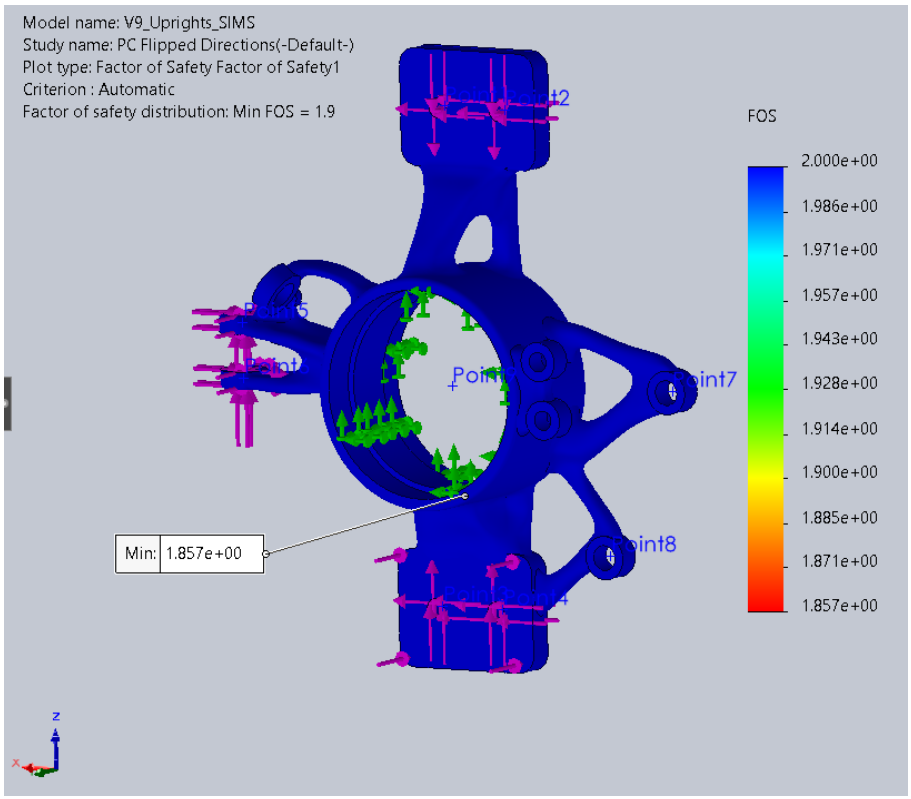
Results

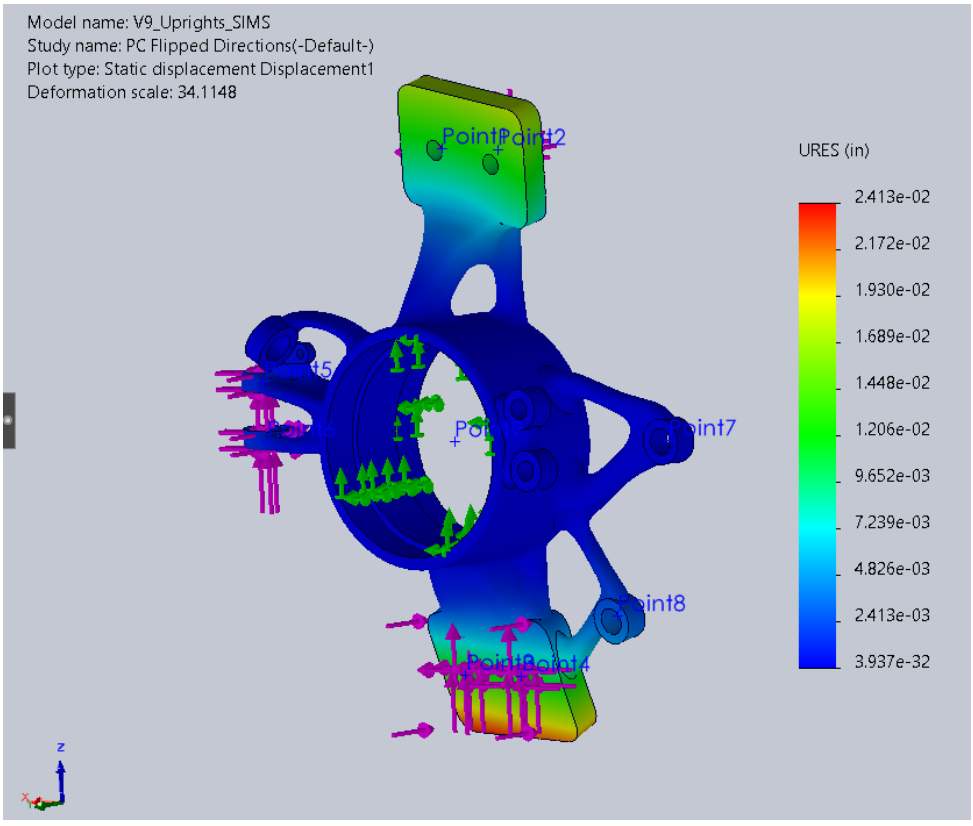
Loading Case 4: Combined Cornering and Braking (Trail Braking)



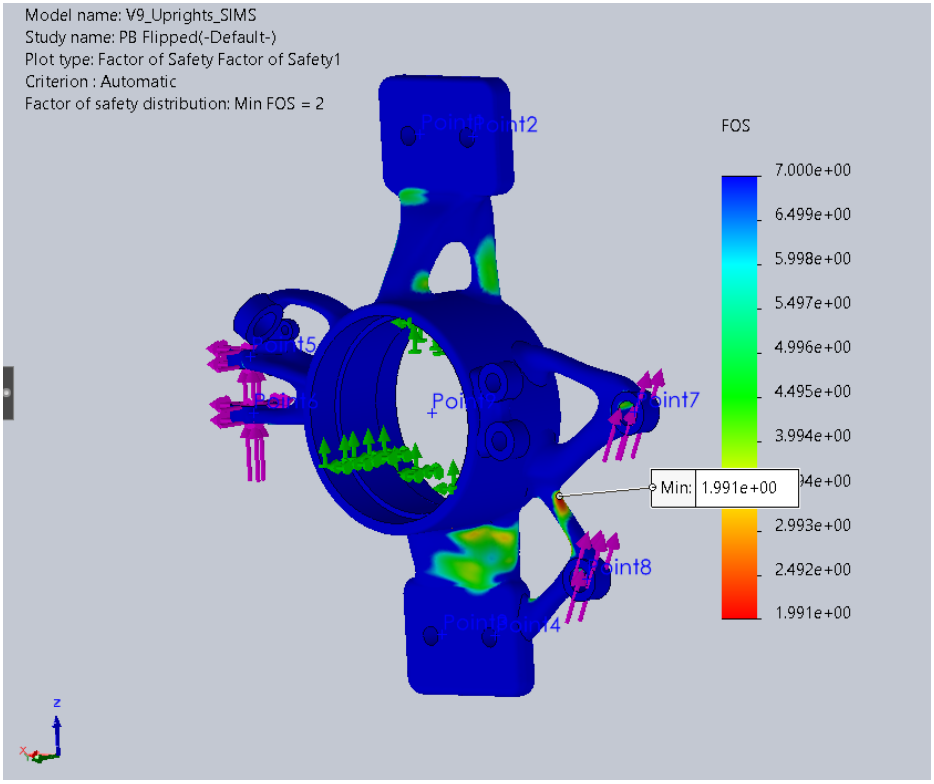


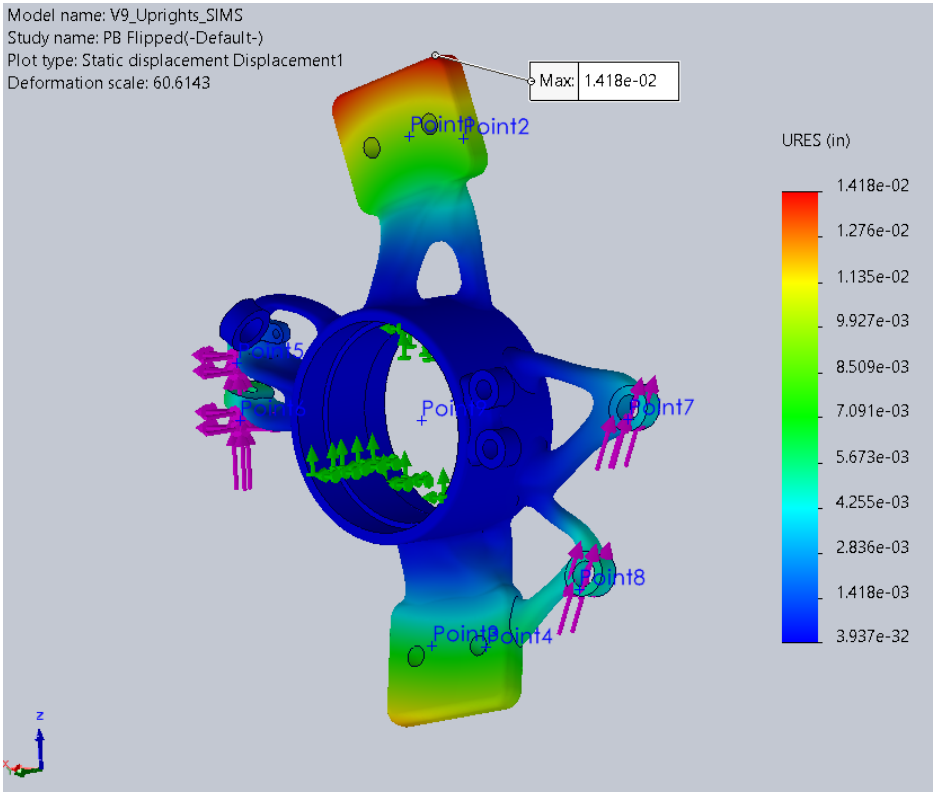
Loading Case 0: Pure Cornering



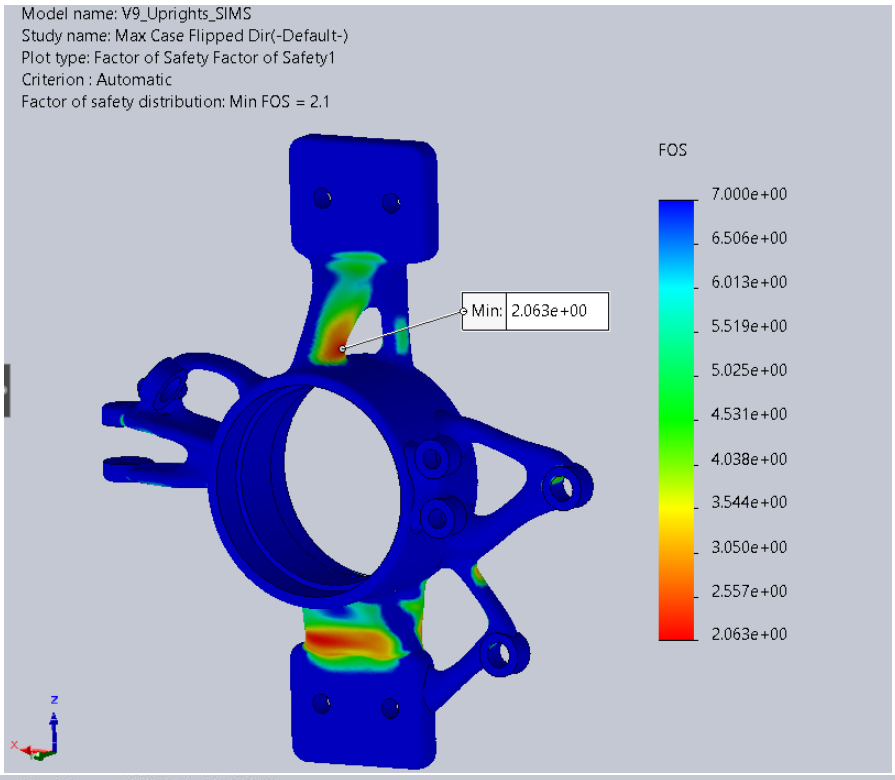


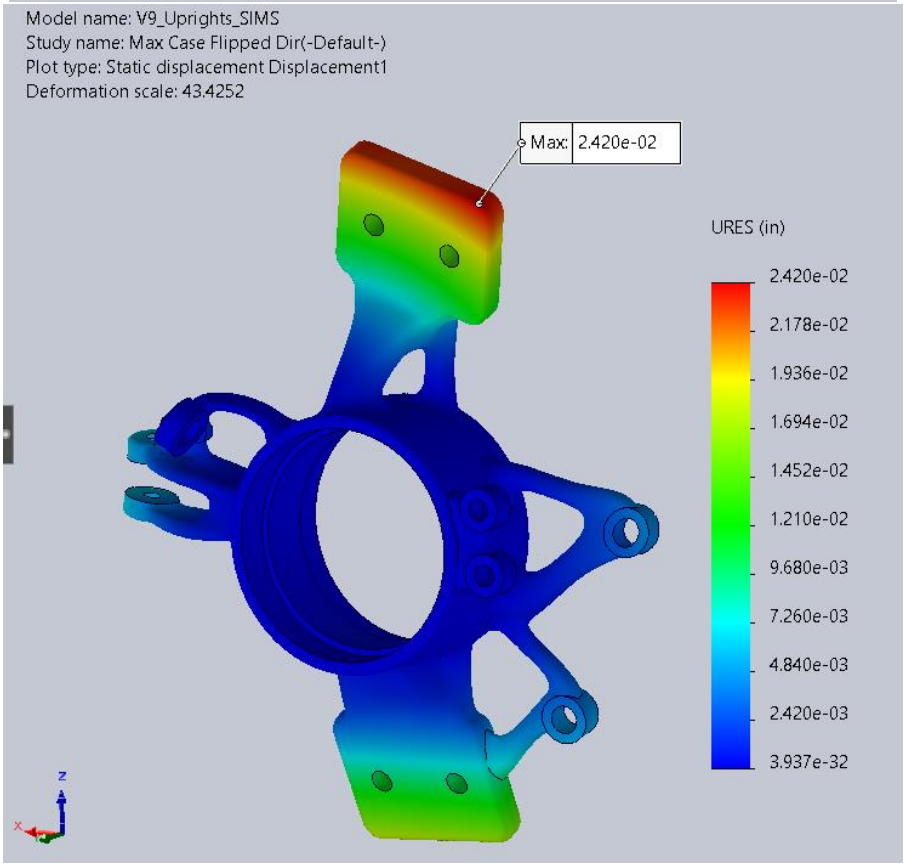
Case 2: Pure Braking





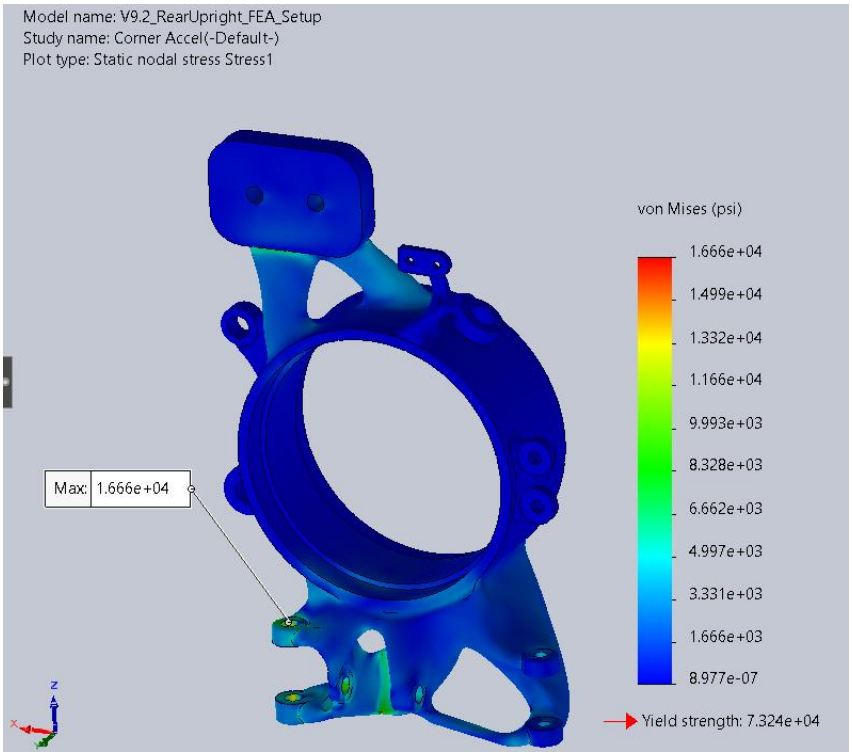
Maximum Loads

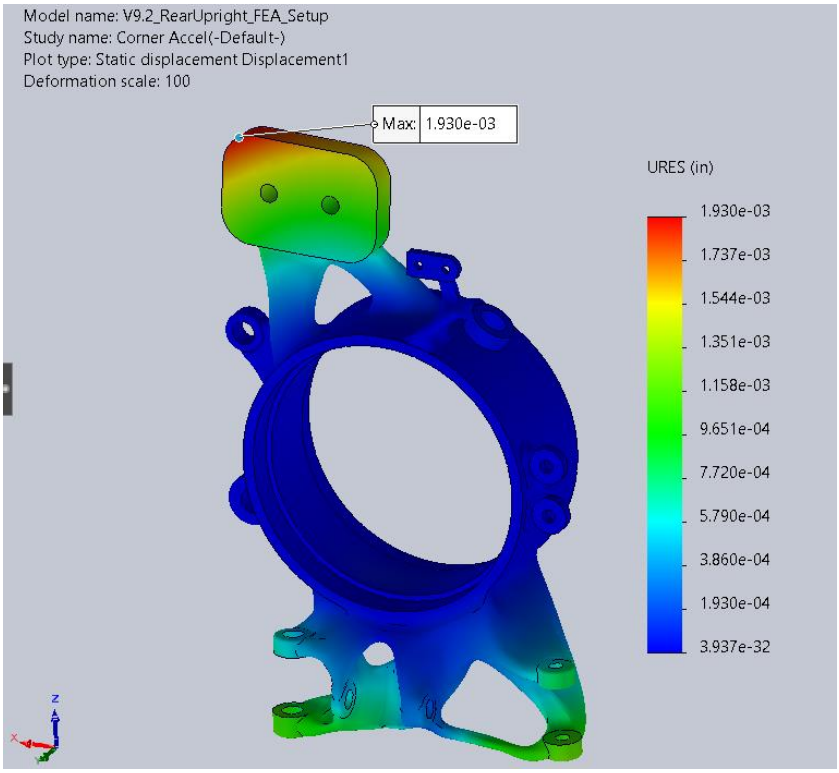




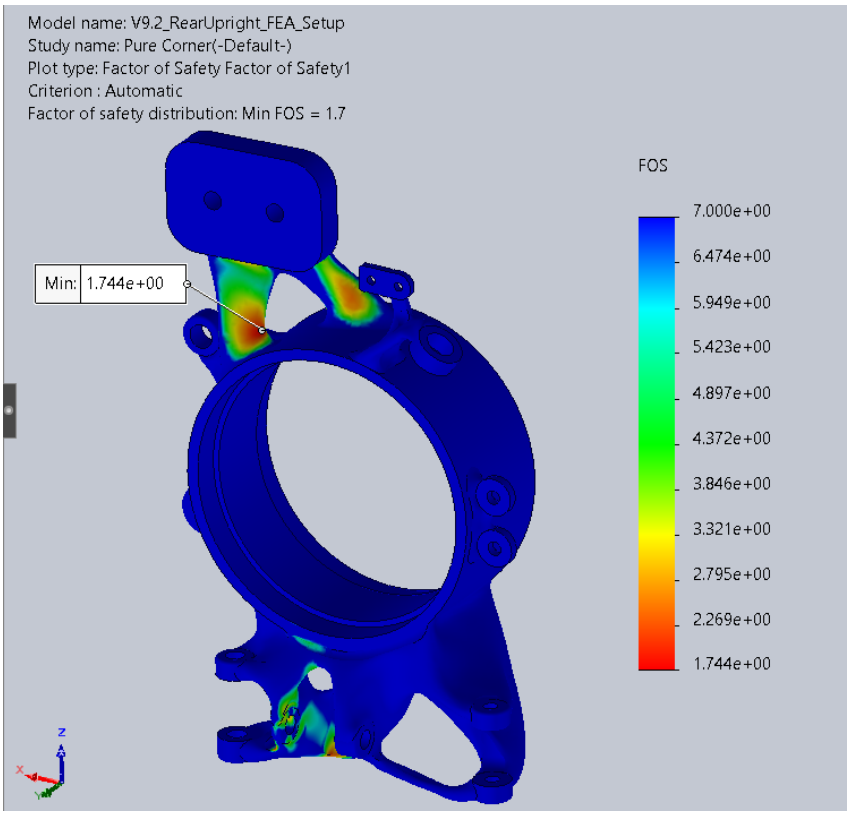
Rear Uprights

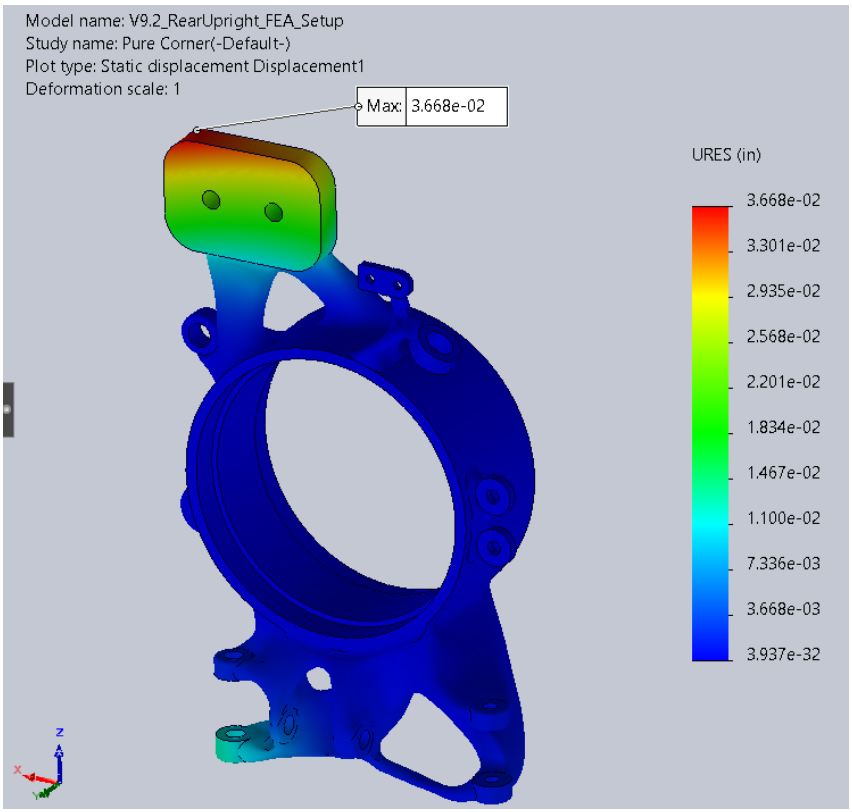
Case 7: Combined Cornering and Acceleration



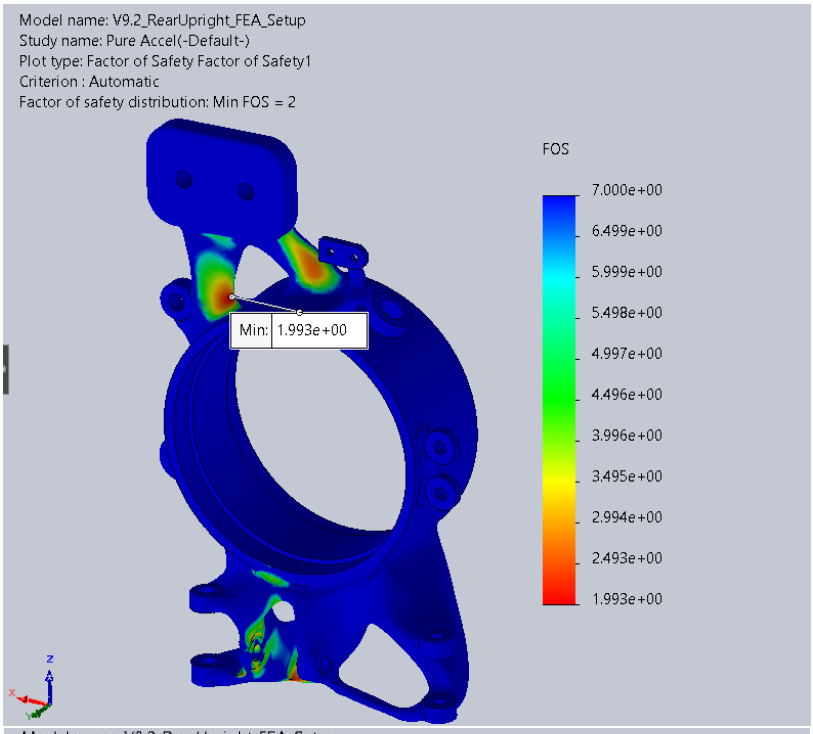


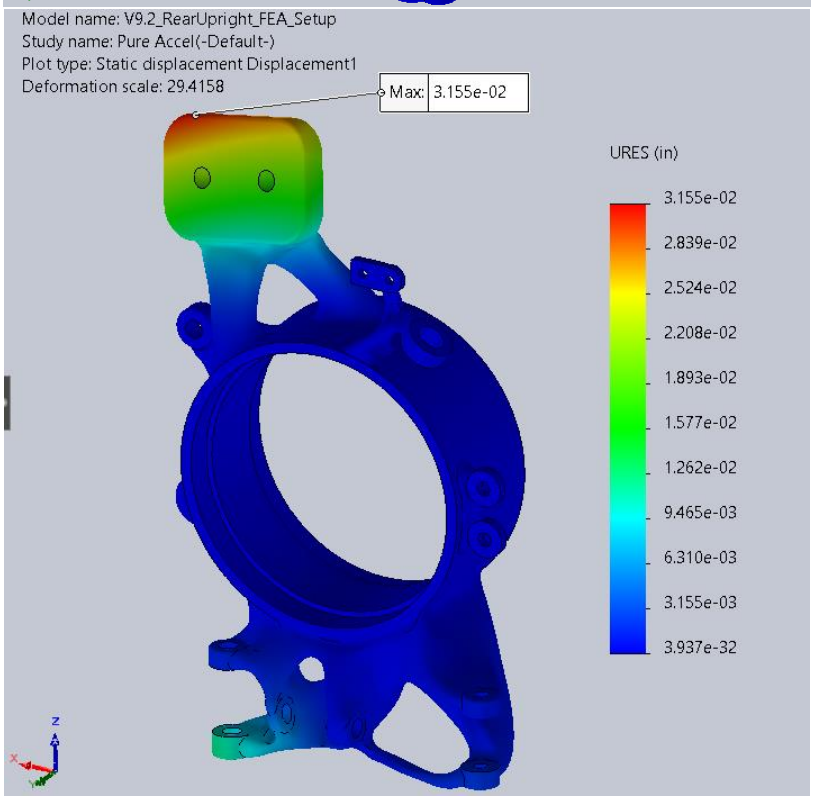
Case 0: Pure Cornering





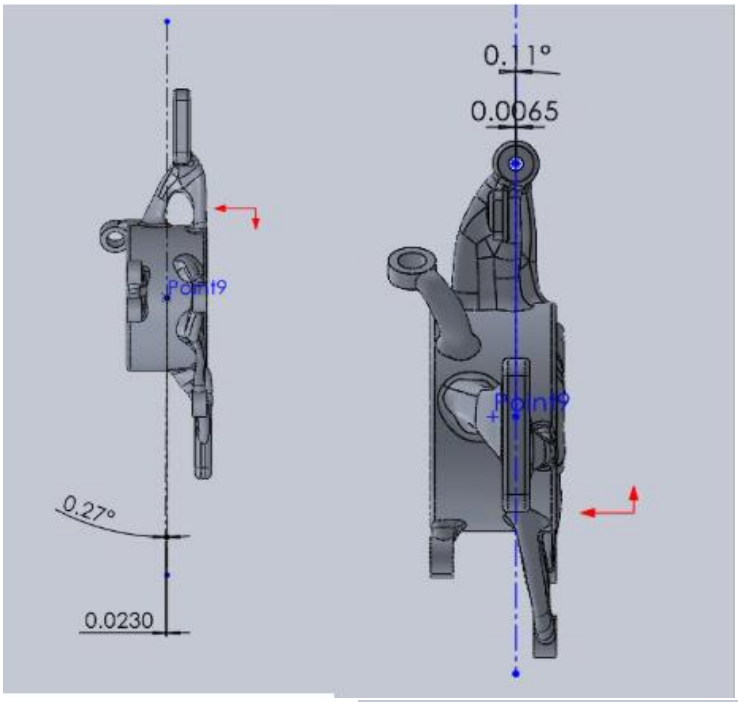
Case 1: Pure Acceleration

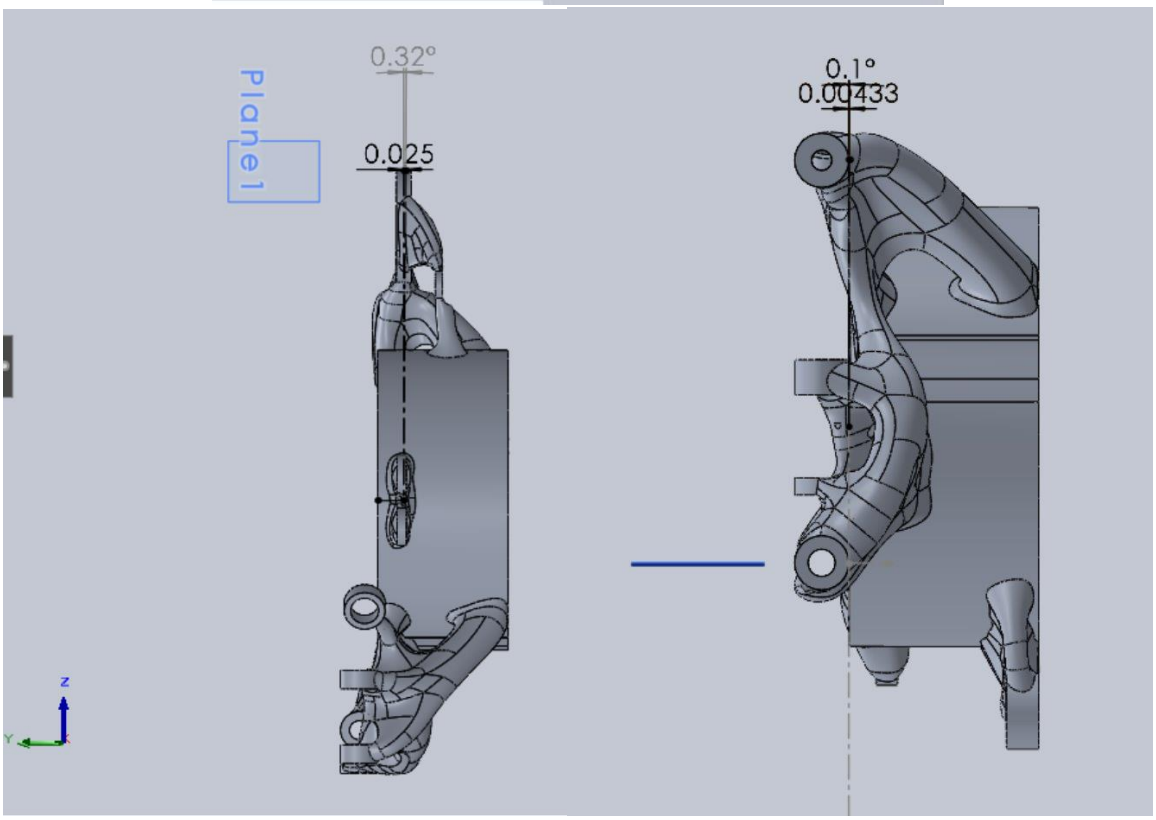

Model name: V9.2_RearUpright_FEA_Setup
Study name: Pure Accel(-Default-)
Plot type: Factor of Safety Factor of Safety1
Criterion : Automatic
Factor of safety distribution: Min FOS = 2
FOS
7.000e+00
6.499e+00
5.999e+00
5.498e+00
4.997e+00
4.496e+00
3.996e+00
3.495e+00
2.994e+00
2.493e+00
1.993e+00
Min: 1.993e+00
Z
X


Model name: V9.2_RearUpright_FEA_Setup
Study name: Pure Accel(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 29.4158
Max: 3.155e-02
URES (in)
3.155e-02
2.839e-02
2.524e-02
2.208e-02
1.893e-02
1.577e-02
1.262e-02
9.465e-03
6.310e-03
3.155e-03
3.937e-32
Z
X

Deformation Analysis

When using the generative design tool to create the upright geometry, a minimum deflection target was not input so that it could solely focus on weight. This is because the thousandths of an inch of deflection that takes place at max loads minimally affect performance.





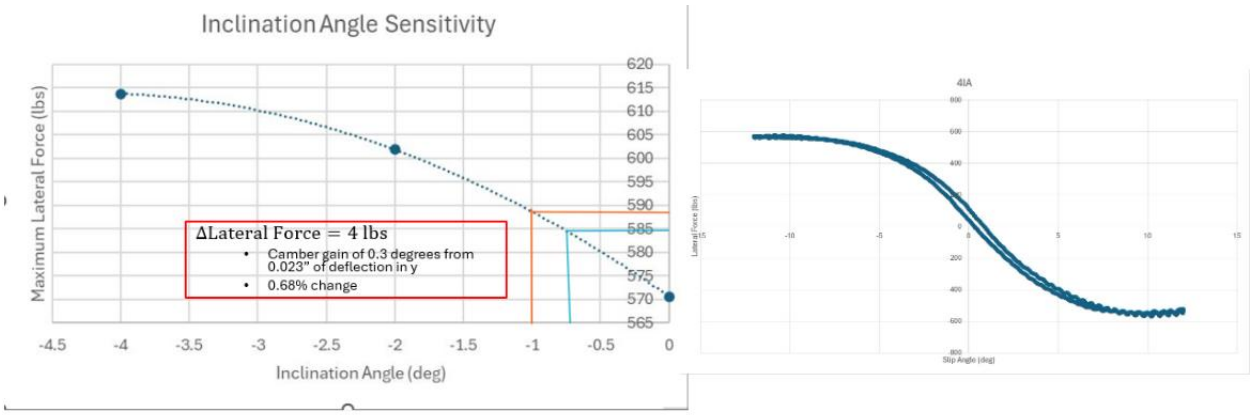


Figure: Tire Camber sensitivity and slip angle sensitivity graphs

From the tire sensitivity graphs, the maximum deflection of the upright in the x-y plane of the car results in just a 0.68% change in lateral force produced by the car.

Fatigue Analysis

Since the material of choice is 7075 Al, a fatigue study is necessary due to aluminum's susceptibility to fatigue. To make an accurate fatigue study, Motec data was used to estimate the number of instances in certain loading cases occurred.

In time intervals of 0.01s, a list of corresponding lateral, longitudinal, and vertical g's are recorded. Then a MATLAB script sifts through the g values, categorizing them into ranges (ex. 1.3-1.4g). MATLAB then calculates the number of times certain ranges occur in an endurance. A medium load case using the lower g-forces is created, and a new FEA static study is created with the smaller loads. Finally, a fatigue study referencing the medium load FEA study is made, inputting the amount of instances as a reference to calculate the damage. These values are multiplied by 60 to meet the total endurance goal.

Cornering + Braking Cases		
Lateral Acceleration Range (g)	Long. Acceleration Range (g)	Instances
1.10 - 1.25	0.45 - 0.60	816
1.35 - 1.50	0.45 - 0.60	475

Values from half endurance
30 Expected Endurances

Pure Cornering Cases	
Lateral Acceleration Range (g)	Instances
1.35 - 1.5	390
1.85 - 2.0	9

Values from half endurance
30 Expected Endurances

- From MoTec logs: number of times certain load cases appeared in endurance
- Multiply sum of instance by 2*30 (half endurance values & 30 endurances

This fatigue approximation allows us to use a constant amplitude, fully reversible loading scenario since one medium load case is used for each fatigue study.

Since the built-in SN curve for 7075 Al in SolidWorks is known to be unreliable, a curve from MMPDS-15 is used. A Kt value of 1.6 is used since the upright has areas of stress concentration, including small fillets and areas where preserve geometry meets with generative geometry. The black curve corresponding to R=-1 is used since we are assuming fully reversed loading.

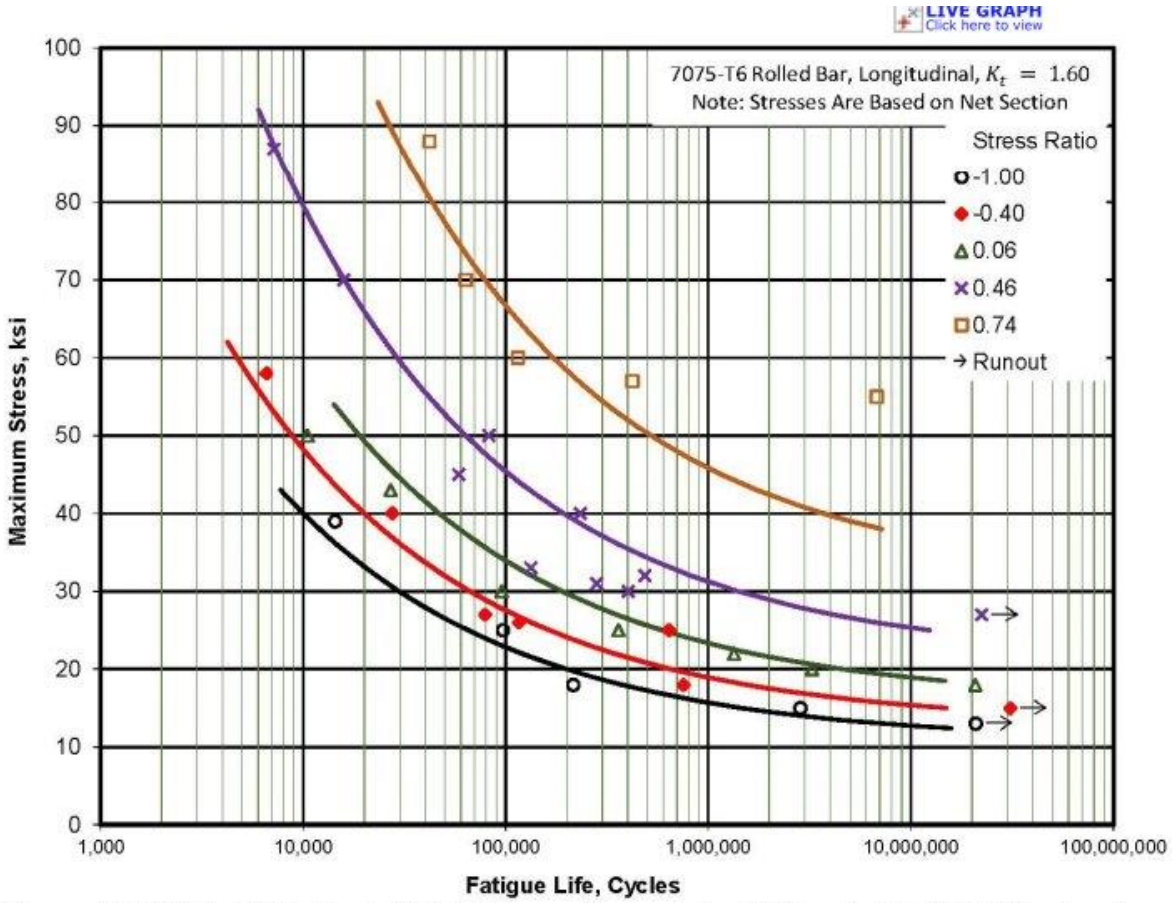
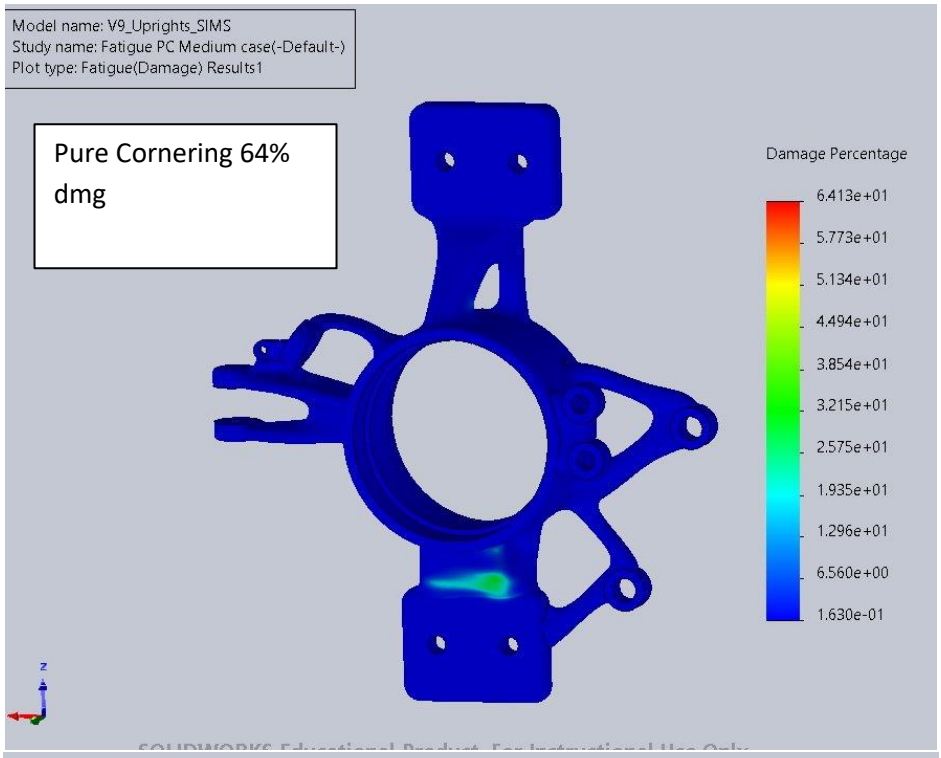
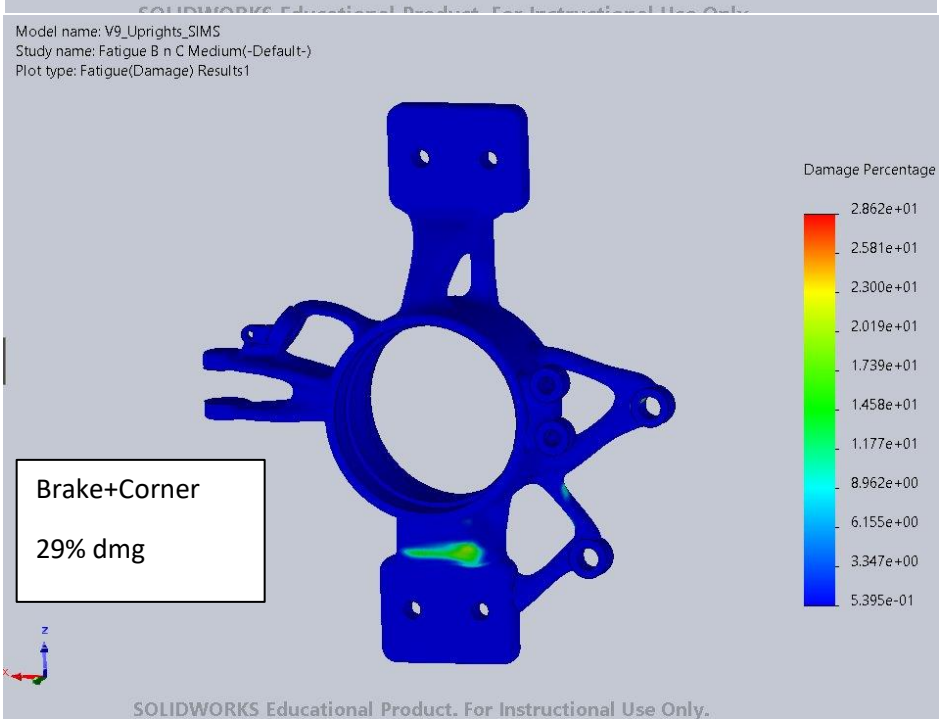
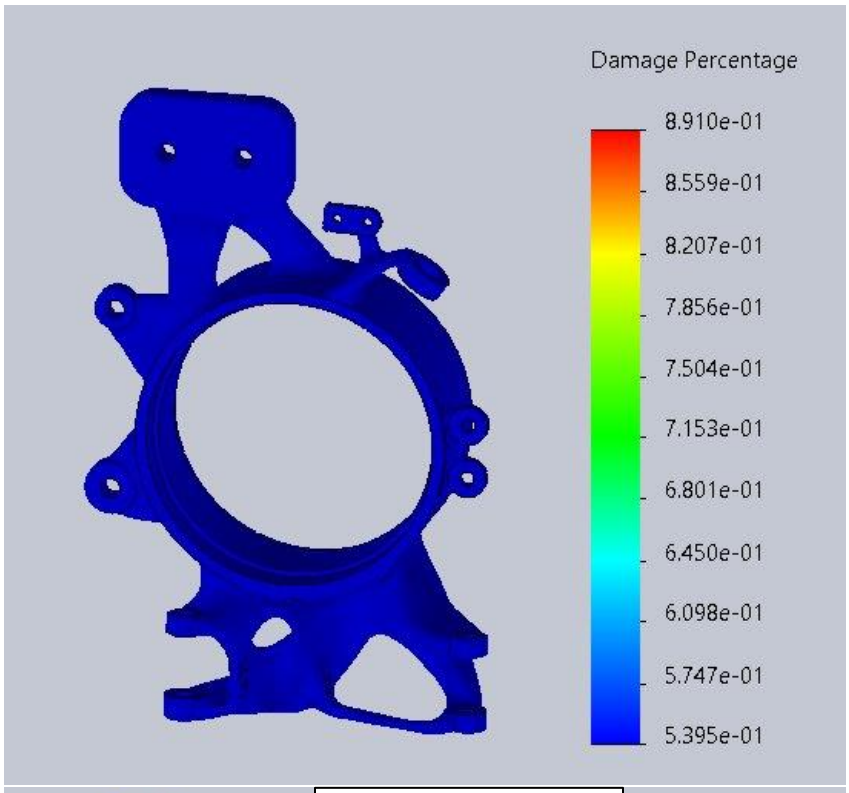


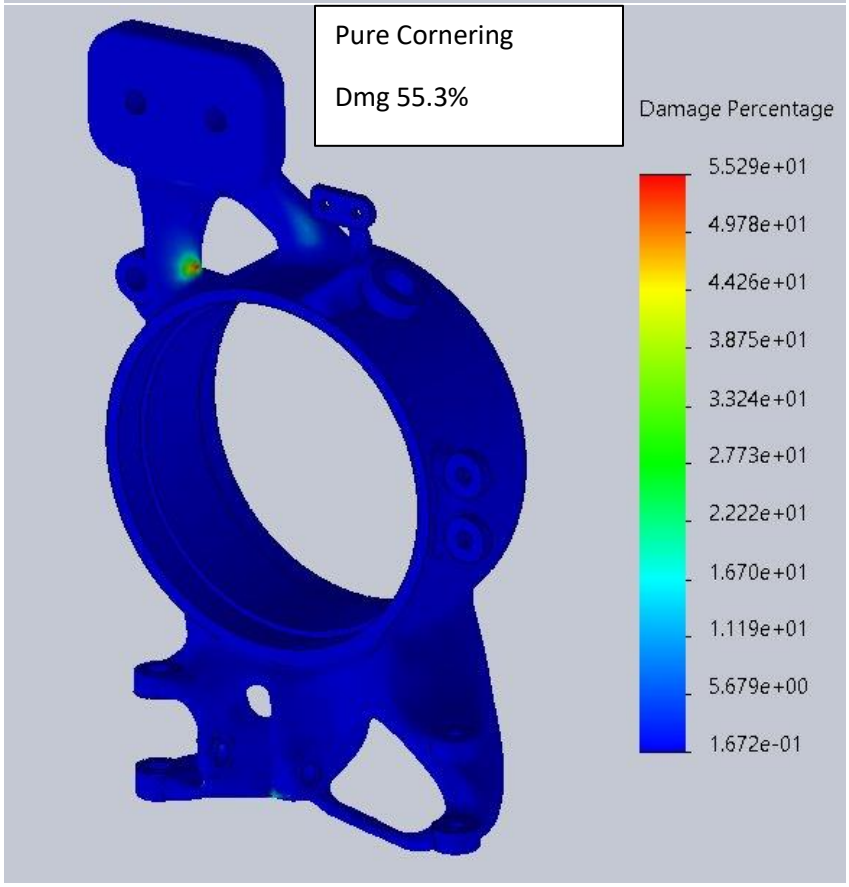
Figure 3.7.10.1.8(b). Best-fit S/N curve for notched, K_t = 1.6, 7075-T6 aluminum alloy rolled bar, longitudinal direction.

Results


Model name: V9_Uprights_SIMS
Study name: Fatigue PC Medium case(-Default-)
Plot type: Fatigue(Damage) Results1
Pure Cornering 64% dmg
Damage Percentage
6.413e+01
5.773e+01
5.134e+01
4.494e+01
3.854e+01
3.215e+01
2.575e+01
1.935e+01
1.296e+01
6.560e+00
1.630e-01


Model name: V9_Uprights_SIMS
Study name: Fatigue B n C Medium(-Default-)
Plot type: Fatigue(Damage) Results1
Brake+Corner
29% dmg
Damage Percentage
2.862e+01
2.581e+01
2.300e+01
2.019e+01
1.739e+01
1.458e+01
1.177e+01
8.962e+00
6.155e+00
3.347e+00
5.395e-01
SOLIDWORKS Educational Product. For Instructional Use Only.


Damage Percentage
8.910e-01
8.559e-01
8.207e-01
7.856e-01
7.504e-01
7.153e-01
6.801e-01
6.450e-01
6.098e-01
5.747e-01
5.395e-01

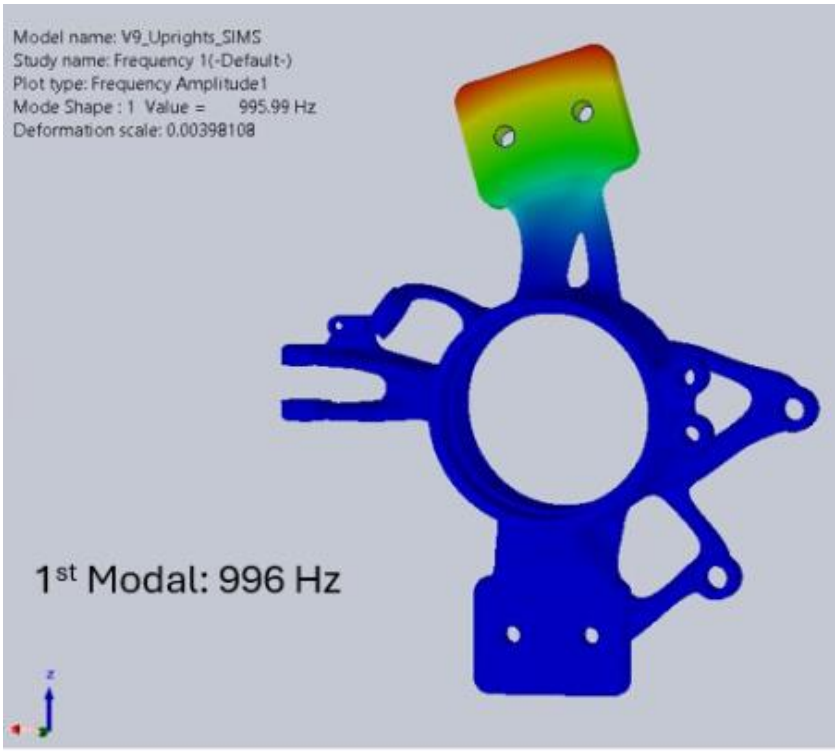

Pure Cornering
Dmg 55.3%
Damage Percentage
5.529e+01
4.978e+01
4.426e+01
3.875e+01
3.324e+01
2.773e+01
2.222e+01
1.670e+01
1.119e+01
5.679e+00
1.672e-01

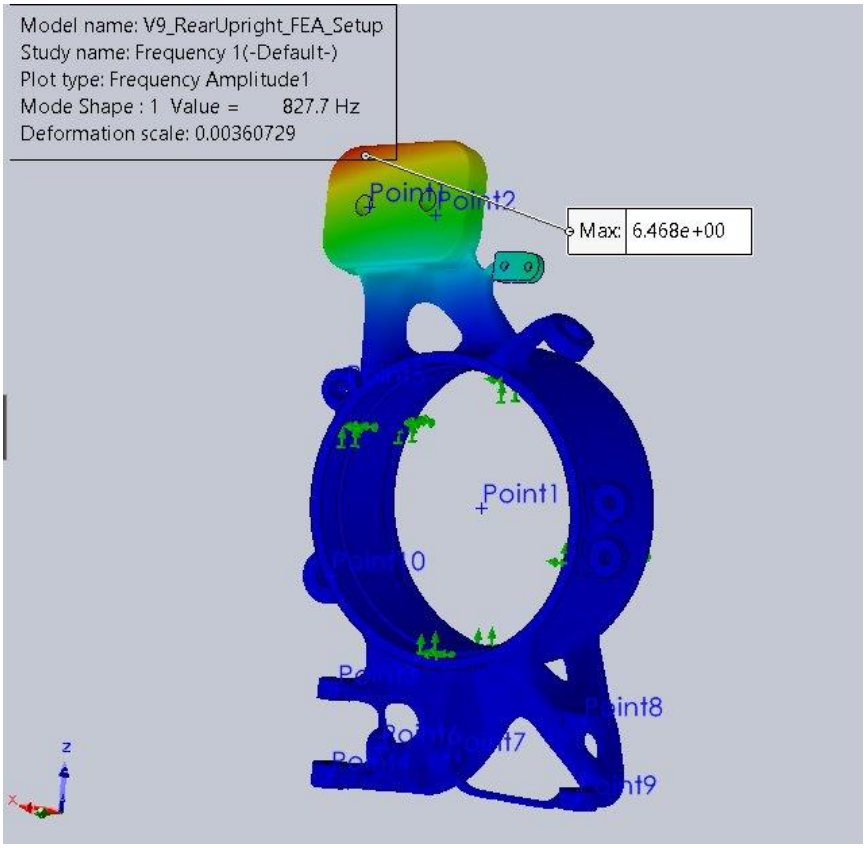
Frequency Analysis

To ensure that the upright resonant frequency is not similar to any major vibrations in the car experiences, a minimum resonance is inputted into the generative design software. One obvious source of vibrations to the upright is the tires as they roll. Based on a ISMA 2010 study, a slick tire rolling on a smooth surface has a natural frequency of around 80-150 Hz. Another source of vibrations is the engine since it is solidly mounted to the chassis with no damping sources other than the material damping of the steel tubes. As a result, this vibration travels through the uprights. Using the following formula, the engine's maximum frequency is just around 400 Hz, much higher than that of the tire. To avoid any mechanical resonance in the upright, we must ensure that the first or minimum natural frequency of the upright is above $400 \times \sqrt{2} = 567$ Hz

- **Calculation:** The firing frequency depends on the engine speed (RPM), the number of cylinders, and the engine cycle (2-stroke or 4-stroke).
  - **Firing Frequency (Hz) = (Engine RPM × Number of Cylinders) / (120 × Number of Strokes per Cycle)**
  - *For a 4-stroke engine:* Firing Freq (Hz) = (RPM × N_cyl) / 120

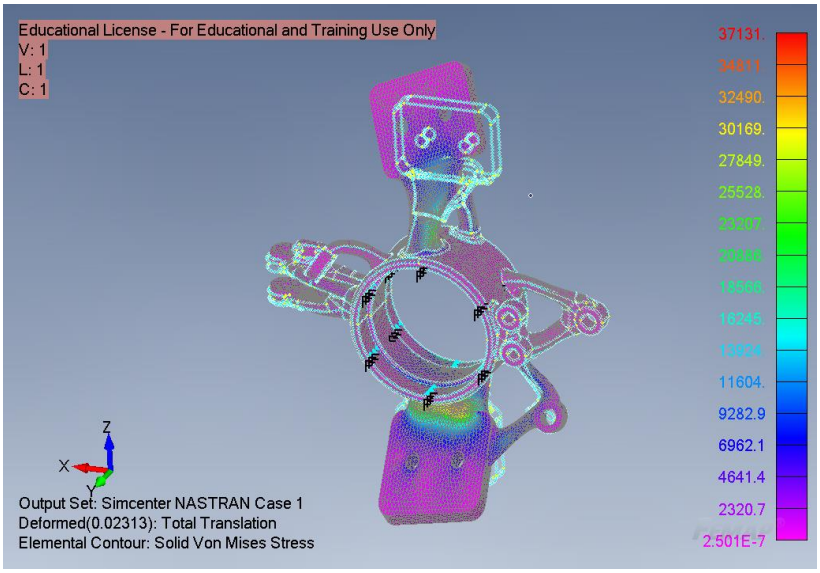
Figure: Equation to determine engine's frequency at maximum RPM (12,000)





Using FEMAP

Many structural engineers in industry point out that SolidWorks FEA is not the most accurate due to the inflexibility for meshing and boundary conditions. Even though this is a relatively simple simulation, we will use FEMAP to verify the accuracy of the SolidWorks study. To start off, a tetrahedral mesh is used with a minimum size of .03 in, like the SolidWorks mesh. The BCs are the same, fixed at the bearing holders with a combination of bearing loads and force loads at the corresponding pickup points. The FEMAP study could only be done on the front upright due to the rear uprights having unsolvable imported geometry issues, leading to meshing errors in FEMAP. I was unable to figure out how to fix the issue.

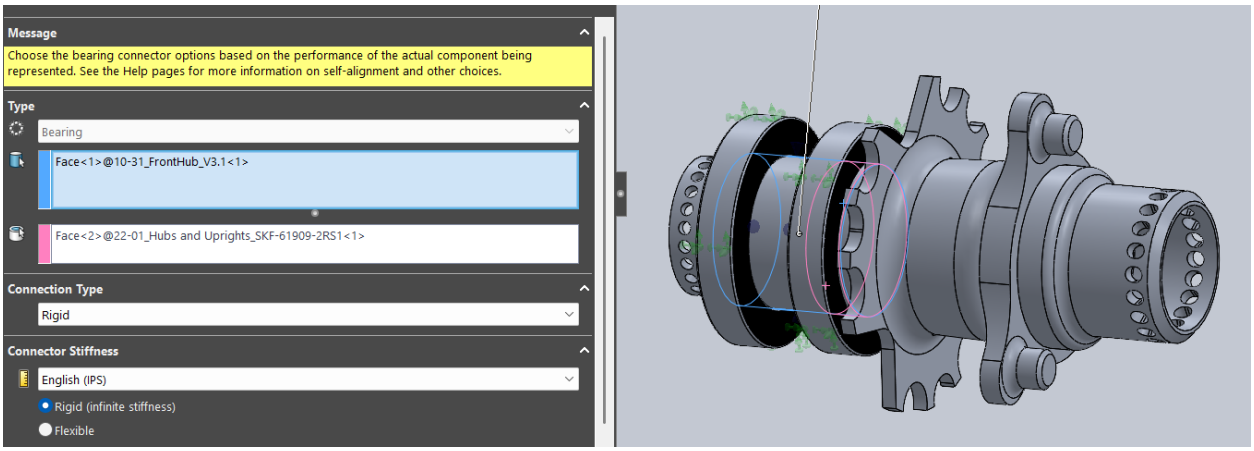


Hubs

Load Cases Analysis for OptK Force Module									
Set #	Name	Vertical Loads [lbf] (Fz)				Lateral Loads [lbf] (Fy)			
		Front Left	Front Right	Rear Left	Rear Right	Front Left	Front Right	Rear Left	Rear Right
0	Pure Cornering (Left Turn)	24.53	274.73	11.18	319.58	48.07	538.46	21.90	626.37
1	Pure Acceleration	42.98	42.98	272.03	272.03	0.00	0.00	0.00	0.00
2	Pure Breaking	256.51	256.51	58.50	58.50	0.00	0.00	0.00	0.00
3	Cone Strike (FR)	149.63	386.03	165.38	165.38	0.00	0.00	0.00	0.00
4	Combined Corner and Brake (Left Turn)	121.92	287.72	8.09	212.29	152.39	359.64	10.11	265.36
5	Combined Corner and Cone (FR)	251.83	488.23	39.48	291.28	290.39	563.00	45.52	335.88
6	Combined Brake and Cone (FR)	256.51	492.91	58.50	58.50	0.00	0.00	0.00	0.00
7	Combined Corner and Accel (Left Turn)	(-) OFF	163.65	159.62	358.62	(-) OFF	207.14	202.04	453.93
8	Combined Corner Brake and Cone (Left Turn & FR)	121.92	524.12	8.09	212.29	140.95	605.97	9.35	245.43
Set #	Name	Longitudinal Loads [lbf] (Fx)				Self Aligning Torque [lbf*ft]			
		Front Left	Front Right	Rear Left	Rear Right	Front Left	Front Right	Rear Left	Rear Right
0	Pure Cornering (Left Turn)	0.00	0.00	0.00	0.00	N/A			
1	Pure Acceleration	49.85	49.85	315.55	315.55				
2	Pure Breaking	256.51	256.51	58.50	58.50				
3	Cone Strike (FR)	0.00	0.00	0.00	0.00				
4	Combined Corner and Brake (Left Turn)	73.15	172.63	4.85	127.37				
5	Combined Corner and Cone (FR)	0.00	0.00	0.00	0.00				
6	Combined Brake and Cone (FR)	227.55	437.27	51.89	51.89				
7	Combined Corner and Accel (Left Turn)	0.00	161.78	157.80	354.53				
8	Combined Corner Brake and Cone (Left Turn & FR)	55.85	240.10	3.70	97.25				

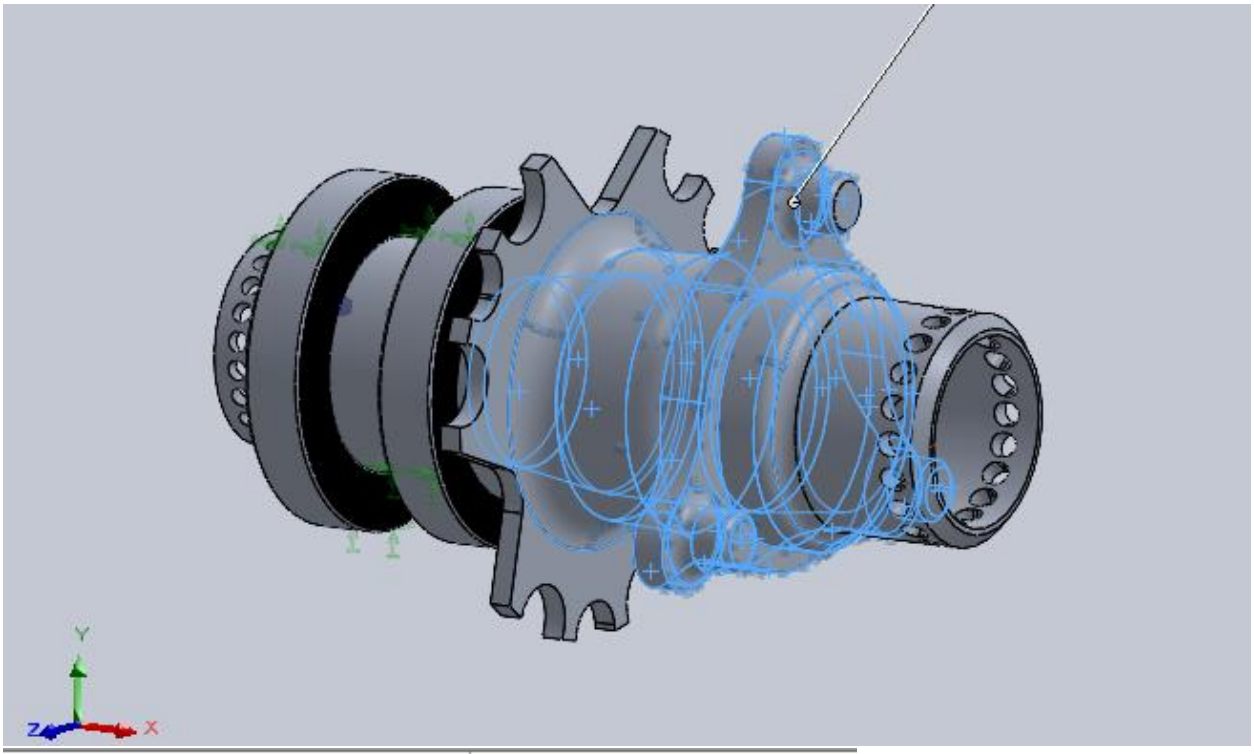
FEA Boundary Conditions

Fixtures and Connections



The outer bearing races are both fixed as they are both pressed into the uprights. For the inner bearing surface, there is a “Bearing” connection with the hub. This is assumed to be a rigid connection since the bearings are secured from sliding via a nut which rests on the inner race. Additionally, there is a “bonded” global interaction between both bearings and the hub. This ensures that the bearings will be fixed relative to the hub and that the assembly is properly constrained.

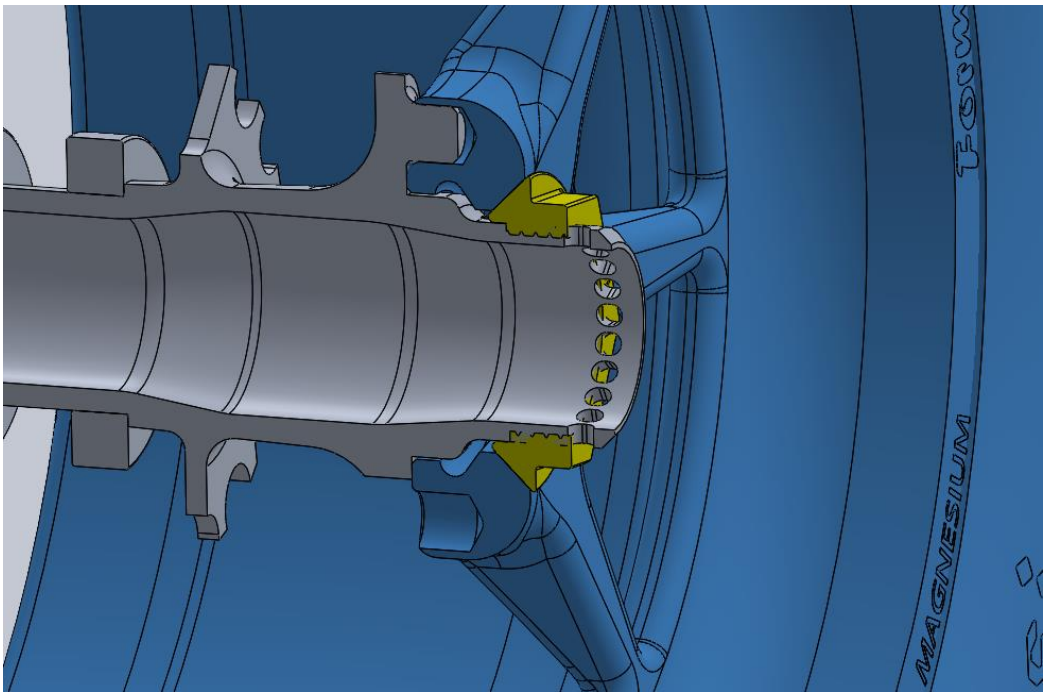
Mesh

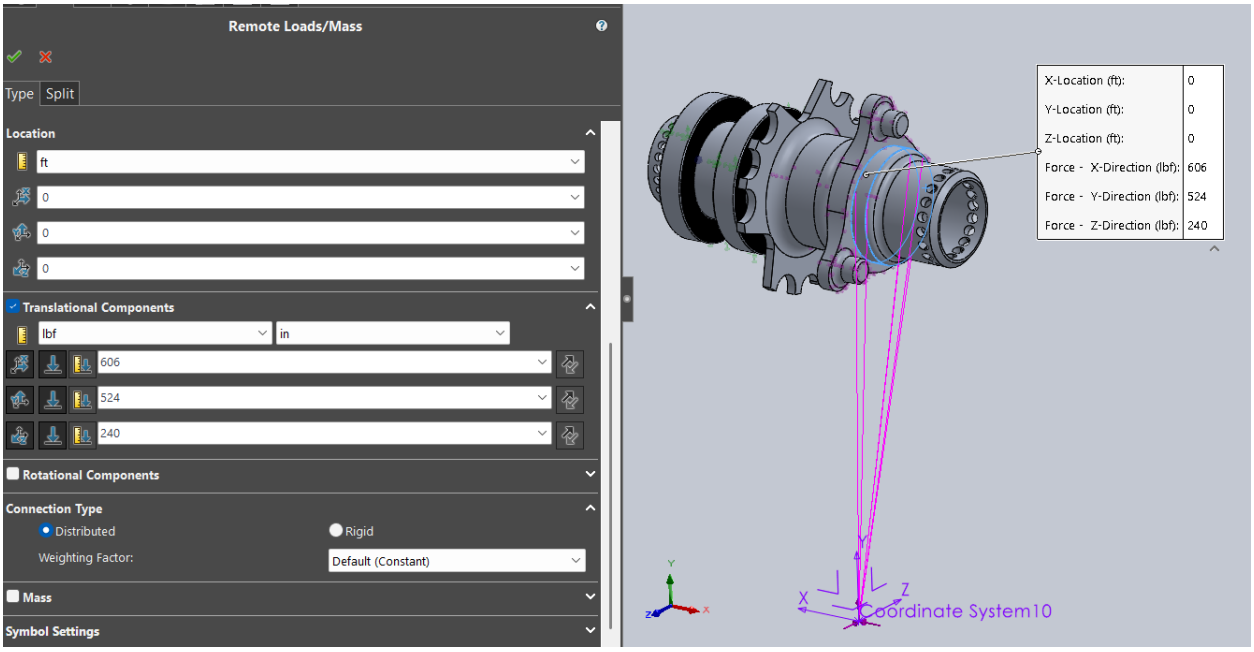


Study name	Fillet C B C Remote* (-Default-)
Total nodes	963462
Total elements	638861
Maximum Aspect Ratio	8.9374
Percentage of elements with Aspect Ratio < 3	99.6
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Number of distorted elements	0

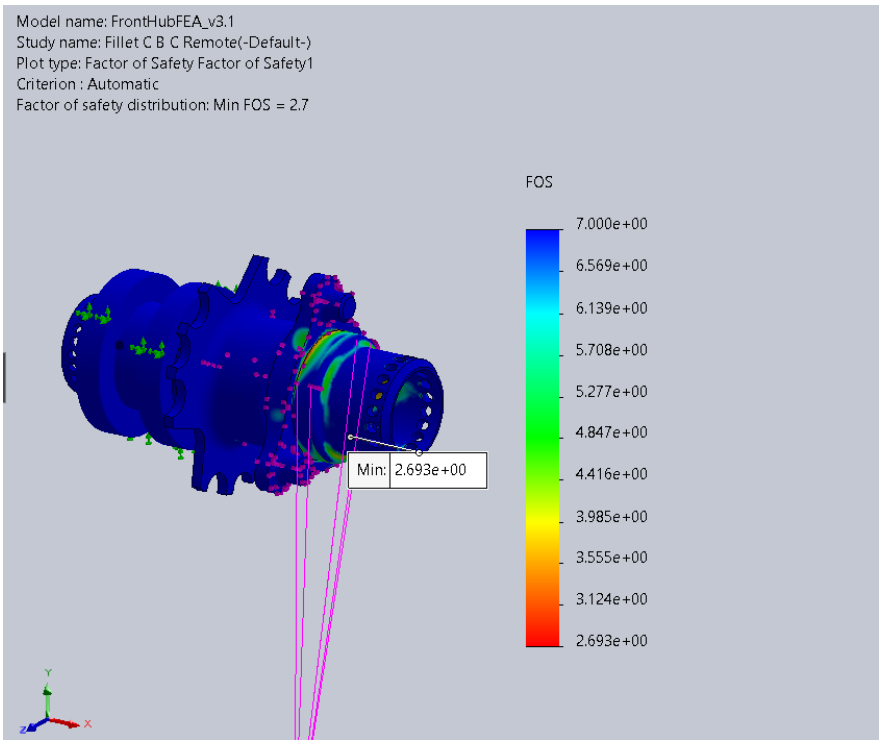
Loading

Due to the wheel nut securely fastening the wheel to the hub, we could treat the entire assembly as one piece. So, using the lateral, longitudinal, and vertical forces at the tire, a remote load is placed at the center of the contact patch. The surface on the hub is the face underneath the dowel alignment pin where the wheel and hub contact each other.

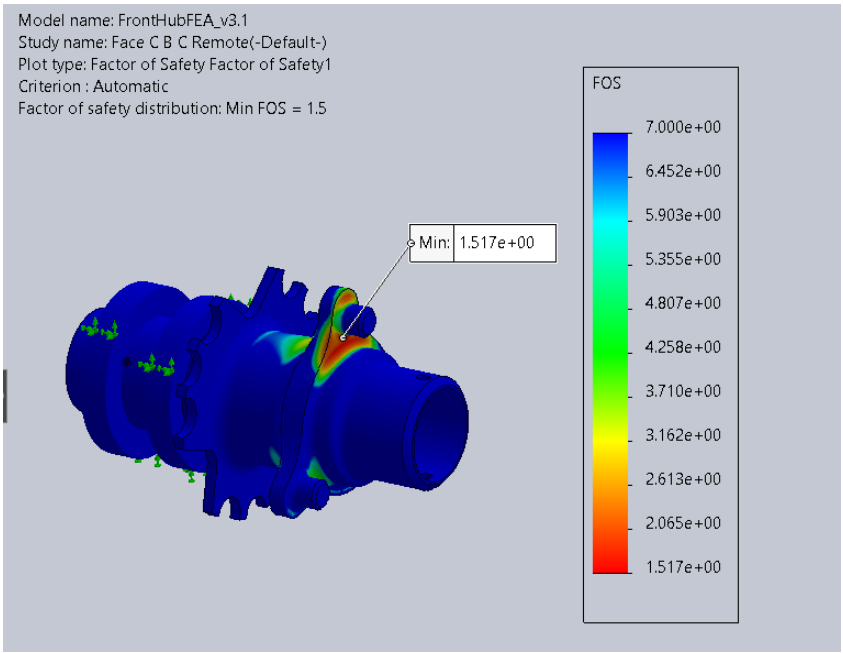


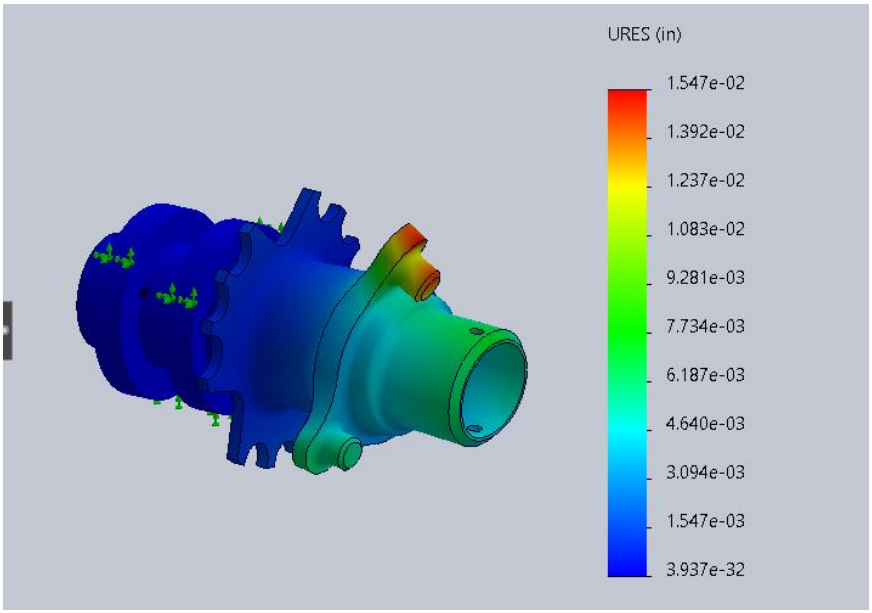


Case 8: Combined Corner, Brake, and Cone

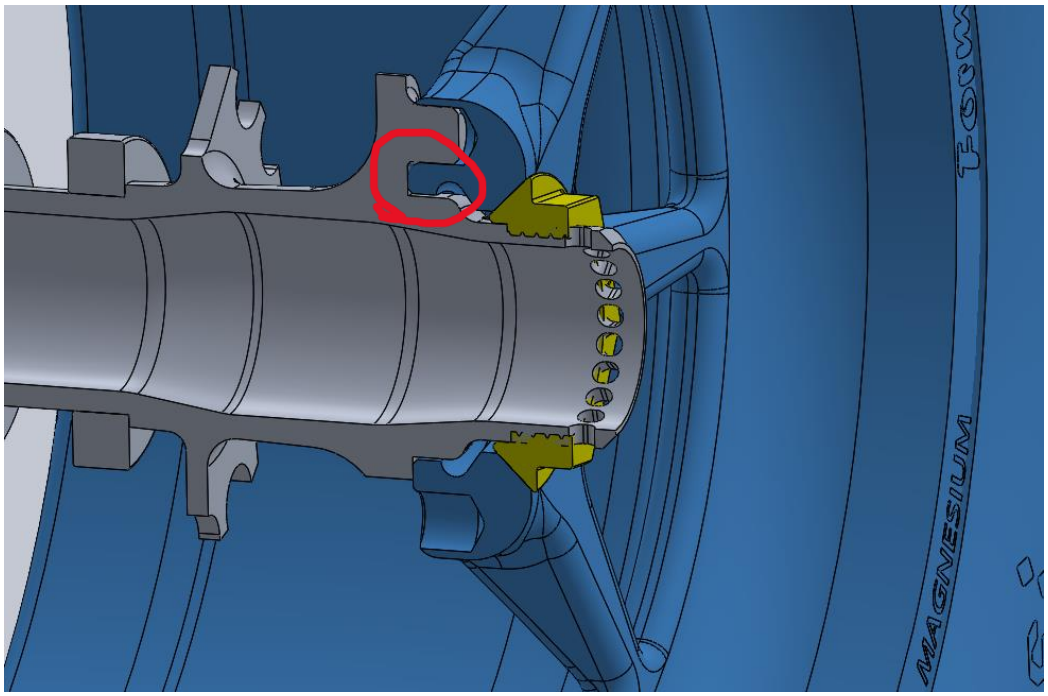


Interestingly, when changing the face that is affected by the remote load to the vertical face, the stress experienced increases significantly. However, applying the remote load to this face is likely not representative. Although the wheel does contact this face, the wheel is not physically connected to it, so it can't pull the face like the deformed view shows.





Upon a visual inspection, it was realized the hub's "shoulder" does not actually touch the wheel. Only the 3 dowel pins and the wheel nut contact the wheel. So, the prior FEA load setups are not representative of real life.

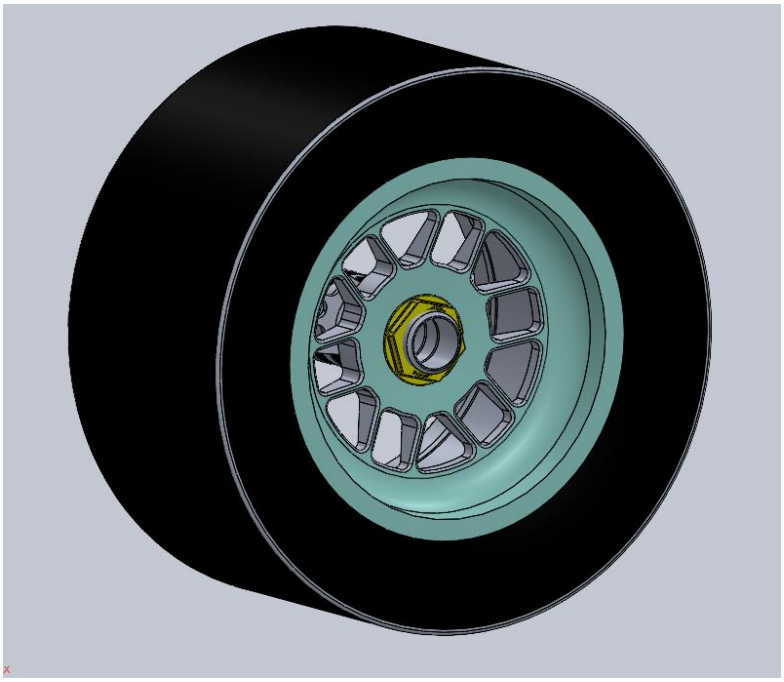


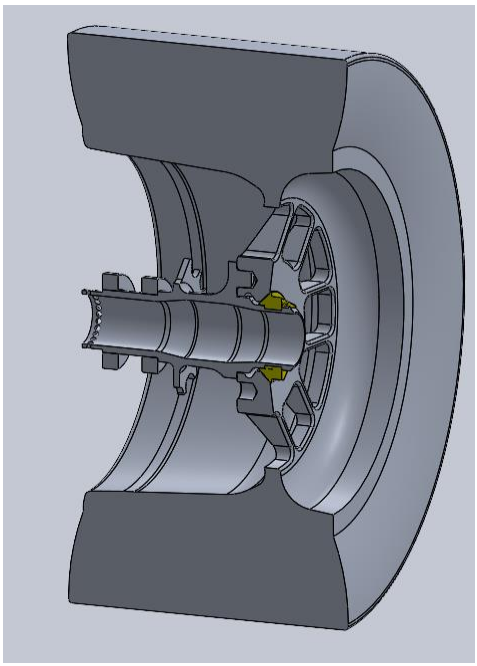
New FEA Hub Setup

This new simulation was unfortunately done after the hub had already been sent off to be manufactured due to the accelerated timeline. The main goal of this simulation is a sanity check to verify that the hub is indeed structurally sound. If the simulation shows the part fails, we can use the old hub design which is thick and has a high FOS.

There also is a new set of loads calculated from the previous year's endurance race log data. These loads are the actual loads seen by the car during the race and were generally lower than the theoretical loads being used in the past.

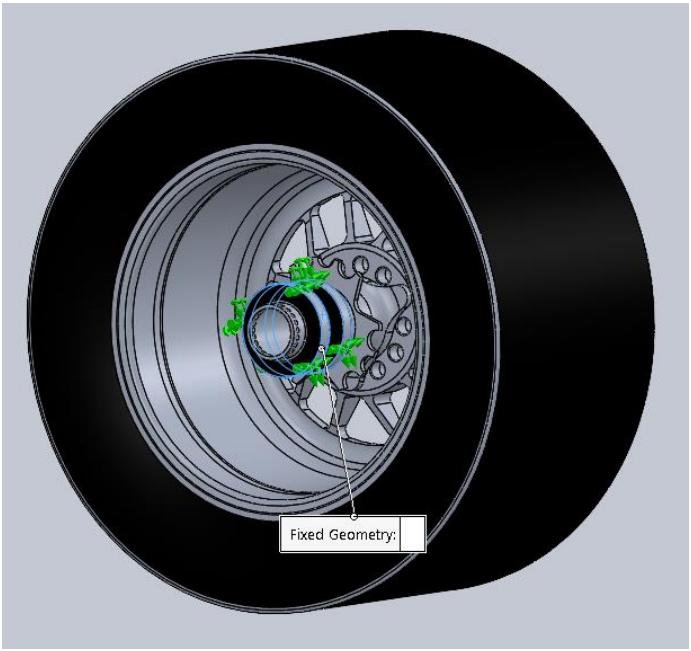
Load Cases Analysis for OptK Force Module									
Set #	Name	Vertical Loads [lbf] (Fz)				Lateral Loads [lbf] (Fy)			
		Front Left	Front Right	Rear Left	Rear Right	Front Left	Front Right	Rear Left	Rear Right
0	Pure Cornering (Left Turn)	62.21	201.68	21.48	265.46	135.19	438.27	46.68	576.87
1	Pure Acceleration	122.32	126.06	216.35	225.71	0.00	0.00	0.00	0.00
2	Pure Braking	194.64	232.50	136.80	178.68	0.00	0.00	0.00	0.00
3	Combined Corner and Brake (Left Turn)	163.47	268.45	93.98	247.63	87.87	144.30	50.52	133.11
4	Combined Corner and Accel (Left Turn)	78.64	160.11	148.91	235.00	72.41	147.42	137.11	216.37
Set #	Name	Longitudinal Loads [lbf] (Fx)				Self Alligning Torque [lbf*ft]			
		Front Left	Front Right	Rear Left	Rear Right	Front Left	Front Right	Rear Left	Rear Right
0	Pure Cornering (Left Turn)	0.00	0.00	0.00	0.00	N/A			
1	Pure Acceleration	109.38	112.72	193.46	201.83				
2	Pure Breaking	199.80	238.66	140.43	183.42				
3	Cone Strike (FR)	109.17	179.28	62.76	165.38				
4	Combined Corner and Brake (Left Turn)	74.00	150.66	140.12	221.13				





Fixed Geometry:

The new setup has the same fixture setup as the original: fixed at the wheel bearings. There is also a bearing connection between the bearings and the hub, like the original.



Connections:

Since the new setup also includes the wheel nut and wheel, we need new connections to constrain these entities. To do so, rigid connections were made between the nut and hub, nut and wheel, and all

three of the hub dowels and the holes they go into on the wheel. A rigid connection is used since it most accurately simulates the real-life characteristics of the system, as the nut preload effectively fixes everything in place. There are bearing connections on the bearings like the original simulation setup.

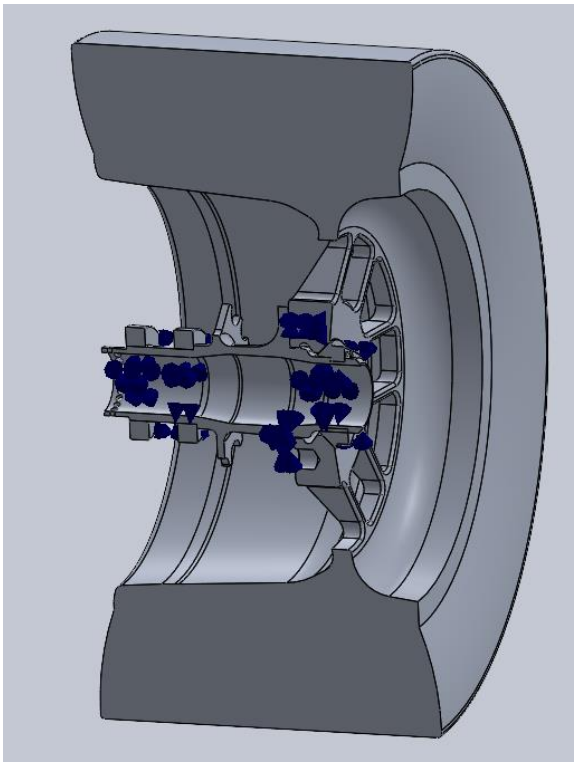
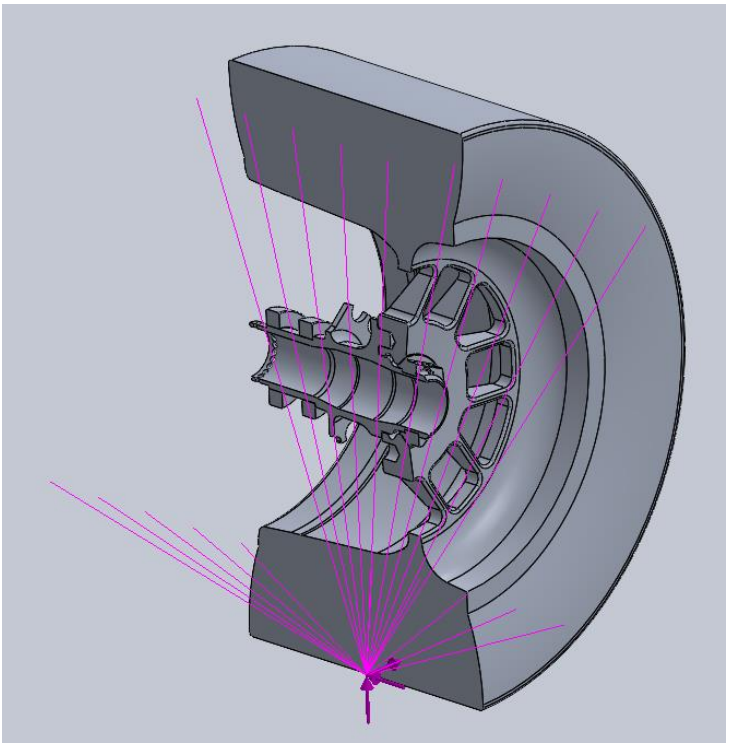
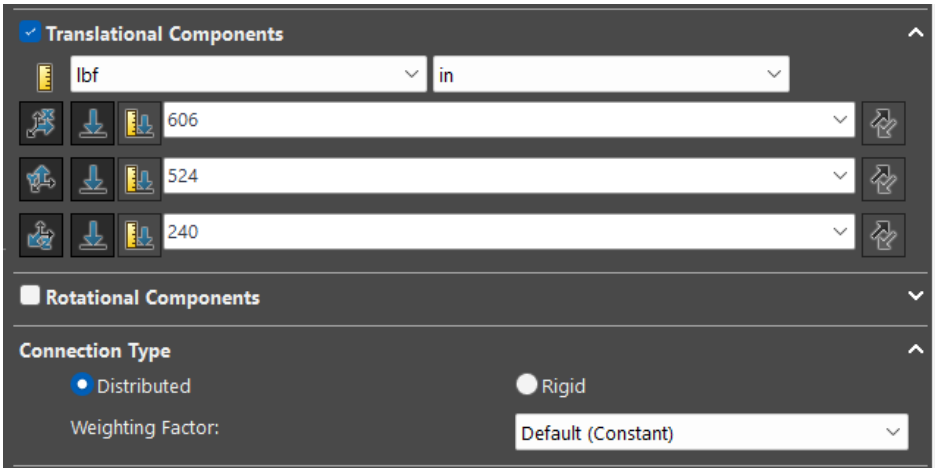


Figure: Connectors marked by the blue arrows

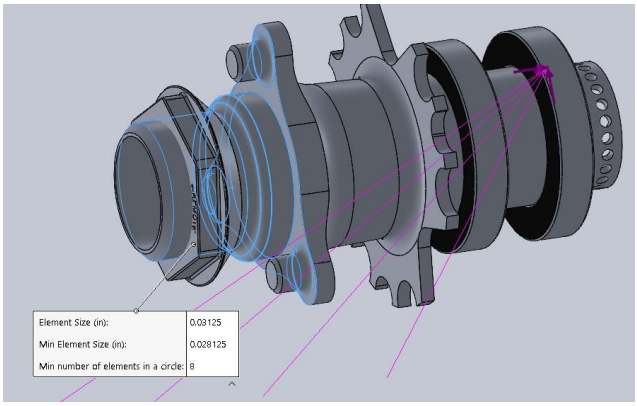
Loads:





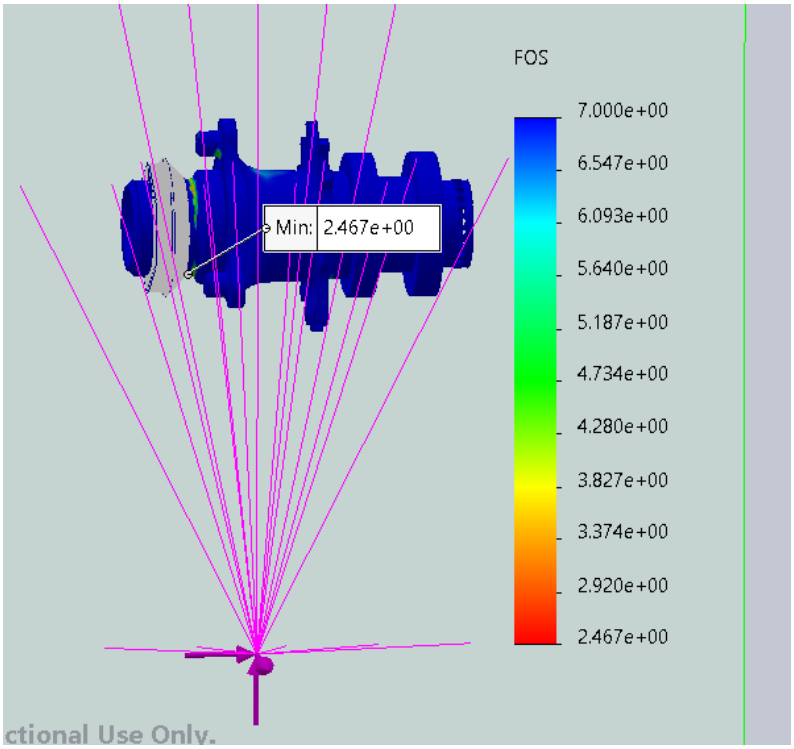
A remote load was used to simulate loads that are happening at the contact patch of the tire. The loads shown in the image are from a cornering, braking, and cone strike scenario happening, which is the maximum load the front hubs will see. This remote load is applied onto the tire in the model. The idea of this model is to make it the same loading path as real life: the tire force at the contact patch creates moments that want to pull the wheel in different directions. The hub's dowels and wheel nut resist these forces, which are the exact forces acting on them.

Mesh:

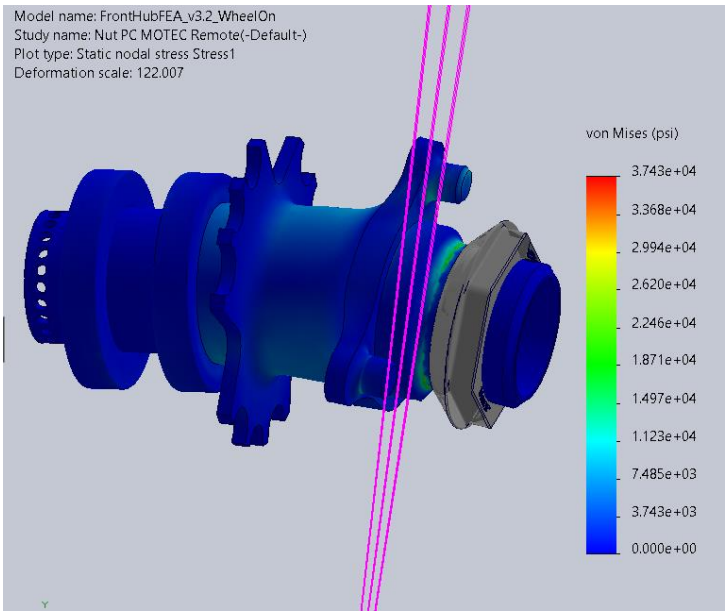


The overall global mesh is a relatively coarse mesh since we are not interested in the wheel, tire, and wheel nut. The coarse mesh speeds up the simulation. Mesh control is applied to the area of interest: the area where the wheel nut sits on the hub. This area sees the highest stress, so a 0.03125in minimum element size is applied to this area. The logic behind this is because there are no features that are smaller than this size since the minimum end mill size that is used will likely be larger than 1/32in.

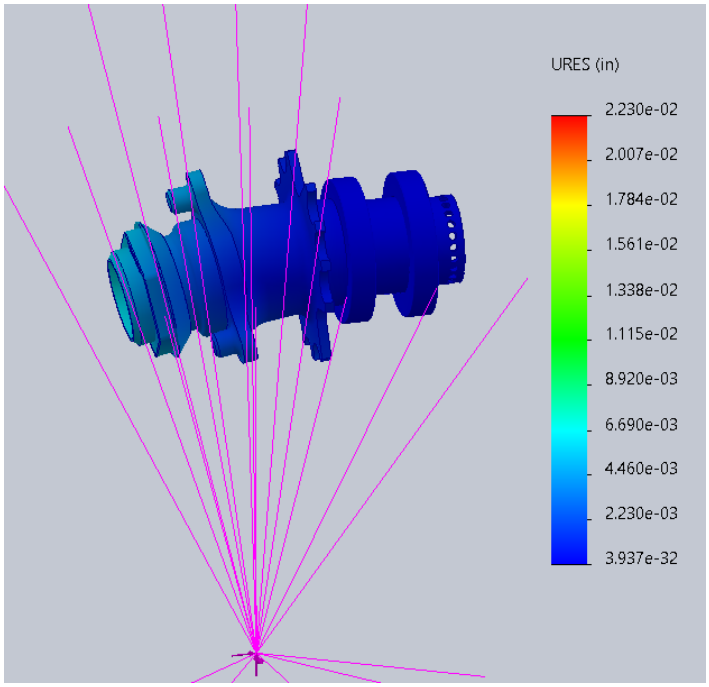
Results:



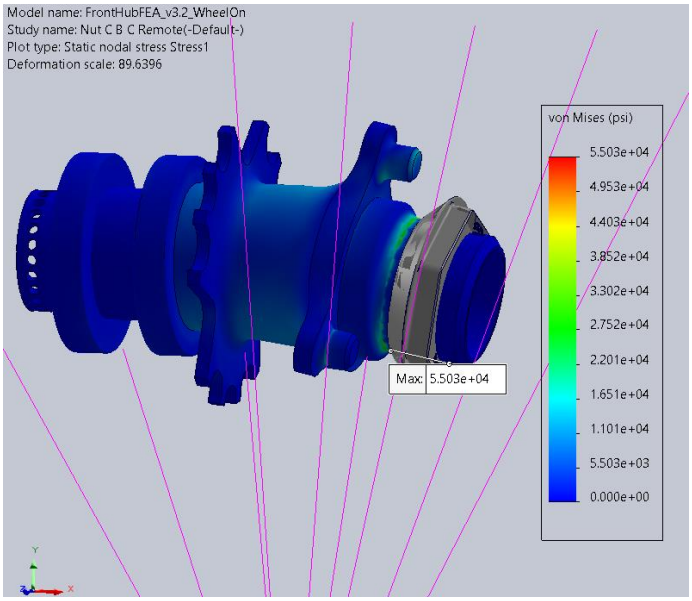
Min FOS = 2.467



Pure Cornering: Max Stress = 37 ksi; Yield Stress: 73 ksi



Max Deflection: 0.0223

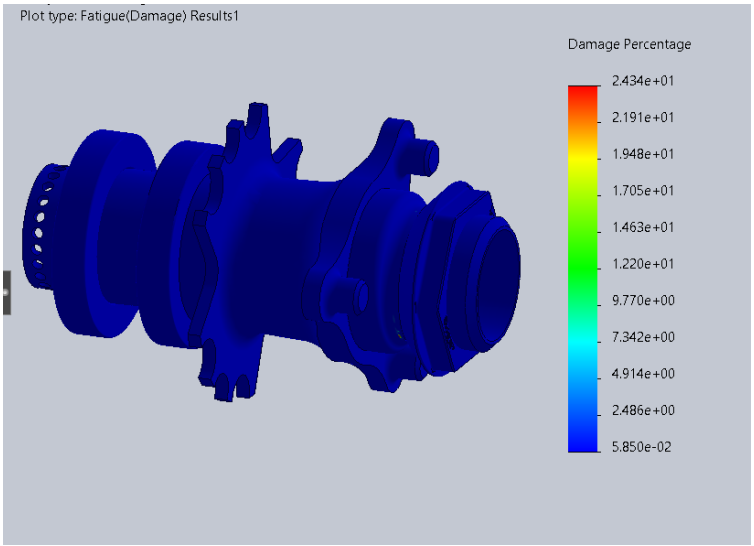


Cornering + Braking + Cone Strike: FOS = 1.3

With a minimum factor of safety of 1.3, we now know that the previous weak point of the hub was not in the middle; it is the base of where the wheel nut sits on the hub. The part of the hub where the walls became thinner still has a factor of safety of 6.09. This means that theoretically, the walls could have gotten thinner along the middle part of the hub, shaving off more weight.

Hub Fatigue

Using the same fatigue table as shown earlier, the hub fatigue results are as follows. A loading ratio was used, which is the ratio of the stress of the inside hub and the outside hub. The hub material properties are the same as the uprights as well.



Pure Cornering: 24% damage

Manufacturing Callout Error



Due to an error of me not calling out the thread runout to the manufacturer, the wheel nuts threads end, then have a gap between the ending threads and the hub body. Due to this, a stress concentration notch is created. To prevent this in the future, one must specify the thread runout on the part drawings. The thread runout controls how the thread tapers off. The correct way for the threads to end is to gradually get shallower and shallower until it blends into the outside surface.

Since this stress concentration is introduced, FEA is done on a new hub with a notch to see if the stress concentration will be a failure point.

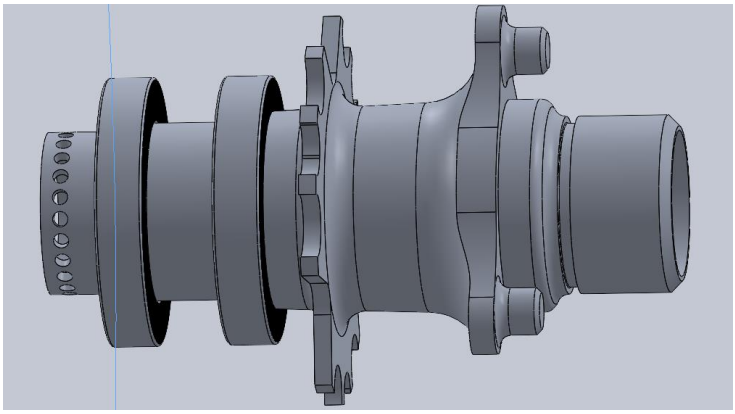


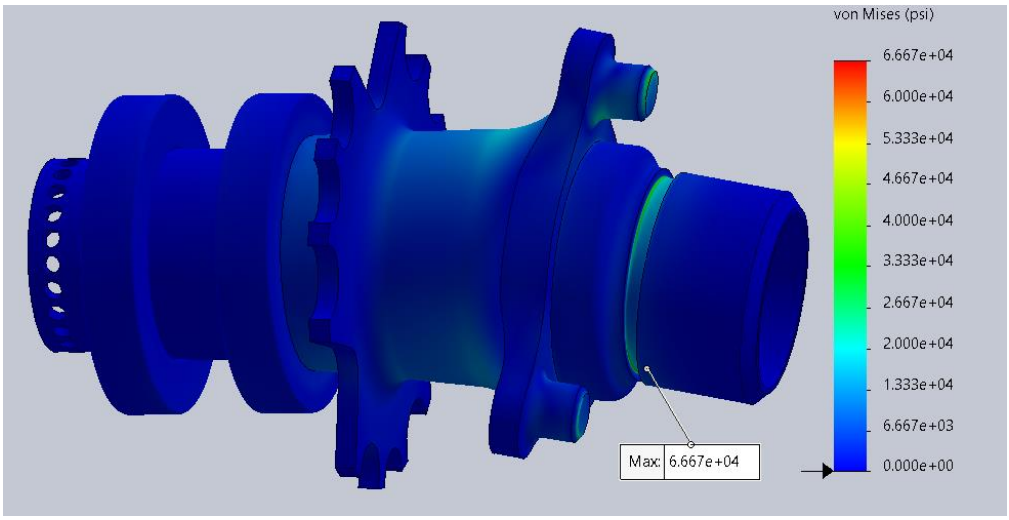
Figure: CAD model of notch

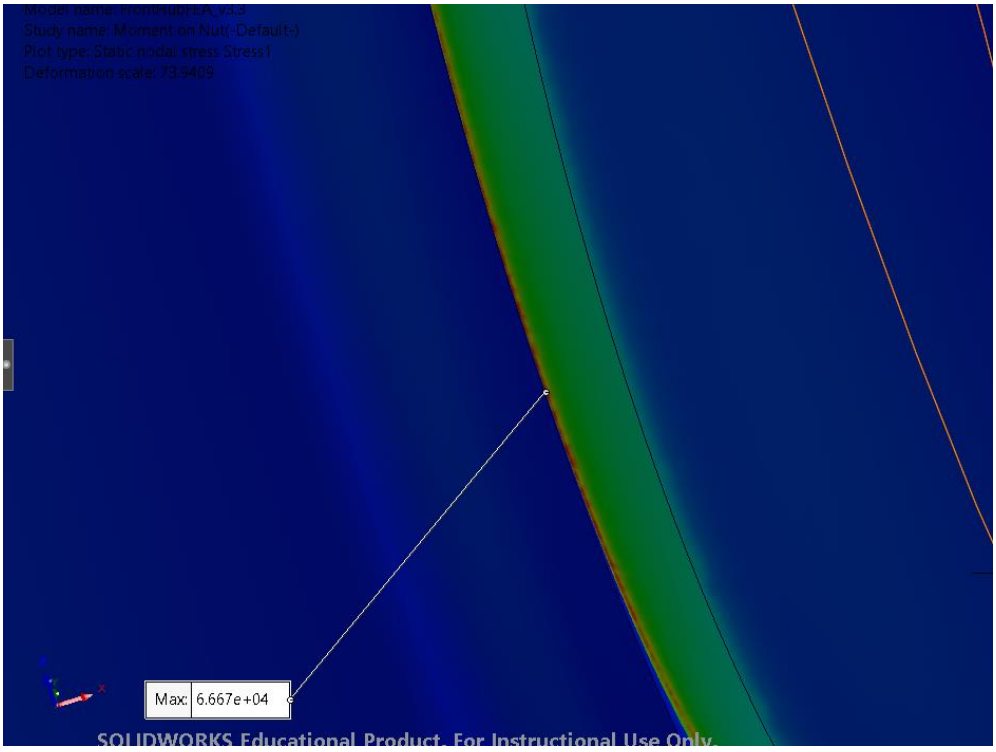
The notch simulation will have the same setup as previously stated, which includes the wheel and wheel nut in the simulation assembly.

Mesh:

The mesh is similar to the previous simulation where it is coarse by default, but mesh controlled in the area of interest. Since triangular elements exhibit constant stress, we must make the elements small enough to be able to pick up stress concentrations. If the element is too large over a small stress concentration, the stress will not be picked up in the tiny area. To help with this, a 0.01in fillet is made inside the groove to help the stress concentration to be recognized in the element.

Results:





There is a significant stress concentration in the groove, dropping the FOS all the way down to 1.1. This load case is the same as the previous simulation, being the max case of an already conservatively

calculated set of loads. Realistically, braking and cornering at maximum G, then hitting a cone is not going to happen, as the driver will likely attempt to either dodge the cone or make the impact less significant. These factors will result in less forces in the event of a cone strike.

New Force Results:

Pure Cornering

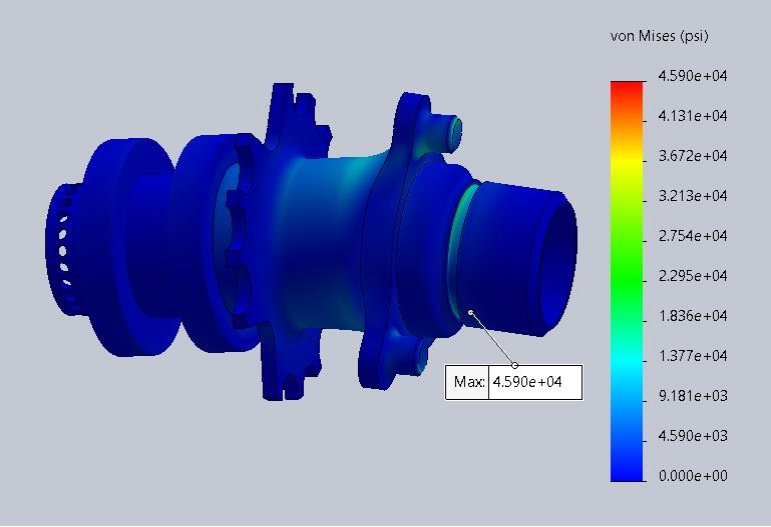


Figure: Max Stress = 45,900 psi

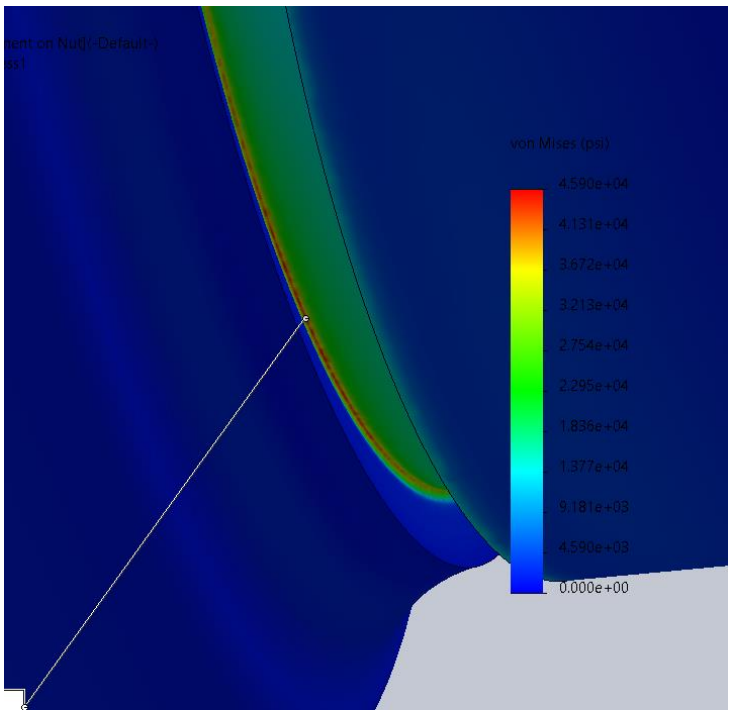


Figure: Max Stress Location

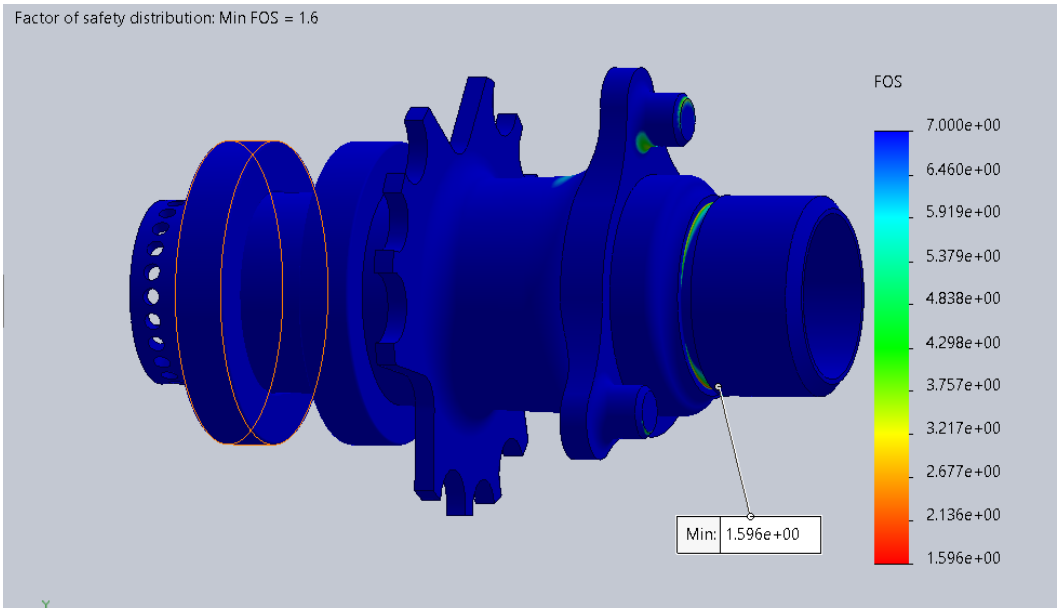
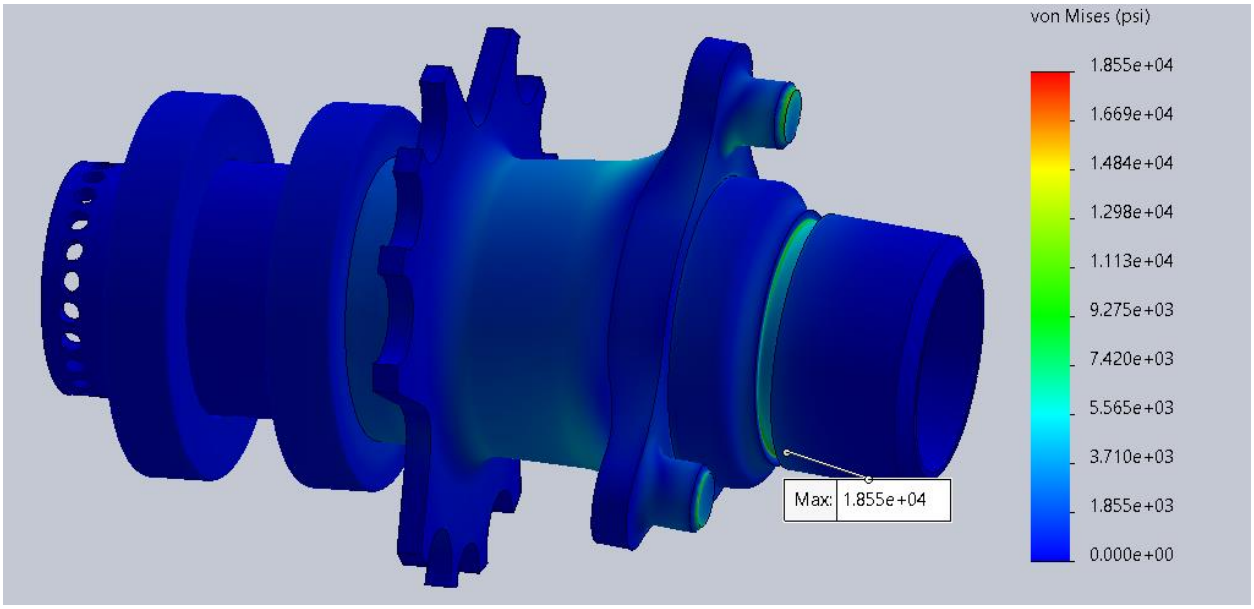
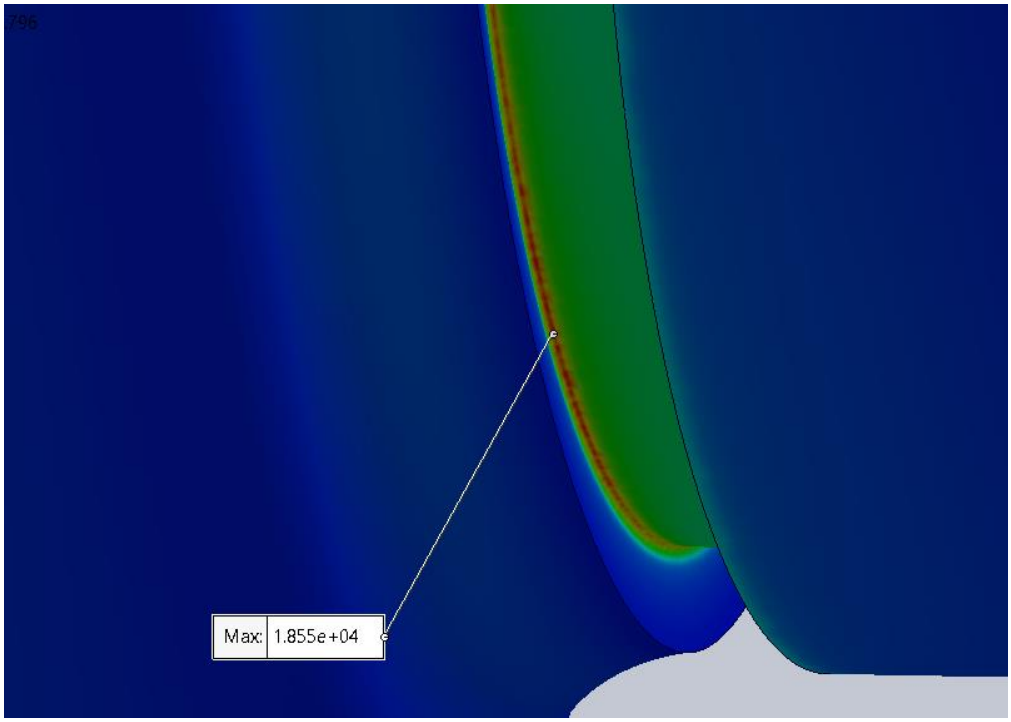
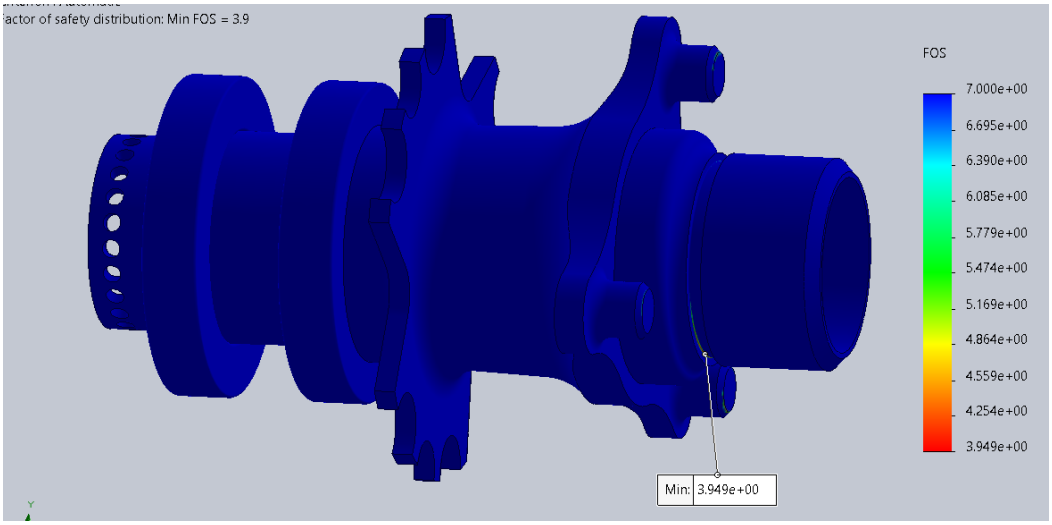


Figure: Min FoS = 1.595

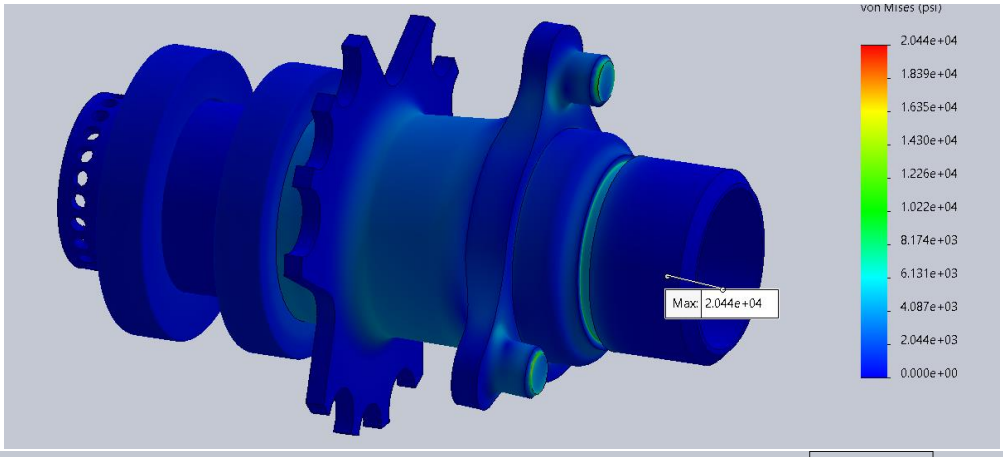
Combined Cornering and Braking

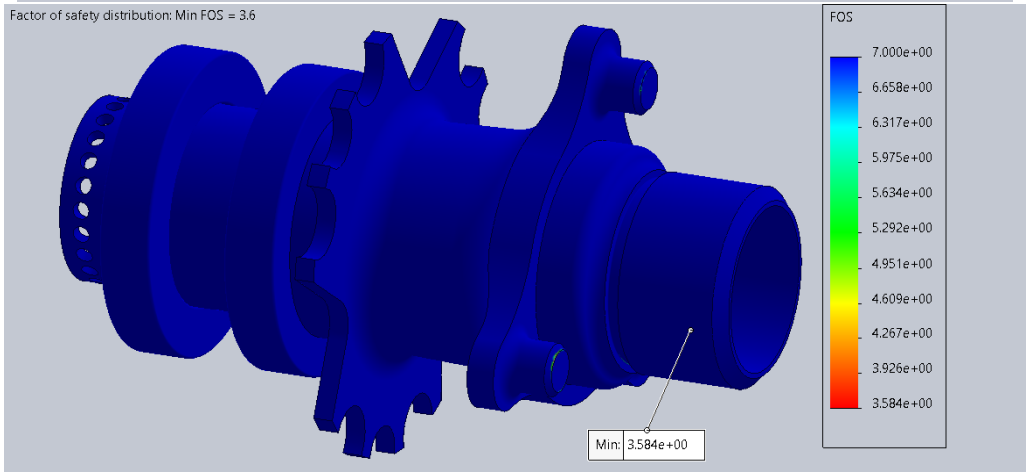






Pure Braking





Fatigue

Cornering + Braking Cases		
Lateral Acceleration Range (g)	Long. Acceleration Range (g)	Instances
1.10 - 1.25	0.45 - 0.60	816
1.35 - 1.50	0.45 - 0.60	475

Values from half endurance
30 Expected Endurances

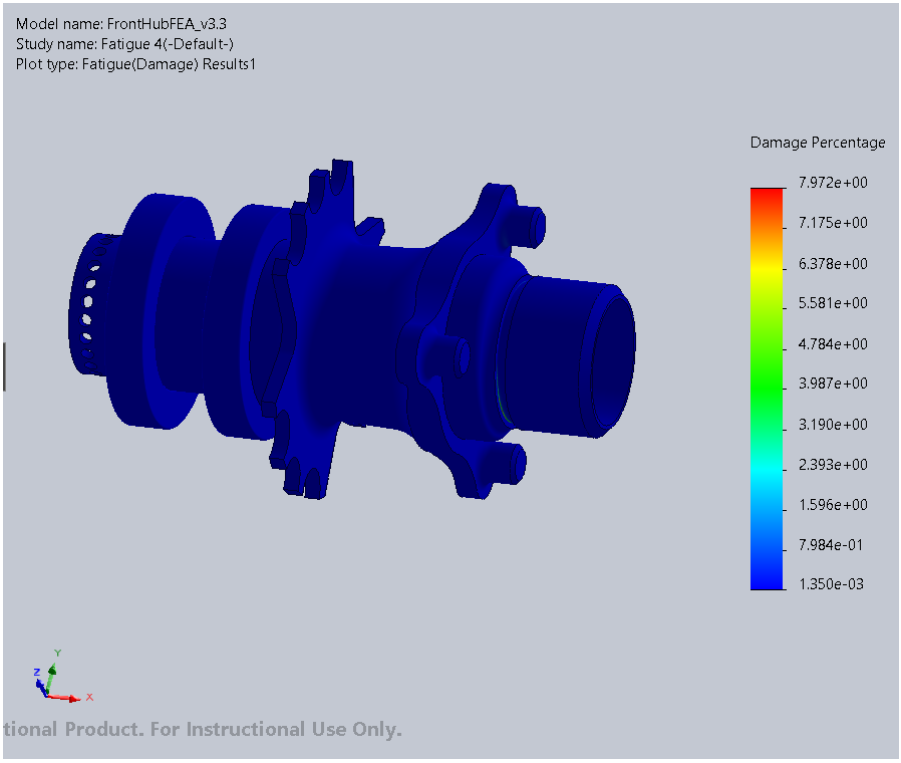
Pure Cornering Cases	
Lateral Acceleration Range (g)	Instances
1.35 - 1.5	390
1.85 - 2.0	9

Values from half endurance
30 Expected Endurances

- From MoTec logs: number of times certain load cases appeared in endurance
- Multiply sum of instance by 2*30 (half endurance values & 30 endurances

Using the same fatigue setup as the uprights, we now will do a study on the hubs. Every parameter should be the same: SN Curve, R=-1 fully reversed loading, Goodman correction factor, Von Mises stress. Since the pure cornering case sees the highest stress, we will do that one first. Based on

the fatigue scenario table, maximum pure cornering loads only occur 9 times per half endurance, so in total there are 540 occurrences.



The plot shows a maximum damage of 7.9% given the load case. This is very safe. The maximum life for this loading case ends up to be 6,774 cycles, well above the 560 target. However, the maximum loads don't happen very often.

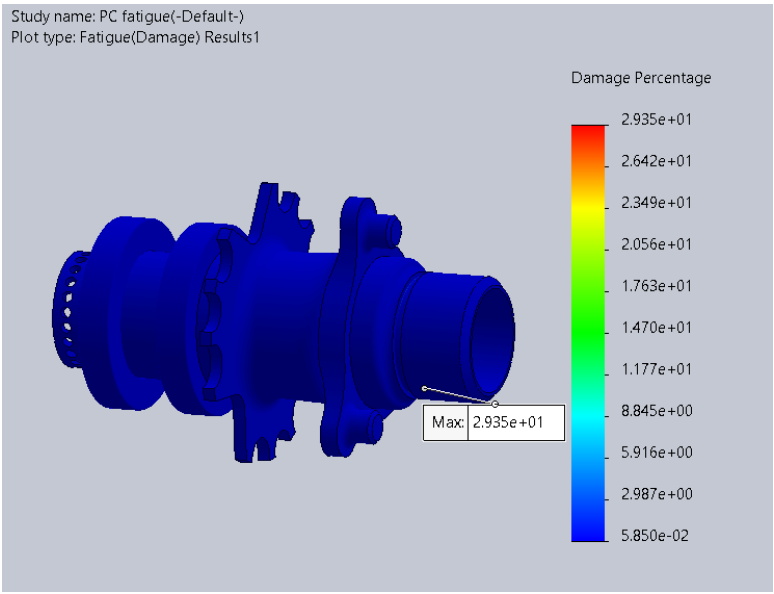
Calculating the loads for pure cornering cases with lower accelerations, we get:

Lat Gs	Inside	Outside
1.9	135.19	438.27
1.75	124.51	403.67
1.6	113.84	369.07
1.5	106.73	346.00
1.4	99.61	322.93
1.3	92.50	299.87

Units are in lbf

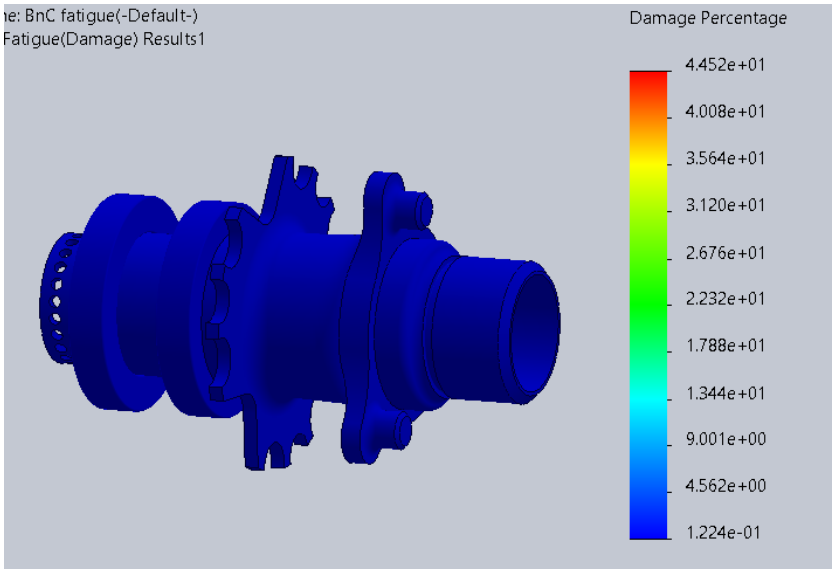
Finding the loads for the inside tire is important in order to get an accurate stress fluctuation. By dividing the stress seen in the inside wheel by the larger stress on the outside, we get a stress loading ratio of –0.278. This is more accurate than assuming fully reversed load with RL = -1 or 0 based loading RL = 0.

For a medium case of 1.5 lateral g's and a goal of 23,400 cycles:

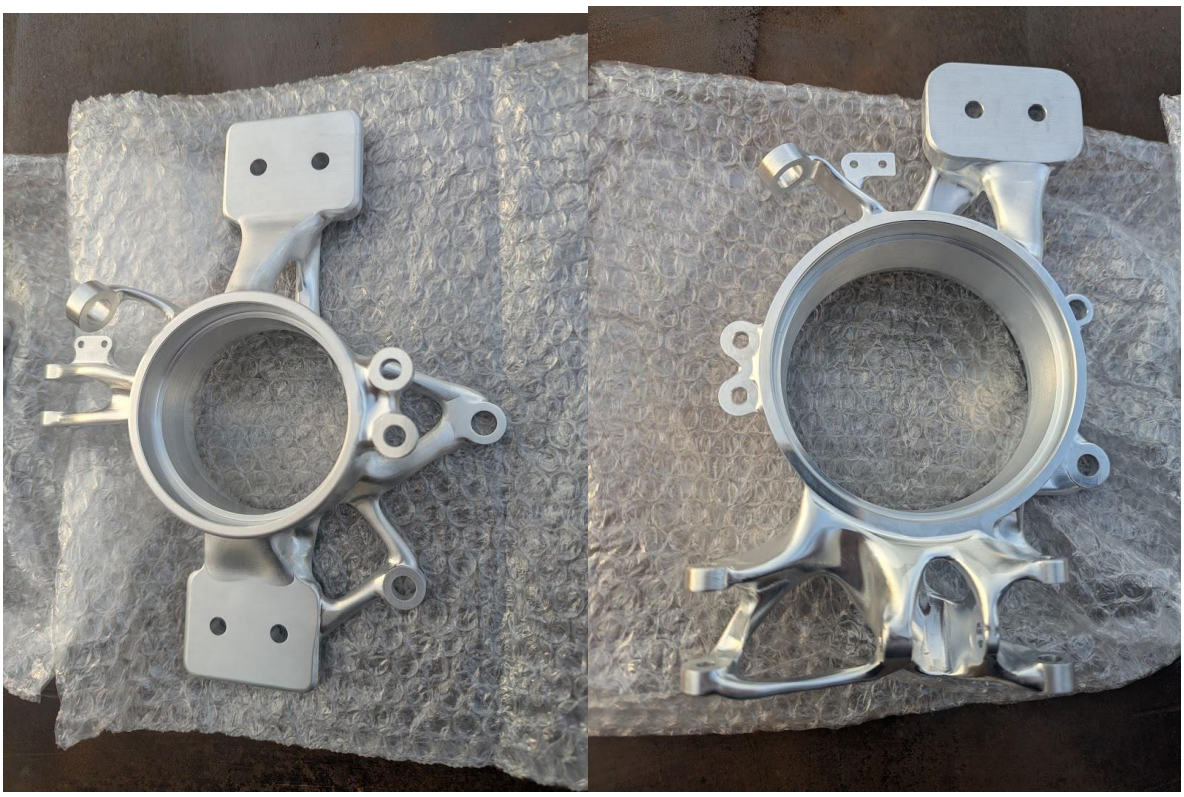


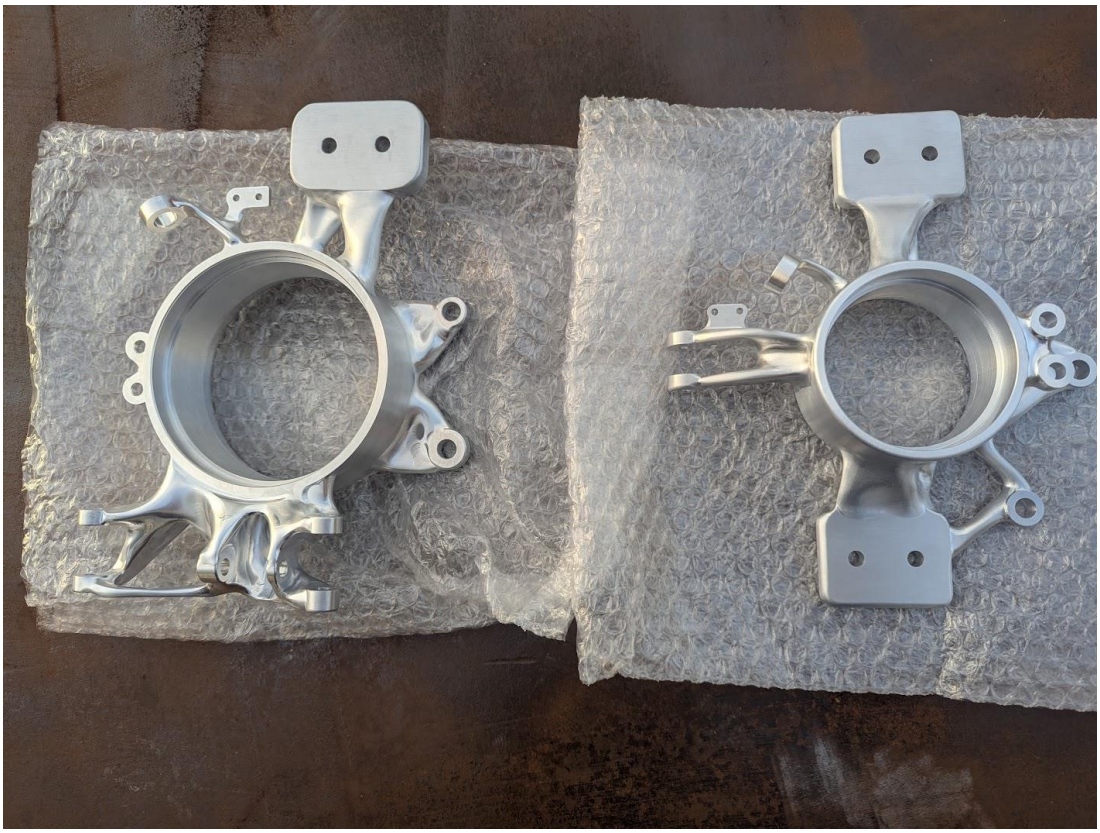
A damage of 29.35% over the endurance goal is below 1.5 FOS. However, due to reliability issues, the amount of testing days the car will go through has been significantly reduced as of March. Therefore, this low damage amount is likely to be much lower during competition.

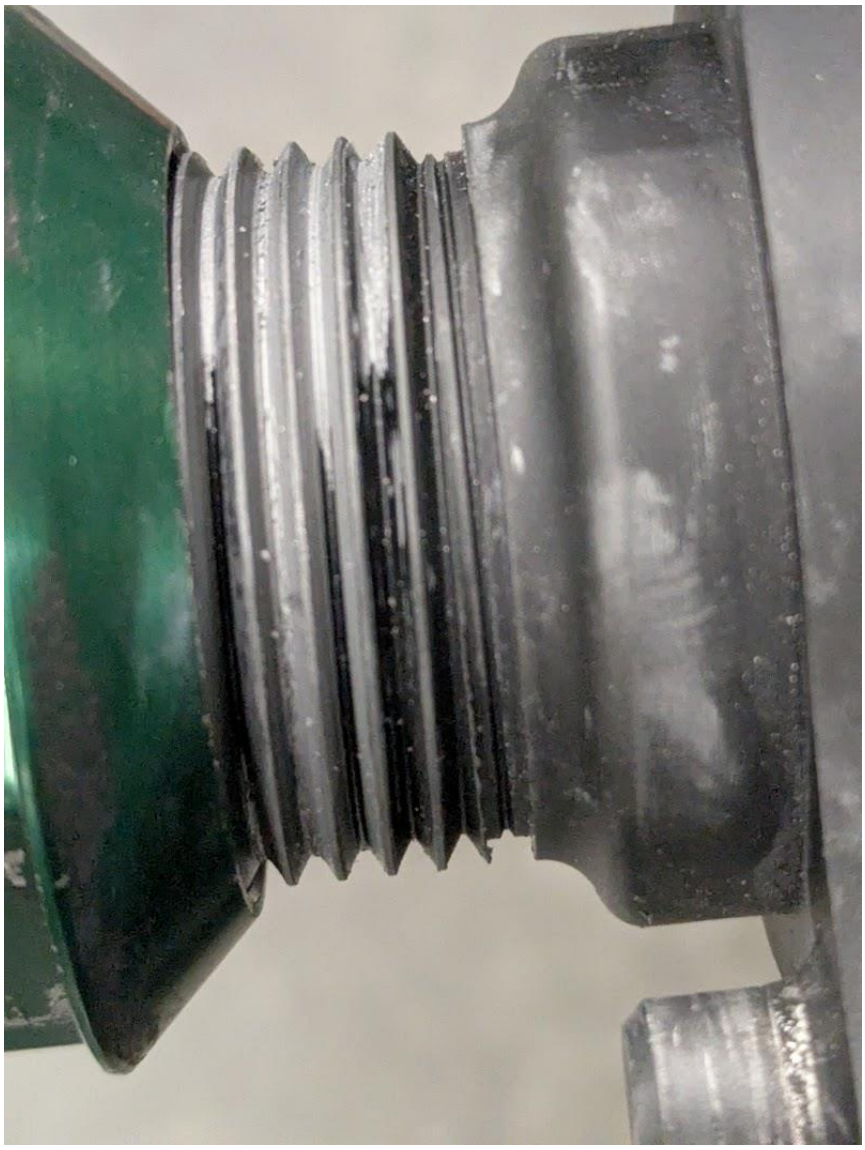
For the combined braking and cornering load case, the damage is around 45%. This case used a loading ratio of –0.61 and a cycle goal of 48690 calculated from the fatigue table shown earlier.



Manufacturing







The hub manufacturer created threads with an abrupt end, presenting a risk for shear failure due to the notch created on the cylindrical hub.